

29

Chronic Medical Conditions: Pulmonary Disease, Organ Transplantation, and Diabetes

JONATHAN H. WHITESON, JEFFREY COHEN, AND SOFIYA PRILIK

SUMMARY PEARLS

- As the US population ages, an increasing number of individuals are living with chronic disabling medical conditions requiring the perspective and skill set of a physiatrist and rehabilitation team.
- Chronic obstructive pulmonary disease is the leading cause of chronic disabling lung disease and the third leading cause of death globally.
- Pulmonary rehabilitation is indicated for individuals physically and functionally limited by chronic lung diseases and encompasses exercise (aerobic, strength, posture, breathing), education, nutrition, and supportive therapies.
- Pulmonary rehabilitation should be delivered through the continuum of care from the intensive care unit, in the acute medical units, in inpatient rehabilitation and skilled nursing facilities, and in the outpatient and community setting.
- Pulmonary rehabilitation before and after lung transplant addresses the pulmonary, musculoskeletal, and neurologic functional deficits, which is integral to the success of lung transplantation, an increasingly viable and utilized surgical treatment for individuals with chronic lung diseases.
- The use of prehabilitation and the introduction of an intensive early rehabilitation program for liver transplant recipients has been found to be well tolerated and feasible in the intensive care unit.
- When prescribing a rehabilitation program for potential kidney transplant patients (preoperative or postoperative), one must consider their markedly impaired cardiorespiratory and musculoskeletal status.
- Kidney transplant candidates are uniquely suited for rehabilitation because they experience a profound loss of functional capacity while waiting years on dialysis for a donor kidney; hence, exercise training in kidney transplantation recipients results in significantly improved exercise tolerability.
- In the immediate and early stages following heart or lung transplant, the physiatrist plays a crucial role in identifying and managing functional deficits to optimize recovery and transition transplant recipients to the most appropriate outpatient rehabilitation programs.
- Outpatient cardiac rehabilitation is essential in optimizing the altered cardiac physiology noted following heart transplantation, including resting tachycardia, a delayed rise in heart rate in response to exercise, and marked physiologic deconditioning as evidenced by a significantly reduced maximal oxygen consumption.
- Exercise is a key component in the prevention and management of diabetes mellitus, and physiatrists are *ideally* trained to prescribe rehabilitation programs to manage associated diabetic complications.

Introduction

Over the past several decades, the prevalence of chronic disease in US adults has continued to rise.²⁵⁶ According to data from the 2018 National Health Interview Survey, over half the adult US population is living with at least one of ten chronic conditions, 24.6% are living with one chronic health condition, and 27.2% are living with two or more chronic conditions.³³ These figures likely underestimate the true number of US adults living with chronic disease because an additional 6.9% of US adults reported

ever having long COVID, and 3.4% had long COVID in 2022.⁵ While 13% of the total adult population live in the community with a disability,²⁵⁷ the rate of disability is higher in rural areas (14.7%) compared to urban (12.6%) and significantly increases with age rising to 46% of people over age 75 years. As the American and global population ages, there is an increasing number of people living with chronic medical conditions and disability, often with multiple comorbidities and functional limitations impacting quality of life. The physical medicine and rehabilitation (PM&R) physician or physiatrist and the rehabilitation medicine

team play an ever-increasing role in the continuum of care in our evolving health care system in the management of such patients. Individuals with chronic pulmonary and cardiovascular complications, diabetes mellitus, renal and hepatic diseases, and those undergoing solid organ transplants (SOTs) require rehabilitative services and the expertise of a PM&R physician. It is essential that the rehabilitation medicine specialist be knowledgeable in the management of such conditions and potential complications.

Pulmonary Rehabilitation

Definition

Pulmonary rehabilitation (PR) is defined as an evidence-based, multidisciplinary, comprehensive intervention that can be integrated into the management of individuals with chronic lung disease across the continuum of care.²²⁷ PR is a multifaceted program, which involves optimization of medical management, physical conditioning, education, nutritional counseling, coping skills, and psychosocial support with the goal of reducing symptoms, increasing strength exercise tolerance, improving functional status, promoting psychological well-being and quality of life, and reducing health care resource use.²³⁰

Classification of Pulmonary Diseases

Pulmonary diseases can be classified into four major categories: obstructive lung disease, restrictive lung disease, pulmonary vasculature disease, and hypoventilation syndromes. Obstructive lung diseases, the most common diagnostic presentation for PR, are characterized by expiratory airflow limitation on pulmonary function testing (PFT) and include chronic obstructive pulmonary disease (COPD), chronic bronchitis and emphysema, bronchiectasis, cystic fibrosis (CF), and asthma. Expiratory airflow limitation results from a decrease of the diameter of the bronchial lumen due to mucous blockage, inflammation and edema of the bronchial walls, or dynamic expiratory collapse due to loss of supportive structure. Restrictive lung diseases, characterized by a reduction in all lung volumes on PFT, are classified as *intrinsic* with lung parenchymal involvement, as in interstitial lung diseases, or *extrinsic* to the lung, as in obesity and neuromuscular disorders. These conditions cause a physical reduction in lung size/volume or increased lung stiffness, resulting in a decreased volume of inspired air. Disorders of the pulmonary vasculature include pulmonary embolism (PE), pulmonary hypertension (PH), and pulmonary venoocclusive disease. Hypoventilation syndromes include obstructive sleep apnea and obesity hypoventilation syndrome. Many respiratory diseases fit into one of these categories, but individuals may present with a combination or mixed pattern of disease. The most common reason for referral to PR is COPD.

Differentiating Chronic Bronchitis and Emphysema

Chronic bronchitis is a clinical diagnosis based on the volume of daily mucous production over months and years cleared by cough and typified by acute exacerbations, often infective, and relatively quiescent remissions. Excess mucus, resulting from an increased number and size of mucus-secreting goblet cells stimulated by elements in tobacco smoke and by infection, blocks airways causing obstruction. Mucous plugging leads to ventilation/perfusion (V/Q) mismatching and can progress to significant hypoxemia,

marked hypercapnia, PH, and right heart failure (i.e., cor pulmonale, the classic **blue bloater**, with dyspneic, cyanotic, and volume overload). Airway inflammation and bronchospasm contribute to airflow obstruction. Emphysema is a pathologic diagnosis based on evidence of airway destruction distal to the terminal bronchiole on high-resolution lung computed tomography (CT) scan or lung biopsy (now rarely done). Elements in tobacco smoke activate proteases that cause destruction of elastins and supportive tissues in the lung interstitium leaving bronchial walls unsupported and prone to collapse on expiration resulting in airway obstruction. There is no V/Q mismatching due to equal destruction of respiratory epithelium and vasculature resulting in significant hypoxemia, mild hypercapnia, and air trapping (the classic **pink puffer**, with dyspneic, noncyanotic, and barrel chest).

Epidemiology of Lung Disease

COPD, the most common form of lung disease and the third leading cause of chronic morbidity and mortality in the United States and worldwide, is primarily attributable to smoking.^{51,159}

Approximately 25% of adults with COPD have never smoked, and workplace exposures likely contributed to their disease.²⁴² Worldwide, other significant contributors include fossil-fuel burning furnaces (for home heating and cooking) and environmental pollution or smog. Encouragingly, current smoking rates among US adults has declined from 20.9% in 2005 to 14% in 2017 to 11.5% in 2021, according to the US Centers for Disease Control and Prevention (CDC).⁶⁵ However, each day in the United States, about 3200 youth under 18 years of age smoke their first cigarette, and more than 400 youth under 18 years of age become daily cigarette smokers.²⁴⁰ Cigarette smoking accounts for more than 480,000 deaths every year, or about one in five deaths. Approximately 80% of COPD deaths are caused by smoking. According to the CDC, in 2021, there were 14.2 million American adults (6.5% of the adult population) currently diagnosed with COPD, but an estimated 12 million more were living with undiagnosed COPD.¹⁵⁴ COPD facts and statistics are shown in [Box 29.1](#).^{167,221}

Restrictive pulmonary disease is most commonly caused by the following:

- Interstitial lung diseases: conditions that cause pulmonary fibrosis (autoimmune related: scleroderma, rheumatoid arthritis; inorganic dust exposure: asbestosis, silicosis; idiopathic: pulmonary fibrosis, sarcoidosis)

• BOX 29.1 Facts Regarding Chronic Obstructive Pulmonary Disease (COPD)

- The majority of patients with COPD have asthma.
- Asthma is the most common childhood chronic disease.
- Causes of COPD are asthma, chronic bronchitis, and emphysema, singly or in combination.
- Smoking is the primary risk factor for COPD. Approximately 80% of COPD deaths are caused by smoking.
- Alpha₁-antitrypsin deficiency is underrecognized by physicians.
- COPD is the third leading cause of death worldwide.
- In 2019, there were 3.23 million deaths worldwide from COPD.
- In 2021, in the United States, >72,000 females died compared to >66,000 males.
- In 2020, the cost to the United States for COPD was \$49 billion and projected to rise to \$60.5 billion by 2029.

- Neuromuscular disorders (e.g., amyotrophic lateral sclerosis, spinal cord injury [SCI], muscular dystrophy)
- Lung/thoracic volume reduction (e.g., pleural disorders: mesothelioma and effusion, lung resection; abnormalities of the thoracic spine: scoliosis/kypnosis; abnormalities of the chest wall: pectus excavatum)

In SCI, lesions above C3 paralyze the diaphragm (C3–C5) as well as the intercostal muscles of respiration, and patients will require mechanical ventilation. Injury below C5 leaves the diaphragm intact, and respiratory impairment depends on the level of injury and extent of intercostal weakness. Below T6, no significant inspiratory muscle impairment is expected because external (and internal) intercostals are innervated by T1 to T6 roots. Expiration is typically gravity assisted, but forced expiration (as in coughing) requires force against a closed glottis using the abdominal muscles. Lower thoracic lesions will impair abdominal muscle wall control and effective coughing, and rectus abdominis and external/internal obliques are innervated by T5 to L1 roots. According to the National Spinal Cord Injury Statistical Center, as of 2018, there were an estimated 255,000 to 383,000 people with SCI or spinal cord dysfunction living in the United States.¹⁴⁴ Of these patients, 59.7% had tetraplegia (incomplete 47.1% or complete 12.5%) with significant involvement of the muscles of respiration. Duchenne muscular dystrophy is one of the more common neuromuscular diseases that cause restrictive pulmonary dysfunction, with an incidence of 1 per 3500 male infants in the United States.¹⁶³

Acute PE ranks third among the most common types of cardiovascular diseases (CVDs), with an annual incidence estimated to be 60 to 70 per 100,000. Untreated, the mortality rate is 30%, dropping to 8% if appropriately treated. Risk factors include age, personal history of deep vein thrombosis or PE, active malignancy, disabling conditions such as heart or respiratory failure, coagulation disorders, hormone replacement therapy (HRT), and oral contraception.²⁶

Primary, or idiopathic, PH has no clear etiology but is likely related to genetic susceptibility. Approximately half the cases of PH occur in association with other diseases, such as connective tissue disease, heart disease (including congenital heart disease), coronary artery disease, systemic hypertension, liver disease (cirrhosis), and lung diseases (including blood clots to the lungs and emphysema). The exact incidence of PH is not known, but about 200,000 related hospitalizations occur annually in the United States. The incidence increases with age and is more common among females and non-Hispanic Blacks. PH is associated with hypoxia and right heart failure. Appropriate oxygen supplementation in patients with COPD can prevent or reverse PH and is part of the management of cor pulmonale.¹⁹¹

Between 5% and 10% of the US adult population have diagnosed obstructive sleep apnea, 7% have breathing pauses during sleep that put them at risk for more severe sleep events, and 23% to 59% snore. Sleep-disordered breathing is more likely to be reported by males than females. However, over 80% with signs and symptoms of moderate to severe sleep-disordered breathing remain undiagnosed. Sleep-disordered breathing is associated with increased risk for coronary artery disease, heart failure, and stroke.²²³ As a novel cause of pulmonary disease and disability, acute infection with SARS-CoV-2 (the virus that causes COVID-19) is noted to result in acute injury to the lung parenchyma, large and small airways, and pulmonary vasculature. While most with acute lung injury recover with no lasting symptoms or pathologic findings, some go on to develop lung fibrosis and progressive respiratory

symptoms and functional limitations.¹²⁶ Approximately 7% of patients hospitalized with COVID-19 have been noted to have persistent lung radiologic abnormalities for 12 months postinfection.²³⁷ While this pathologic process may be part of the symptom complex of postacute sequelae of SARS-CoV-2, or long COVID, many with mild acute COVID-19 and persistent respiratory symptoms, including shortness of breath at rest and/or on exertion, have normal pulmonary radiologic, vascular, and function testing. A number of theories have been put forward to account for this, including hyperventilation, cardiac dysfunction, physiologic deconditioning, and anxiety.²⁷³ Although the exact etiologic mechanism(s) has not been elucidated to date, it is likely to be multifactorial.

Symptoms of Pulmonary Disease

Typical presenting symptoms of pulmonary disease include a history of worsening shortness of breath on progressively less intense exertion, cough with or without sputum production, progressive weakness, and fatigue. Functional decline is evidenced by limitations with activities of daily living (ADLs), balance disorder and falls, progressive difficulty with steps, and limitations with community ambulation. Often patients present late with COPD due to a progressive accommodation/restriction in lifestyle to avoid symptoms. As such, social isolation and depression are often present. Shortness of breath is multifactorial, and it is important to rule out other significant causes, including cardiac disease. Anxiety and cognitive decline are also common. Quality of life is universally impacted in later, more advanced stages of chronic pulmonary diseases. It is worth stating that many people with PFT or radiologic changes consistent with early chronic obstructive or restrictive lung disorders will be asymptomatic. However, treatment should not be delayed until symptoms develop because prompt management can impact the course of disease and maintain function.

Treatment Options in Pulmonary Disease

Because the majority of patients referred for PR have COPD, treatment options presented here will focus on this disease. Treatment involves a multifaceted approach, including optimal medical management, oxygen therapy, and PR. With advanced pulmonary impairment, other treatment options, including mechanical ventilation, can be considered. Surgical options, including partial lung resection (lung volume reduction surgery [LVRS]) and lung transplant, may be helpful. A comprehensive summary of treatment options in COPD is found in the Global Initiative for Chronic Obstructive Lung Disease (GOLD) *Pocket Guide to COPD Diagnosis*.⁹⁴

General Medical Management

General medical management starts with making/confirming the correct diagnosis. A detailed history and physical exam, in combination with radiologic testing and PFTs, will assist in making an accurate diagnosis and establishing the severity of disease. On PFT studies, COPD is diagnosed by a value of less than 0.7 for the ratio of forced expiratory volume in 1 second (FEV₁) to forced vital capacity (FVC). The GOLD classifies the severity of COPD based on FEV₁ percentage compared to normal (Table 29.1). Life expectancy in COPD is strongly correlated with PFTs. The FEV₁ is the best predictor: A 30% mortality rate within 1 year is associated with an FEV₁ less than 750 mL, and 50% within 3 years.

TABLE 29.1 GOLD Classification of Airflow Limitation Severity in COPD

Class	Severity	FEV ₁ % Predicted
GOLD 1	Mild	FEV ₁ ≥80% predicted
GOLD 2	Moderate	FEV ₁ <80% but ≥50% predicted
GOLD 3	Severe	FEV ₁ <50% but ≥30% predicted
GOLD 4	Very severe	FEV ₁ <30% predicted

COPD, Chronic obstructive pulmonary disease; FEV₁, forced expiratory volume in 1 second; GOLD, Global Initiative for Chronic Obstructive Lung Disease.

Survival is also negatively impacted by ongoing tobacco use, failure to vaccinate as recommended against influenza, poor nutritional status, and limited mobility.

Elevated dyspnea rating, although more subjective, has also been shown to correlate with poor survival, as have features of cor pulmonale and more extensive disease on chest CT.

Tobacco/smoking *cessation* is key for those who continue to use tobacco products despite a diagnosis of COPD. Assessing readiness to quit is important because motivation and a strong commitment to “stick with it” is essential for long-term success. A simplified treatment plan encompasses five *As*:

- **Ask** if they are smoking and want to quit
- **Advise** patients to quit, including the benefits of quitting and harms of continuing
- **Assess** readiness and willingness to quit
- **Assist** in setting a quit date and nicotine replacement therapy (NRT) plan
- **Arrange** follow-up for support

A comprehensive approach includes supportive counseling (individual and/or group), behavioral change strategies, centrally acting medications (bupropion, varenicline), and NRT. With this approach, long-term quit rates are 25% to 40%. Centrally acting medications should be prescribed in discussion with other treating physicians because of side effects, medication interactions, and the need for close follow-up. The US Food and Drug Administration (FDA)-approved NRTs include patches, gum, lozenges, inhalers, and nasal spray. Enhanced success is found combining daily 24-hour use of the patch and slowly decreasing the dosage from 21 mg to 14 mg to 7 mg, then off, with the gum, lozenge, inhaler, or nasal spray used as needed.¹⁵⁰ Use of electronic cigarettes as an aide to tobacco cessation has been discussed in the literature but is not FDA approved at this time.¹⁶ Vaccination annually for influenza, per protocol for pneumococcal pneumonia, and per current guidelines for COVID-19 and respiratory syncytial virus significantly decreases COPD exacerbations and associated infections, including pneumonia. Hospitalizations and all-cause mortality are similarly reduced.^{119,182}

Although the PM&R physician may not be the primary medication-prescribing physician for patients with COPD, it is essential to fully appreciate the spectrum of medications available.⁴² Understanding of the anatomy and physiology of the pulmonary system helps guide pharmacologic therapy. Vagal parasympathetic stimulation of cholinergic receptors on bronchial smooth muscle cells increases their contraction leading to bronchoconstriction. Vagal stimulation also increases goblet cell mucous production. Sympathetic activation has the reverse effect. Other smooth muscle receptors include beta₂-adrenergic

stimulation, which relaxes smooth muscle. Cellular receptors include histamine and leukotriene. Cellular enzymes include cyclic nucleotide phosphodiesterase (PDE), which inactivates cyclic adenosine monophosphate (cAMP); cAMP inhibits inflammation and relaxes smooth muscle. Other cells involved in the local inflammatory response underlying COPD exacerbations include mast cells and lymphocytes.

Bronchodilators include inhaled or nebulized short- or long-acting (or combined) quaternary anticholinergic (vagolytic) and/or beta₂-adrenergic agonists (sympathomimetic). Intravenous, oral, or inhaled⁴⁹ corticosteroids (alone or in combination with bronchodilators) decrease the bronchial inflammatory response. Mast cell stabilizers and antileukotrienes reverse the allergic inflammatory response. Methylxanthines given orally or intravenously support the respiratory effort by increasing the force of diaphragm contraction and central respiratory stimulation. PDE-4 inhibitors inhibit PDE-4, leading to increased cAMP levels and downregulating the inflammatory response. Antibiotic therapy for treatment of infectious exacerbations should be guided by results of sputum culture and sensitivities. Long-term prophylactic antibiotic use may reduce COPD exacerbations. Mucolytics (N-acetylcysteine) and expectorants (guaifenesin) help to thin and clear tenacious mucous secretions. Intravenous alpha₁-antitrypsin infusion may be of benefit in patients with deficiency-related emphysema.²²⁶ The rehabilitation team plays a pivotal role in vaccine and medication compliance, as well as in the correct technique and timing of use of a patient's medical regimen. Poor technique and compliance are associated with increased COPD exacerbations. In addition, several treatments prolong life or reduce progression in this disease,⁵⁰ and these include smoking cessation, noninvasive mechanical ventilation, LVRS, combined long-acting beta agonists with inhaled corticosteroids, and oxygen therapy.

Oxygen Therapy

Oxygen therapy reduces mortality when used appropriately in patients with COPD and hypoxemia with PO₂ less than 55 mm Hg or O₂sat less than 88%.^{155,183} Increased life expectancy is due to reduction in PH and right heart failure and is directly correlated with duration (>15 hours/day) of use. Other benefits include improved strength, reduced fatigue, and improved cognitive function. Oxygen levels may be normal at rest but decrease with exertion (tracking oxygen saturations with home/community pulse oximetry helps determine the need for portable oxygen use with activity). Nocturnal hypoxemia detected by overnight oximetry should be managed judiciously with supplemental oxygen in patients with documented hypercapnia. Long-term oxygen therapy is also needed if the patient's Sao₂ is less than 89% and there is evidence of PH or peripheral edema, suggesting congestive cardiac failure, or polycythemia.¹⁹⁶

There is no universally ideal oxygen delivery system. The pros and cons need to be considered in the context of each individual patient. Wall plug-in oxygen concentrators are the most efficient method of providing oxygen in the home; some models provide 8 to 10 L of continuous nasal cannula flow oxygen. Battery-powered portable models (ranging between 6 and 20 pounds) are available, but limitations include a relatively short battery life (4 to 6 hours), a pulsed/intermittent oxygen flow with lowered oxygen delivery (1 to 4 L), and a device weight that can negatively impact compliance. Portable concentrators can be used indoors with an alternating or direct current inverter. Devices can be

recharged in 2 hours, and backup batteries can be carried. Portable concentrators can also be carried on a pull cart. Some companies have produced models that provide up to 8 L/min of oxygen and have two flow meters so that two individuals can use oxygen from the same concentrator if the total use is no more than 8 L/min. Other systems incorporate a refill station as part of the concentrator so that portable oxygen cylinders can be filled over 2 hours. An oxygen-conserving regulator allows the oxygen to last four times as long. Oxygen cylinders are a viable alternative to portable concentrators and provide continuous flow, but size and weight often preclude them from being carried, and they can run out quickly at high flow rates. Cylinders can be carried in a push cart or safely mounted on a standard or motorized wheelchair if the motors and batteries are sealed and both are covered by a rigid housing.

At altitude and when traveling by air, there is reduced partial pressure of oxygen, which decreases blood oxygen saturation. Airlines are typically pressurized to 8000 to 10,000 feet above sea level, and patients with COPD and borderline oxygen saturations at ground level may require supplemental oxygen inflight; the need is greater for flights over 5 hours. If saturations fall below 92%, oxygen should be started. Typically, in those not using supplemental oxygen at baseline, 1 L is sufficient to maintain adequate O_2 sat. Otherwise, the rule of thumb is 1 to 2 L over normal flow rates for patients on chronic oxygen. Of note, patients cannot take oxygen cylinders on planes in the United States, and therefore a portable oxygen concentrator is required.

Pulmonary Rehabilitation

PR is the comprehensive approach to optimizing the medical, physical, and functional capacity of patients limited by respiratory diseases. The PM&R physician must understand medical and oxygen management to optimize the PR plan. PR engages a multidisciplinary team, including chest therapy, physical and occupational therapy, nutrition, and psychology.

Chest physical therapy

A good understanding of PFTs (Table 29.2 and Fig. 29.1), as well as the mechanics and work of breathing in normal and diseased states, helps in planning an effective PR program.¹⁰² A normal inspiratory to expiratory breathing time ratio is 1:2. Disordered breathing patterns with COPD include rapid and shallow ventilation with inadequate expiration (the ratio changes to 1:1 due to

shortened expiratory time). Air trapping due to dynamic expiratory airway collapse contributes to a failure to breathe normal volumes of air and results in abnormal gas exchange (a significant volume of inspired air oscillates in the respiratory dead space). In conjunction with abnormal diaphragmatic excursion, this contributes to an increased work of breathing. Shortness of breath at rest and/or on exertion are common complaints of individuals with long COVID; etiology of respiratory symptoms in long COVID is often multifactorial, and disordered breathing patterns have been noted.⁷¹

Although patients are often unaware of their abnormal breathing patterns, they often experience respiratory distress and air hunger (dyspnea). Breathing exercises under the guidance of a physical therapist raise the breathing pattern to a level of awareness that when combined with relaxation techniques facilitates retraining of that pattern. Retraining techniques for persons with COPD include pursed lip breathing, head-down and bending-forward postures, slow deep breathing, and localized expansion exercises or segmental breathing. These techniques maintain positive airway backpressure during exhalation and help reduce airway collapse and hyperinflation. Although diaphragmatic breathing is widely taught, it has been shown to increase the work of breathing and dyspnea compared with the natural pattern of breathing in the patient with COPD.⁵⁰ Respiratory muscle endurance training focusing on inspiratory resistance training is also used to reduce breathing fatigue. Training of the expiratory muscles has also been found to be of some value.¹⁷³

Airway clearance strategies are indicated for persons with (1) abnormal cough mechanics (e.g., muscle weakness), (2) altered mucous rheology (e.g., CF), (3) structural airway defects (e.g., bronchiectasis), and (4) altered mucociliary clearance (e.g., primary ciliary dyskinesia).¹⁷¹

Clearance of secretions is mandatory to reduce the work of breathing, improve gas exchange, and limit infection and atelectasis. For secretion clearance through chest physical therapy to be effective, mucoactive medications should be given.²¹⁴ These include expectorants, mucolytics, bronchodilators, surfactants, and mucoregulatory agents that reduce the tenacity and volume of mucous secretion and support bronchodilation. Use of antitussives for uncontrolled coughing can help limit dynamic airway collapse, bronchospasm, or syncope. Cough suppressant use should be limited to avoid secretion retention. Use at night to support sleep is a consideration.

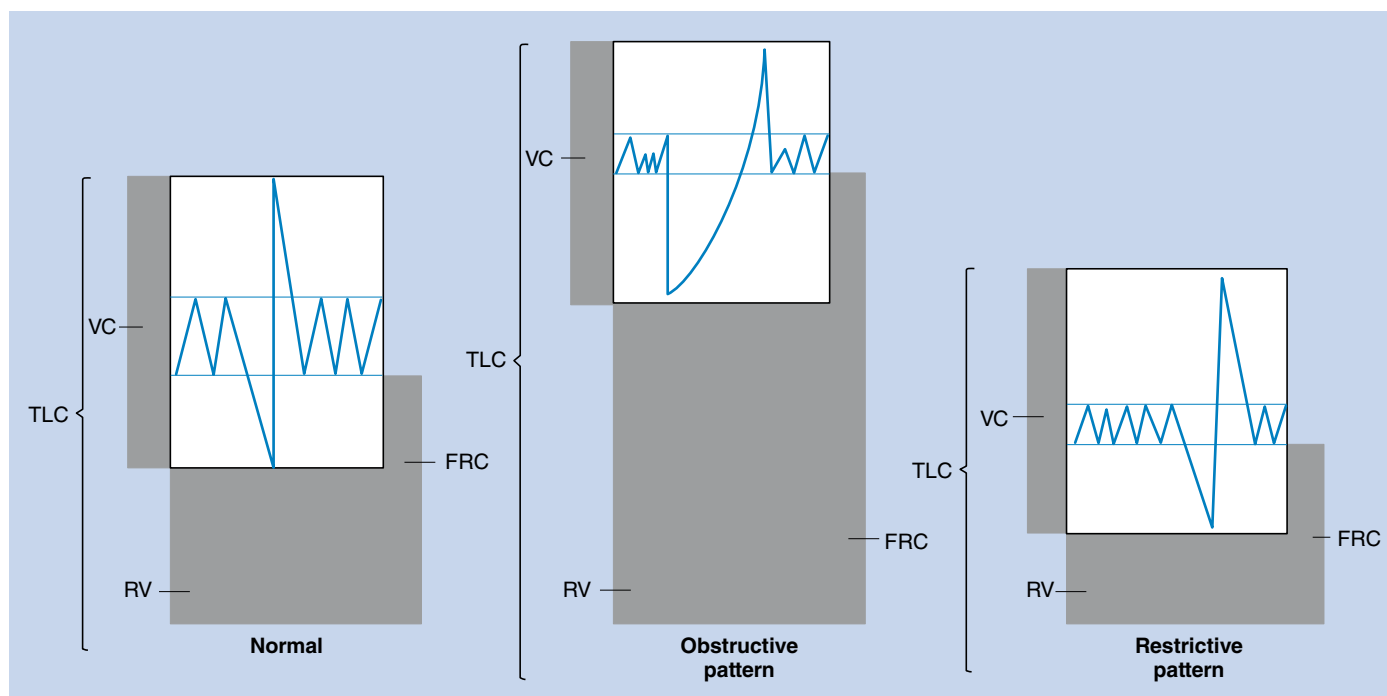
TABLE 29.2

Characteristic Physiologic Changes Associated With Pulmonary Disorders

Measure	Obstructive Disorders	Restrictive Disorders	Mixed Disorders
FEV ₁ /FVC	Decreased	Normal or increased	Decreased
FEV ₁	Decreased	Decreased, normal, or increased	Decreased
FVC	Decreased or normal	Decreased	Decreased or normal
TLC	Normal or increased	Decreased	Decreased, normal, or increased
RV	Normal or increased	Decreased	Decreased, normal, or increased

FEV₁, Forced expiratory volume in 1 second; FVC, forced vital capacity; RV, residual volume; TLC, total lung capacity.

Data from the Merck Manual Professional Version (Known as the Merck Manual in the US and Canada and the MSD Manual in the rest of the world), edited by Robert Porter. Copyright (2015) by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Kenilworth, NJ. Available at <http://www.merckmanuals.com/professional>. Accessed March 31, 2015.



• **Fig. 29.1** Lung volumes in health and disease. The breathing pattern is read from right to left. The obstructive pattern demonstrates prolonged expiration. *FRC*, Functional residual capacity; *RV*, residual volume; *TLC*, total lung capacity; *VC*, vital capacity.

Techniques for clearing secretions include postural drainage, manual or device-induced chest percussion and vibration, device-induced airway oscillation, incentive spirometry, and other devices and measures that improve the ability to cough. Head-down tilt positions for postural drainage should be used with caution in persons with severe heart disease.¹⁸¹

In a manually assisted cough, the patient's abdomen is compressed while the patient controls the depth of inspiration and the timing of closing and opening of the glottis. Noninvasive intermittent positive-pressure ventilation (NIPPV) with air stacking or glossopharyngeal breathing (GPB) is used to increase the depth of inspiration when inspiratory muscles are too weak to produce a deep breath. Air stacking is holding a portion of two or more breaths to fully inflate the lungs before exhalation. In GPB, the person uses the tongue to breathe. The technique is described more fully later in this chapter. When an upper motor neuron lesion above the midthoracic level has paralyzed the abdominal muscles, functional electrical stimulation of these muscles can produce a cough.¹¹⁶

Breathing techniques to support secretion clearance include controlled coughing, huffing (gentle repeated forced expiratory flows through a partially closed glottis), active cycle of breathing (a three-phase technique), and autogenic drainage (a self-drainage technique using different speeds of breathing to move mucus).

Hands-on chest PT includes percussion techniques (tapping, cupping, vibration) and postural drainage utilizing gravity and positioning to assist drainage from specific lung segments.

Simple respiratory devices that aid secretion clearance include an incentive spirometer to promote deep breathing/chest expansion and oscillating positive expiratory pressure devices (Flutter, Acapella, Aerobika, and Lung Flute).

Positive expiratory pressure mask therapy followed by huffing is a useful technique when other methods of mobilizing secretions

are not tolerated.¹⁶⁴ Using a mechanical insufflator (exsufflator cough-assist machine) (Fig. 29.2) provides deep inspiration through a face mask or mouthpiece, or with an adapter, to the patient's endotracheal or tracheostomy tube, followed rapidly by controlled suction. It has been shown to provide highly effective secretion removal.¹⁹ The device must be used with caution and is contraindicated in patients with bullous emphysema or a history of pneumothorax or pneumomediastinum in the recent past, especially as a result of barotrauma. The chest vest is another advanced respiratory device delivering high-frequency chest wall oscillation using an inflatable vest.



• **Fig. 29.2** Using a mechanical insufflator-exsufflator cough assist machine. (From CANVent. Canadian Alternatives in Noninvasive Ventilation. Available at https://www.youtube.com/watch?v=rOvR8ZKxI_M. Accessed Oct 29, 2024.)

Exercise therapy: general considerations

Aerobic exercise is the goal in any PR program to overcome the fatigue from physiologic deconditioning that accompanies chronic lung diseases. However, in more advanced chronic lung disease, the rehabilitation need may be more basic, including range of motion (ROM) exercises, strengthening, transfers, balance, posture, and gait training, to get the patient ready for PR. To be appropriate for PR, a patient must demonstrate a decrease in functional exercise capacity as a result of pulmonary disease and be able to participate safely in a progressive cardiorespiratory endurance training program. Patients should be medically stable, oriented to the process and goals of the PR program, be motivated to exercise on a consistent basis, and ideally be free of tobacco products. Following referral for PR, a comprehensive evaluation should be completed by the PR team, including a thorough review of the history, a detailed patient examination, and review of all diagnostic testing. Ideally, and if available, a pre-participation cardiopulmonary exercise test (CPET) quantifies baseline physiologic and functional parameters, assists in risk stratification to minimize cardiopulmonary risk of exercise, and provides a basis for the initial exercise PR prescription. Indications for CPET have been adopted by the American Thoracic Society (ATS) and the American College of Chest Physicians (CHEST) (Box 29.2).¹³ Patients who cannot tolerate exercise testing without supplemental oxygen are not candidates for CPET, and a standard exercise stress test (EST) is acceptable. Patients unable to perform standard ESTs should complete a 6-minute walk test (6MWT).¹⁷

Exercise prescription for pulmonary rehabilitation

It is imperative that the personnel supervising the exercise program are appropriately trained in the implementation of exercise prescriptions in people with chronic illnesses and special needs. The American College of Sports Medicine (ACSM) offers certification programs to those with appropriate backgrounds, such as exercise science or kinesiology, nursing, occupational therapy, physical therapy, and respiratory therapy. PR exercise prescription

• BOX 29.2 American Thoracic Society/American College of Chest Physicians Indications for Cardiopulmonary Exercise Testing

- Evaluation of exercise intolerance
- Unexplained dyspnea
- Evaluation of patients with cardiovascular disease
- Evaluation of patients with respiratory disease
- Chronic obstructive pulmonary disease
- Interstitial lung disease
- Chronic pulmonary vascular disease
- Cystic fibrosis
- Exercise-induced bronchospasm
- Preoperative evaluation
- Preoperative evaluation for lung cancer resectional surgery
- Lung volume reduction surgery
- Evaluation for lung or heart-lung transplantation; preoperative evaluation of other procedures
- Exercise prescription for pulmonary rehabilitation
- Evaluation of impairment or disability

From American Thoracic Society, American College of Chest Physicians. ATS/ACCP statement on cardiopulmonary exercise testing. *Am J Respir Crit Care Med*. 2003;167(2):211-277.

guidelines are focused on patients with COPD but are appropriate for patients with other pulmonary diseases as well. The ATS,²¹³ CHEST,²¹¹ and ACSM¹² have provided recommendations for PR exercise. In general, PR exercise programs should include endurance training of larger muscle groups. Treadmill walking is the preferred method because it is generally well tolerated, effective for physiologic conditioning, and translates into real-world functional improvement. Walking in a hallway with assistance of a therapist is acceptable for those too limited to safely tolerate the treadmill. Leg and arm cycling, stair stepping, rowing, and elliptical exercise machines can also be incorporated. Arm exercises can increase ventilatory drive at lower workloads and can worsen the patient's dyspnea during the activity. Resistance/strength training and flexibility exercises also contribute to functional improvements in patients with pulmonary diseases. The most specific exercise prescription guidelines have been provided by the ACSM (Table 29.3).

Exertional dyspnea is among the most frequently experienced symptoms of pulmonary disease and significantly contributes to physical disability and functional impairment. Cardiorespiratory exercise training is effective for decreasing exertional dyspnea. During a program of progressive aerobic training, cardiovascular, respiratory, and neuromuscular system adaptations occur leading to an increased ability to sustain physical activity. Following a PR program, the efficiency of oxygen delivery (cardiac output) and muscle oxygen extraction (arteriovenous oxygen difference), combined with increased mitochondrial oxidative activity, lowers blood lactate levels at any given level of strenuous exercise. Lower blood lactate lessens carbon dioxide production and the ventilatory requirement (or sense of dyspnea) for a given workload.

In conjunction with the PR supervising physician, the initial exercise session intensity should be established between patient and supervising professional. While patients tend to opt for activity intensities that minimize fatigue and shortness of breath, it is the skill, support, and encouragement of the PR professional that enables a patient to push gradually beyond that comfort level to an exercise intensity that promotes physiologic change. Close monitoring and adjustment of the intensity and duration of the session should be made according to the patients' perception of exertion and physiologic responses. The progression of exercise intensity and duration over subsequent sessions should reflect the individual patient's functional progress and tolerance of the program. Although beneficial results have been observed with PR programs of 6 to 10 weeks in some individuals, those with more advanced pulmonary disease may require longer periods of participation to show significant results. Home exercises should be prescribed during the rehabilitation program, and patients are encouraged to complete them daily. In anticipation of graduating from the monitored PR program, patients should be educated in the ongoing need for lifelong exercise. On completion of the PR program, the CPET (or standard EST or 6MWT) should be repeated to quantify physiologic and functional gains and recalibrate the exercise training intensity for the ongoing self-monitored program.

Exercise in COPD

Exercise in patients with COPD considers the severity of their respiratory disability (Table 29.4).¹⁶⁵ Numerous studies have been carried out on the effects of exercise on patients with COPD (Box 29.3).^{43,46,74,82,91,194,284} In patients with COPD, cardiorespiratory endurance exercise therapy has been shown to improve

TABLE 29.3 Summary of the American College of Sports Medicine’s Guidelines for Exercise Prescription

Component	Cardiorespiratory Endurance Training
Activity	Dynamic exercise of large muscle groups
Mode	Overground or treadmill walking Stationary leg cycling or outdoor bicycling Stair climbing
Frequency	3–5 days/week
Duration	20–60 min/session
Intensity	50–85% heart rate reserve 65–90% maximum heart rate RPE = 12–16 (category scale) RPE = 4–8 (category-ratio scale) ^a
Component	Strength and Muscle Endurance Training
Activity	Resistance training: low resistance with high repetition
Mode	Variable resistance or hydraulic weight machines Isotonic weight machines Free weights
Frequency	2–3 days/week
Duration	One set of 3–20 repetitions on 8–10 exercises that include all of the major muscle groups
Intensity	Volitional exhaustion on each set, or Stop 2–3 repetitions before volitional exhaustion
Component	Flexibility
Activity	Static stretching of all major muscle groups
Frequency	Minimum of 2–3 days/week Ideally 5–7 days/week
Duration	15–30 sec/exercise, 2–4 stretching exercise sets
Intensity	Stretch to tightness at the end of the range of motion but not to pain

RPE, Rate of perceived exertion.
^aWarburton DE, Nicol CW, Bredin SS. Prescribing exercise as preventive therapy. *CMAJ*. 2006;174(7):961-974.
Modified from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM’s Guidelines for Exercise Testing and Prescription*, 11th ed. Wolters Kluwer; 2021.

maximum or symptom-limited aerobic capacity, timed walk distance, and health-related quality of life. Interval training when tolerated, incorporating brief episodes of higher intensity aerobic exercise, has been shown to be superior to lower intensity steady-state exercise.¹⁷² Adding resistance training to the rehabilitative regimen can provide additional benefits, such as increased fat-free mass and muscle strength.

Several adjunct modalities might reduce the extreme breathlessness and peripheral muscle fatigue that prevent patients with severe COPD from exercising at higher intensities.¹¹ Continuous positive airway pressure and NIPPV during exercise may reduce the perception of dyspnea. Taking part of the work of breathing away from

TABLE 29.4 Classification Scheme for Chronic Obstructive Pulmonary Disease Severity: National Heart, Lung, and Blood Institute–World Health Organization Global Initiative for Chronic Obstructive Lung Disease Criteria

Stage	Criterion ^a
0	Normal lung function
1 (Mild)	FEV ₁ ≥80% of predicted
2 (Moderate)	50–79% of predicted
3 (Severe)	30–49% of predicted
4 (Very severe)	FEV ₁ <30% of predicted or presence of respiratory failure or clinical signs of right-sided heart failure

FEV₁, Forced expiratory volume in 1 second; FVC, forced vital capacity.
^aIn the presence of FEV₁/FVC ratio <70%.

• BOX 29.3 Improvements Seen in Exercise Reconditioning in Moderate Chronic Obstructive Pulmonary Disease

- Inspiratory Muscle Training**
- Increased maximal inspiratory mouth pressure
 - Increased strength of the diaphragm
- Pulmonary Rehabilitation With or Without Inspiratory Muscle Training**
- Increased maximal workload
 - Improved activities of daily living scores
 - Improved anxiety and depression scores
 - Increased 6- or 12-min walking distance
- Pulmonary Rehabilitation (Cycle Ergometry, 70 W)**
- Minute volume decrease of 2.5 L/min per blood lactate decrease of 1 mEq/L (normal: minute volume decrease of 7.2 L/min per blood lactate decrease of 1 mEq/L)

the respiratory muscles and reducing intrinsic positive end-expiratory pressure are considered two mechanisms by which these techniques relieve dyspnea. Nocturnal NIPPV in selected patients can improve their ability to exercise during the day. Adding electrical stimulation to strength exercises for peripheral muscles has been shown to further improve muscle strength in patients with COPD. Interval training with rest periods is capable of producing training effects in those who cannot tolerate a sustained period of exercise. Oxygen supplementation, even in patients who do not desaturate during exercise, allows for higher exercise intensities and produces a superior training effect. Although high-intensity physical group training in water has been shown to produce benefits for select patients with COPD,²⁶⁶ aqua therapy is not considered a mainstay of standard PR programs. After an initial exercise rehabilitation program, patients must continue to participate in regular exercise to prevent the positive effects of exercise from being lost. Continuous outpatient exercise training, home-based or community-based exercise programs, or exercise training in groups of persons with COPD is necessary to sustain the benefits acquired during the initial rehabilitation program.²³¹

Exercise in asthma

Guidelines for the diagnosis and management of asthma were published in 2007 by the National Heart, Lung, and Blood Institute¹⁷⁹ and revised in 2020.¹ While exercise is supported in the management of asthma, the mainstay of treatment is pharmacologic both in prevention of and recovery from exacerbations. When asthma is quiescent, often little or no functional limitations are seen. Persistent symptoms and progressive functional decline can be seen in asthmatic bronchitis or when asthma overlaps with COPD. In these settings, benefit is seen from pulmonary rehab as for nonasthmatic patients with COPD. Exercise can cause airflow obstruction due to exercise-induced bronchoconstriction (EIB) or as exercise-induced asthma (EIA).⁹² Nearly 90% of those with asthma experience EIB during exercise. Symptoms typically begin during exercise, worsen after completion of exercise, and resolve within 1 hour. A late phase of more minor symptoms can be seen 4 to 12 hours after stopping exercise and resolving within 24 hours. Exercise that is most likely to trigger EIA includes exercise in cold or dry weather, especially outdoor winter sports. Sports requiring constant activity, such as long-distance running, soccer, and lacrosse, also commonly trigger EIA. Environmental triggers include dry and/or cold air, pollution, high pollen or allergen counts, irritants (smoke, fumes), and a recent cold or asthma episode.²³⁶ EIA is diagnosed with normal pre- and abnormal postexercise PFTs indicating EIB. Preventive measures include covering mouth and nose with a scarf when exercising in cold and dry weather, premedicating with bronchodilators before exercising, and incorporating a prolonged warmup (10 minutes) before exercise. Preferred short-acting bronchodilators include beta₂ agonists (albuterol) and anticholinergics (ipratropium). Standard asthma control medications should be continued. Inhaled corticosteroid inhibit the late-phase inflammatory response.³⁴

Two major patterns of asthma (intermittent vs. persistent) are seen; however, bronchial biopsies in patients with intermittent asthma show evidence of ongoing, persistent airway inflammation. A mouse model of asthma¹⁹³ shows exercise decreases the activity of a factor that upregulates the expression of genes that encode inflammatory mediators. Moderate-intensity aerobic exercise training of the mice decreased inflammation and pathologic structural remodeling within the lungs. It is extrapolated that aerobic exercise in people with asthma reduces airway inflammation in a similar manner.

Exercise in CF

The prevalence of CF in 2020 was approximately 40,000 people in the United States⁶⁸ and 70,000 worldwide; it is most prevalent in Europe, North America, and Australia. CF is due to an autosomal recessive genetic disorder and significantly impacts life expectancy. The disease is caused by mutation of a gene that encodes a chloride-conducting transmembrane channel regulating chloride transport and airway mucociliary clearance. This results in mucus retention, chronic airway infection, and local inflammation resulting in bronchiectasis, small airway obstruction, and progressive respiratory impairment. CF transmembrane conductance regulator dysfunction mainly affects epithelial cells, although there is evidence of a role in immune cells. CF affects several body systems, and morbidity and mortality are mostly caused by bronchiectasis, small airway obstruction, and progressive respiratory impairment. Comorbidities include pancreatic malabsorption, liver/biliary cirrhosis, impaired sweat gland function and heat

shock, and infertility due to impaired vas deferens function. Quality of life and clinical outcomes have improved in patients with CF (median life expectancy is now >40 years) due to the development of drugs enhancing the clearance of mucus from the lungs and treatment of chronic infection, the correction of pancreatic insufficiency, and dietary regimens for malnutrition. Small molecule and gene-based therapies have been developed that target the basic defect in CF and are effective in improving lung function and reducing pulmonary exacerbations.^{79,120}

PR for CF is a lifelong commitment to medication adherence, secretion clearance, nutrition, and exercise. Psychological and supportive therapies are essential to embed this complex health plan for patients and for family members who are essential participants in the overall care plan. Compliance through adolescent years is particularly challenging.

The Vest Airway Clearance System (Hill-Rom, St. Paul, MN) helps loosen secretions by providing chest wall percussion and vibration by rapidly inflating (compressing) and deflating (releasing) the vest on the chest wall, a process called high-frequency chest wall oscillation.

Children with CF have reduced anaerobic performance and do not participate in activities requiring short bouts of high-energy expenditure to the same extent as healthy children. A study of anaerobic training on children with CF older than 12 years showed no child had an FEV₁ less than 55% of the predicted level. Children were trained 2 days per week for 12 weeks, with sessions lasting 30 to 45 minutes. Each anaerobic activity lasted 20 to 30 seconds.¹³⁵ Three months after the end of training, anaerobic performance and quality of life were significantly higher in the trained group. A measurable improvement in aerobic performance, however, was not sustained.

The lung function of adults with CF has been studied by measuring FEV₁. Thirty-six percent were found to have normal or mild lung dysfunction (FEV₁, 70% predicted), 39% had moderate dysfunction (FEV₁, 40% to 69% predicted), and 25% had severe dysfunction (FEV₁, <40% predicted). One-third of adults with CF have multiresistance gram-negative organisms. A 3-year study of regular aerobic exercise in adults with CF has found that it reduced the expected decline in pulmonary function throughout that period. Appropriate vigorous aerobic exercise enhances cardiovascular fitness, increases functional capacity, and improves quality of life. Patients with CF should rarely be exercised in close proximity to one another, however, because of the possibility of transmission of *Burkholderia cepacia*, a highly transmissible bacterium that frequently infects patients with CF.

Survival in CF is correlated with maximal oxygen uptake. Blau et al.³² found that an intensive 4-week summer camp program for both children and young adults improved exercise tolerance and nutrition in patients with CF.

Exercise in disorders of chest wall function

There are numerous causes of chest wall dysfunction that exist with both mechanical and neuromuscular components. The chest wall can be viewed as encompassing the rib cage, spine, diaphragm, abdomen, shoulder girdles, and neck. Distortions of the trunk are also reflected in distortions of the heart, lungs, airway, and abdominal contents. Many conditions negatively affect the chest wall, including ankylosing spondylitis, pectus excavatum, obesity, the sequelae of thoracoplasty or phrenic nerve crush for the treatment of pulmonary tuberculosis, neuromuscular diseases with weakness of the respiratory bellows mechanism and

superimposed spinal curvatures, and Parkinson disease. Ventilatory muscle training in these disorders can reduce respiratory muscle fatigue. Persons with Parkinson disease have shown overall improvement with PR.⁴⁷

Patients with developmental disorders are a heterogeneous group in which severe scoliosis is commonly seen. Exercise and other ongoing rehabilitation techniques, such as positioning, and management of spasticity and dysphagia are of value in their treatment.²⁵²

Nutritional issues

Nutrition evaluation and therapy are essential in patients with chronic lung disease presenting for PR. A typical PR program will treat patients with low, normal, and high body mass index (BMI) representing the spectrum from malnourished/cachectic to obese.

Diet and nutrition are important modifiable risk factors for the development, progression, and management of obstructive lung diseases such as asthma and COPD through an immunomodulatory role of key nutrients.⁹⁸ The relationship between dietary patterns, nutrient intake, and weight status in obstructive lung diseases has been well summarized.²⁸

Dietary patterns improving respiratory diseases include high fruit and vegetable intake, Mediterranean-style diet, fish, and omega₃ intake. Fast-food intake and Westernized highly processed dietary patterns have adverse associations. Antioxidants are associated with positive effects on inflammation, clinical outcomes, and respiratory disease prevention, but intervention studies of individual antioxidants have not indicated widespread adoption of supplementation.²⁶³

A whole foods approach to nutrient supplementation with increased intake of fruit and vegetables increases intake of multiple nutrients, including vitamins C and E, carotenoids, and flavonoids, and shows more potential in reducing COPD²²⁴ and asthma exacerbations.²⁷⁴

In moderate to severe COPD, weight loss is common and leads to decreased exercise capacity and health status with increased morbidity and mortality. Weight loss is likely the result of insulin resistance, elevated catecholamine levels, and dyslipidemia. Loss of fat mass is generally caused by energy imbalance, whereas loss of fat-free lean/muscle mass results from substrate and protein catabolism.^{76,216} Nutritional supplementation therapy with daily oral liquid nutritional supplements in nutritionally depleted patients with COPD increases body weight, lean muscle mass, maximal inspiratory mouth pressure, strength, and peak workload. Respiratory symptoms also improve.⁶⁷ However, oral glucocorticosteroid treatment can blunt this response.

Total daily energy expenditure in patients with COPD, irrespective of their weight, is no different than in healthy people.²⁴⁴ Insufficient food/calorie intake is the cause of malnutrition. Weight gain is supported when total energy intake is 1.3 times higher than the resting energy expenditure.¹⁹⁸

Counterintuitively, smaller doses (125 mL) of a dietary supplement promote greater improvements in weight gain (larger volumes increase bloating and early satiety, compromise diaphragmatic movement, and contribute to meal skipping).³⁹ Nutrition counseling and promoting self-management for changes in dietary behavior should start early in the course of the disease, when patients with COPD are still under the care of a primary physician.⁴⁰ Practical suggestions for shopping, food preparation, and eating smaller, more frequent meals should be offered. When fatigue, dyspnea, swallowing, or poor appetite interferes with eating, referral to a

dietitian is recommended. Involuntary weight loss is best addressed preemptively.

Psychosocial support

Biopsychosocial considerations for persons with pulmonary dysfunction include the ongoing education of the patient and family, vocational counseling, and disability evaluation. Occupational therapy can also provide patients with severe respiratory disease counseling on energy conservation and techniques for performing basic ADLs.¹⁵⁸ Depression and anxiety are very common comorbidities associated with COPD¹⁹² because of the often drastic limitations in functional activities and the distress associated with dyspnea. Anxiety and/or depression in patients with COPD is associated with increased mortality, COPD exacerbation rates, length of hospital stay, and decreased quality of life and functional status. Treatment options include psychological (e.g., relaxation therapies, cognitive behavioral therapy, self-management) and pharmacologic interventions (antidepressant and anxiolytic medications). PR improves anxiety and depression, but rehabilitation completion rates are negatively affected by their presence.²⁰²

Summary of recommendations

Table 29.5 presents a summary of recommendations, statements, and grades in the evidence-based guidelines on PR with a strength of evidence/balance of benefits grading system.

Management Options for Individuals With Severe Lung Disease and Long-Term Outcomes

Supported and Mechanical Ventilation

Self-supported ventilation techniques and mechanically supported noninvasive ventilation assist regular respiratory efforts when patients with chronic lung disorders develop worsening respiratory status either through progression of the chronic disease or in an acute exacerbation or infection. Air stacking (adding consecutive quantities of air into the lungs), maximal insufflations (passively inflating the patient's lungs to maximal capacity), assisted coughing, and noninvasive ventilation have been detailed by Bach,²⁰ who also describes an oximetry feedback respiratory aid protocol where the pulse oximeter is used to provide feedback of the patient's oxygen saturation (SpO₂). The goal is to maintain SpO₂ greater than 94%, supported by techniques to maintain effective alveolar ventilation and airway secretion elimination. With declining SpO₂, patients and families are taught to intensify airway clearance techniques and use NIPPV continuously. If these measures fail to bring the SpO₂ back to acceptable levels, then patients must be evaluated by their clinician or in the emergency room and admitted if necessary. If admitted, ongoing efforts to provide noninvasive assisted ventilation and avoid intubation should be made; once intubated, patients with chronic lung disease (especially neuromuscular disorders) are challenging to extubate and may require tracheotomy and chronic/lifelong invasive ventilation.

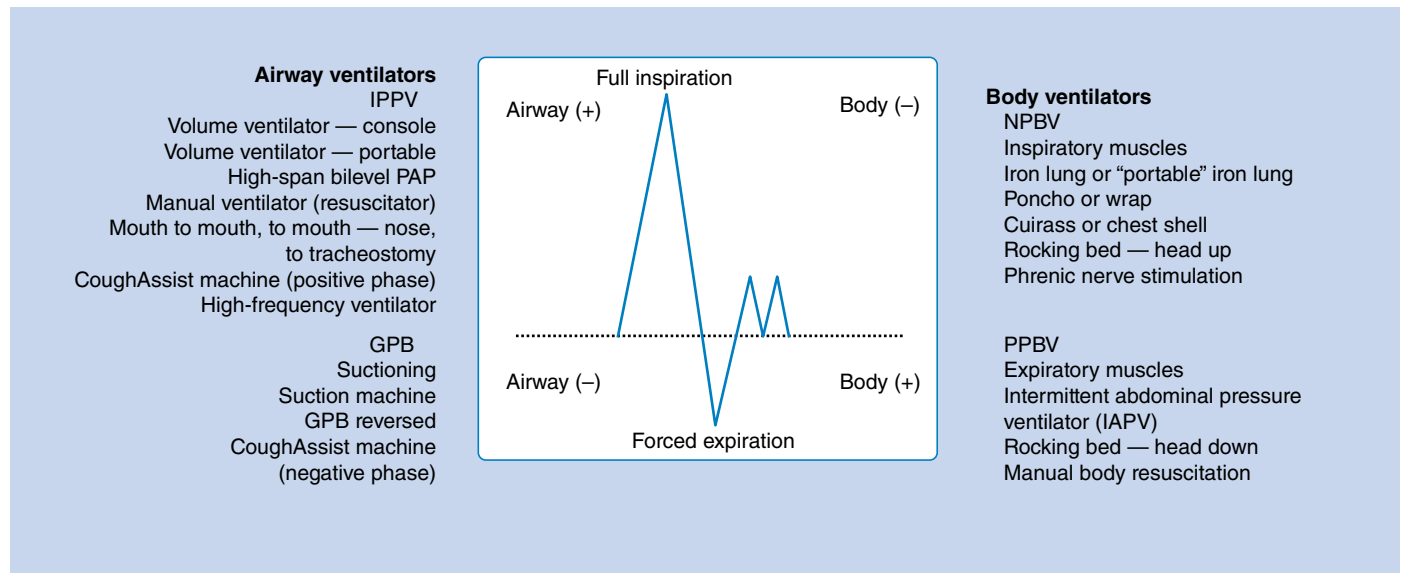
Noninvasive ventilation (NIPPV) delivered via nasal-mouth interfaces is the ventilator method of choice. The equipment is lightweight and portable, powered by either alternating or direct

TABLE 29.5 Summary of Recommendations, Statements, and Grades in the Evidence-Based Guidelines on Pulmonary Rehabilitation (PR) With Strength of Evidence/Balance of Benefits Grading System

Recommendation or Statement		Strength of Evidence/ Recommendation Grade			
A program of exercise training of the ambulation muscles is a mandatory component of PR for patients with chronic obstructive pulmonary disease (COPD).		1A			
PR improves dyspnea in patients with COPD.		1A			
PR improves health-related quality of life in patients with COPD.		1A			
PR reduces the number of hospital days and other measures of health care use in patients with COPD.		2B			
PR is cost effective in patients with COPD.		2C			
Insufficient evidence exists to determine whether PR improves survival in patients with COPD.		No recommendation			
Psychosocial benefits result from comprehensive PR programs in patients with COPD.		2B			
8A. 6–12 weeks of PR produces benefits in several outcomes, but these benefits decline gradually over 12–18 months.		1A			
8B. Some benefits, such as health-related quality of life, remain above control at 12–18 months.		1C			
Longer (>12 weeks) PR programs produce greater sustained benefits than do shorter programs.		2C			
Maintenance strategies after PR have a modest effect on long-term outcomes.		2C			
Lower extremity exercise training at a higher exercise intensity produces greater physiologic benefits than lower intensity training in patients with COPD.		1B			
Both low-intensity and high-intensity exercise training produce clinical benefits for patients with COPD.		1A			
Addition of a strength-training component to PR increases muscle strength and muscle mass.		1A			
Current evidence does not support the routine use of anabolic agents in PR for patients with COPD.		2C			
Unsupported endurance training of the upper extremities benefits patients with COPD and should be included.		1A			
The evidence does not support the routine use of inspiratory muscle training as an essential component.		1B			
Education is an integral component of PR and should include information on collaborative self-management and prevention and treatment of exacerbations.		1B			
Minimal evidence supports the benefits of psychosocial interventions as a single therapeutic modality.		2C			
Although evidence is lacking, current practice and expert opinion support the inclusion of psychosocial interventions for patients with COPD.		No recommendation			
Use supplemental oxygen rehabilitation exercise training in patients with severe exercise-induced hypoxemia.		1C			
In patients without exercise-induced hypoxemia, supplemental oxygen during a high-intensity exercise program can improve gains in exercise endurance.		2C			
In selected patients with severe COPD, noninvasive ventilatory support from a mechanical ventilator might modestly improve exercise performance.		2B			
Insufficient evidence exists to support the routine use of nutritional supplementation in PR of patients with COPD.		No recommendation			
PR benefits some patients with chronic respiratory diseases other than COPD.		1B			
Although evidence is lacking, current practice and expert opinion suggest that PR for patients with chronic respiratory diseases other than COPD should be modified to include treatment strategies specific to individual diseases and patients, in addition to the treatments used for patients with COPD.		No recommendation			
Strength of Evidence	GRADING SYSTEM: BALANCE OF BENEFITS TO RISKS AND BURDENS				
		Benefits Outweigh Risks/Burdens ^a	Risks/Burdens Outweigh Benefits ^b	Risks/Burdens and Benefits Balanced ^c	Uncertain ^d
	High	1A: Strong recommendation	1A: Strong recommendation	2A: Weak recommendation	
	Moderate	1B: Strong recommendation	1B: Strong recommendation	2B: Weak recommendation	
	Low	1C: Strong recommendation	1C: Strong recommendation	2C: Weak recommendation	2C: Weak recommendation

^aBenefits clearly outweigh the risks and burdens (certainty of imbalance).^bRisks and burdens clearly outweigh the benefits (certainty of imbalance).^cRisks/burdens and benefits are closely balanced (less certainty).^dBalance of benefits to risks and burdens is uncertain.

Modified from Ries AL. Pulmonary rehabilitation: summary of an evidence-based guideline. *Respir Care*. 2008;53:1203-1207, used with permission from Respiratory Care and the American Association for Respiratory Care. Originally modified from Guyatt G, Gutterman D, Baumann MH, et al. Grading strength of recommendations and quality of evidence in clinical guidelines: report from an American College of Chest Physicians task force. *Chest*. 2006;129(1):174-181; and Ries AL, Bauldoff GS, Carlin BW, et al. Pulmonary rehabilitation: joint ACCP/AACVPR evidence-based clinical practice guidelines. *Chest*. 2007;131(5):S4-S42.



current, and can be delivered via tracheostomy when needed (Figs. 29.3 to 29.6). NIPPV has replaced the use of body ventilators, including the iron lung, chest shell, intermittent abdominal pressure ventilators, and the rocking bed. The use of portable and home-based noninvasive ventilators for improved physical activity in daily life in patients with chronic respiratory disease is discussed elsewhere.⁸⁸

Diaphragmatic pacing (Fig. 29.7), used effectively for over 40 years, is indicated in respiratory insufficiency with ventilator dependency in patients with high-level SCI and central hypoventilation syndromes (Ondine curse). The goal is to wean from mechanical ventilation but not to remove the tracheostomy, maintenance of which allows mechanical ventilation should the need arise due to diaphragmatic pacer failure or any other cause. Diaphragmatic pacing is achieved through a surgical procedure to place electrodes either directly around the phrenic nerve (intrathoracic diaphragm pacing) or directly into the diaphragm muscle near the insertion of the phrenic nerve through laparoscopy (intraperitoneal diaphragm pacing). Diaphragmatic pacing requires the phrenic nerves to be intact. Diaphragm reconditioning is required after implantation. Perioperative mortality is negligible, and ventilator weaning rate is reported to be 72% to 96%. Although not all patients can be weaned postprocedure, those who are experience enhanced quality of life through improved patterns of physiologic breathing.¹⁴⁶ The diaphragm pacing system consists of an external transmitter and antenna, implanted electrodes, and a receiver. The working life expectancy of the receiver is for the lifetime of the patient, and batteries last for 2 to 3 weeks. A bipolar electrode is available for use in persons who already have demand cardiac pacers. Advanced warning is given of transmitter battery failure, with a gradual decrease in tidal volume over several days. Although thoracic phrenic nerve implants for infants and younger children have required thoracotomy in the past, thoracoscopic placement of the electrodes has now been reported in



Fig. 29.4 Nasal intermittent positive-pressure ventilation in a tetraplegic ventilator user with postpolio syndrome. The ventilator is a typical portable volume ventilator.



• **Fig. 29.5** Mouth intermittent positive-pressure ventilation with angulated mouthpiece in a tetraplegic ventilator user with postpolio syndrome. The mouthpiece is held with teeth and lips.



• **Fig. 29.6** Mouth intermittent positive-pressure ventilation (IPPV) with angulated mouthpiece and without lip seal in a ventilator user with postpolio syndrome. She is taking a nap in her wheelchair. She can hold the IPPV hose with her hands on her chest. She has some ability to breathe on her own and is a good frog breather when she is awake.

children as young as 5 years.²²⁵ Remote monitoring can be used to assess effectiveness and to diagnosis problems.

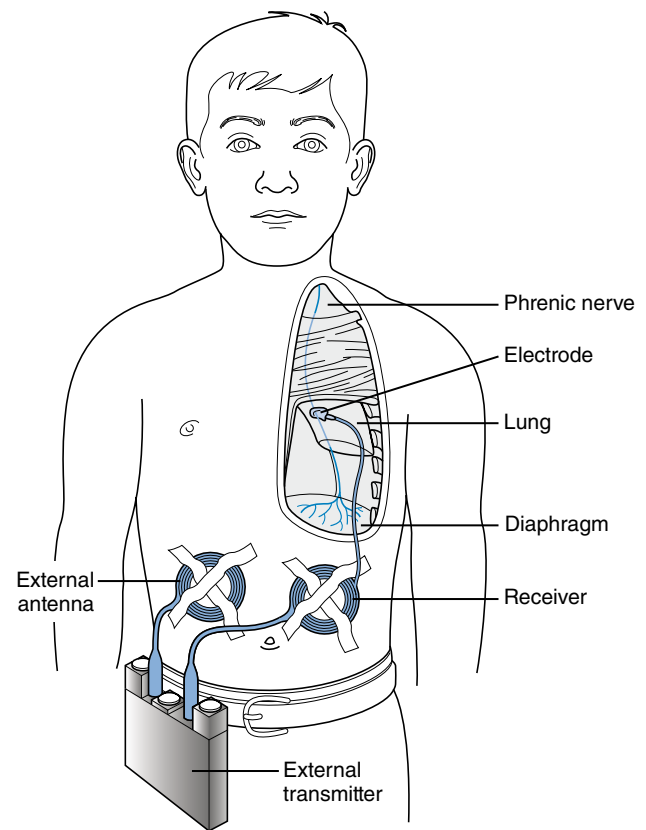
Injury to the phrenic nerve precludes diaphragmatic pacing unless a nerve transfer is performed. Patients with high cervical SCI at the C3 to C5 level develop axonal loss in the phrenic nerves, and pacing is not possible. Nerve transfer, where the fourth intercostal nerve was attached to the distal segment of the phrenic nerve above the diaphragm, has proven successful.¹⁴⁰

In this study, the diaphragmatic pacemaker was implanted on the phrenic nerve distal to the anastomosis. The average interval from surgery to diaphragmatic response to electrical stimulation was 9 months, and all patients were able to tolerate diaphragmatic pacing as an alternative to NIPPV.

In patients having only a single phrenic nerve that can be paced, combined intercostal muscle and unilateral phrenic nerve stimulation has also been shown to maintain ventilatory support.²²⁵

Long-Term Outcomes of Mechanical Ventilation

Many patients surviving significant illness become chronically ventilator dependent. To meet this need, the number of long-term



• **Fig. 29.7** Bilateral diaphragmatic pacemaker use in a young adult with central hypoventilation syndrome.

care hospitals in the United States continues to increase. Survivors of acute lung injury weaned from the ventilator have been noted to have generalized cognitive dysfunction as well as decrements in attention, memory, concentration, and/or mental-processing speed that can persist long term after discharge. Many also have musculoskeletal complications such as weakness and numbness resulting from critical illness neuropathy and critical illness or steroid-induced myopathy. Recovery of physical strength and function has been noted following comprehensive rehabilitation programs to address such deficits.¹²² Acute respiratory distress syndrome survivors weaned from the ventilator have been studied 1 year after hospital discharge.⁶⁹ Emotional dysfunction included the presence of depression, anxiety, and posttraumatic stress disorder. Long-term outcomes of diaphragm paralysis after acute SCI have been reported.¹⁸⁹

Over a 16-year period, 107 patients required assisted ventilation in the acute phase of injury. Of this group, 31% (33 patients) with a level of injury between C1 and C4 (grade A in the American Spinal Injury Association Impairment Scale) had diaphragmatic paralysis at the time of respiratory failure. Twenty-one percent of the 33 patients were able to breathe independently after 4 to 14 months, with a vital capacity greater than 15 mL/kg. Another 15% (five patients) had some recovery that took more time and for whom the vital capacity remained less than 10 mL/kg. With fastidious adherence to a comprehensive NIPPV protocol in a population with neuromuscular disease, long-term respiratory outcomes are very encouraging, including avoidance of tracheotomy in patients with Duchenne muscular dystrophy.²¹ Patients can be maintained on 24-hour mechanical ventilation for

up to 15 to 25 years. Cardiomyopathy has become the limiting factor for survival in this population as respiratory deaths are avoided. Long-term outcomes of inpatients with chronic-assisted ventilatory care are also encouraging, with 36% of patients able to leave an inpatient facility to a more independent community-based environment.²⁷²

Better outcomes were seen among patients with SCI and neuromuscular disease than in patients with COPD and thoracic restriction. Long-term NIPPV in kyphoscoliotic ventilatory insufficiency (thoracic restriction) improves blood gas levels and respiratory muscle performance and reduced hypoventilation-based symptoms and number of hospital days.⁹⁶

Lung Volume Reduction in Emphysema

Lung volume reduction (LVR) in patients with advanced emphysema improves pulmonary function, exercise capacity, and quality of life in a select subset of patients. Patients considered for LVR include those with advanced emphysema who have failed maximal medical therapy, undergone PR, and are not smoking. Surgical and bronchoscopic approaches for LVR have been described. Careful patient review is essential to determine approach and select appropriate candidates. Surgical LVR surgery has been associated with significant morbidity and mortality, prompting the development of bronchoscopic approaches and the deployment of bronchial valves or coils to facilitate LVR.^{207,222}

In surgical LVR, 20% to 30% of one or both lungs (usually at the apices) is removed resulting in reduced hyperinflation (decreased residual volume and total lung capacity) and improved FEV₁ and FVC.⁶³ The 6MWT increased significantly, and quality of life improved. In general, inspiratory muscle function (maximal inspiratory pressure and maximal transdiaphragmatic pressure) also improved. Combining LVRS with PR produces greater benefits than exercise alone.^{25,110} Not all patients will benefit from surgical LVR. Patients with FEV₁ less than 20% predicted and homogeneous disease, or a diffusion capacity less than 20% predicted, experience an increased mortality rate after LVR surgery. Additional studies suggest that patients getting the most benefit from LVR surgery are those with predominately upper lobe emphysema with low postrehabilitation exercise capacity.¹⁸⁰

Conversely, patients with non-upper lobe involvement and high postrehabilitation measures of maximal work received little benefit from LVR surgery. The results of the National Emphysema Treatment Trial suggest that all LVR surgery candidates first enroll in a PR program because some patients can improve to the point that they might not qualify for the surgery.⁸⁴

Furthermore, participation in a PR program before LVR surgery helps to both identify appropriate surgical candidates²¹² and ensure that candidates have appropriate levels of compliance and an understanding of their condition.²⁴³ Bronchoscopic LVR is an alternative treatment option, which is a less invasive and potentially beneficial intervention for patients who may not be candidates for traditional LVRS.⁷ There are many different interventions, including various stents, occluders, polymers, coils, and thermal vapor ablation. Stapling and plication of the most emphysematous tissue are carried out. Atelectasis can also be induced in these areas with the use of endobronchial sealants or valves. Extraanatomic tracts between the major airways and the emphysematous tissue are created and prevented from collapsing with stents to facilitate expiratory airflow.

Long-Term Results for Lung Volume Reduction Surgery

The distribution of emphysematous changes in lung parenchyma (measured by high-resolution CT) and measures of exercise capacity after rehabilitation are important factors in predicting the long-term effects of LVR.¹⁸⁰ Patients with predominantly upper lobe emphysema and lower levels of exercise capacity who received LVR surgery continued to have improved survival rates after 4 years, improved function after 3 years, and improved dyspnea after 5 years. Patients with upper lobe involvement and high levels of exercise capacity after rehabilitation continued to have symptomatic improvements when assessed after 4 years. Patients with non-upper lobe emphysema and low exercise capacity primarily experienced improvements in health-related quality of life that became insignificant after 3 years. Patients with non-upper lobe emphysema and high exercise capacity did not experience significant benefits from LVR surgery and experienced a higher mortality rate during the first 3 years. Importantly, improvements in all groups were no longer significant after 5 years. Overall, LVR surgery in selected patients is effective in improving symptoms, function, and quality of life.¹¹²

Bronchoscopic Endobronchial Valve Treatment

A form of bronchoscopic LVR, the placement of one-way endobronchial valves is a viable option for the management of select patients with advanced emphysema with severe hyperinflation.¹³⁶ Long-term follow-up over 3 years demonstrates sustained improvements in PFTs, walking distance, and quality of life.¹⁰⁵

Lung Transplantation

As a treatment option for patients with end-stage progressive and debilitating lung disease, the rate of lung transplantation (LTx) has been steadily increasing since 2000. Since 2016, approximately 4500 adult lung transplants have been performed globally every year. Further growth is attributed to a shortage of lung donors. The most common indications for LTx in adults include COPD, idiopathic pulmonary fibrosis, CF, alpha₁-antitrypsin deficiency (AATD), and idiopathic pulmonary arterial hypertension. CF and primary PH are the two main diseases of childhood for which lung transplant is performed. Patients with these diagnoses need to meet defined criteria to be eligible for transplant.¹⁴⁷ LTx should not be considered in a patient with a florid infection, recent malignant tumor (within 2 years), continued addictive behavior, smoking, or lack of reliable social support. Relative contraindications include older age, poor functional status, obesity, severe osteoporosis, and correctable coronary artery disease.¹⁵¹ Of programs in North America that have been queried, 58% have a minimum requirement for exercise capacity to be considered for lung transplant. Covering a minimum of 600 feet on a 6MWT is the most common requirement, although some programs require only 250 feet. If an individual can only transfer from bed to chair, 46% of lung transplant centers consider this an absolute contraindication to lung transplant, and 52% consider it a relative contraindication. Survival in LTx continues to improve in part due to an increasing number of bilateral lung transplants (BLTs) being performed.²⁶⁹

Long-term results in lung transplant

The median survival for adult lung transplant recipients since 2010 is 6.7 years and is increasing compared to earlier time periods.⁵⁴ Those undergoing BLTs have a better median survival (7.8 years) than single lung recipients (4.8 years). In 2021, 85.3%

of lung transplant recipients survived 1 year, 67% survived 3 years, 54.3% survived 5 years, and 32.8% survived 10 years.²⁵⁹ Posttransplant survival has improved over time despite the relaxation of selection criteria resulting in patients' characteristics and severity at the time of transplant being significantly worse in more recent times. One of the main determinants of LTx outcome is the underlying disease, with a median survival of 8.9 years for CF patients, 6.7 years for COPD with AATD, 5.6 years for COPD without AATD, 4.8 years for idiopathic interstitial pneumonia, and 2.8 years for retransplantation. Other prognostic factors relate to the recipient (sex, age, 6-minute walking distance, mechanical ventilation, dialysis, or intensive care unit [ICU] stay), the donor (diabetes, age, cause of death), the donor/recipient interaction (human leukocyte antigen [HLA] mismatches, cytomegalovirus [CMV], or sex mismatch), the surgical approach (single vs. bilateral), and the center volume.²⁵⁰ The highest mortality rate occurs during the first year.²⁵⁴ Sepsis and bronchiolitis obliterans remain the two most common causes of death. Cause of death following LTx varies with time posttransplantation. Thirty-day mortality is related primarily to surgical issues, donor lung preservation, and primary graft dysfunction. In the subsequent posttransplant period, infectious causes, malignancy, and chronic lung allograft dysfunction (CLAD) predominate. With significant improvements in short-term outcomes owing to improved donor-lung preservation and surgical technique, the 5-year survival after LTx has increased to nearly 60%,³⁶ but it remains a challenge due to the development of CLAD (i.e., graft dysfunction resulting from multiple etiologies, including bronchiolitis obliterans of the small airways and fibrosis in the peripheral lung tissue).²⁷⁸ These outcomes should be tempered with the knowledge that the mortality rate for those on the transplant waitlist is also high, although it is stable at 18.8 deaths per 100 patient-years.²⁵⁹ Pretransplant mortality rates are higher for males, blood group AB, and those with extremes of stature (e.g., tall or short). The 3-year survival rate for those in the sickest category of the COPD BODE (BMI, degree of airflow obstruction, functional dyspnea, exercise capacity) index is only 50%.⁴⁸

Special Considerations of Pulmonary Dysfunction

Obesity-Related Pulmonary Dysfunction

The prevalence of obesity (BMI >30) continues to increase, reaching 42.4% among adults and 19.3% among youth in the United States in 2017-2018. The percentage of US adults aged 20 years and older who are overweight or obese was 73.1%.⁸⁶ All racial and ethnic groups and both sexes are affected. Primary respiratory complications associated with obesity include obstructive sleep apnea, obesity-hypoventilation syndrome, and asthma.²⁸⁰ Restrictive pattern on PFTs is seen and may be severe with increasing BMI. Chronic hypoxemia related to obesity-hypoventilation syndrome can lead to polycythemia, PH, cardiac dysrhythmias, and right ventricular failure. One study reported that three of four adults who presented to the emergency department with acute asthma were either overweight (BMI = 25–29.9; 30%) or obese (BMI >30; 44%).²⁵¹ Other conditions that can be associated with obesity include atelectasis, pneumonia, venous thrombosis, and PE.¹⁷⁶

Daily physical activity is important in preventing and treating overweight and obesity as many of the aforementioned conditions are improved with weight loss. Obesity increases the work and

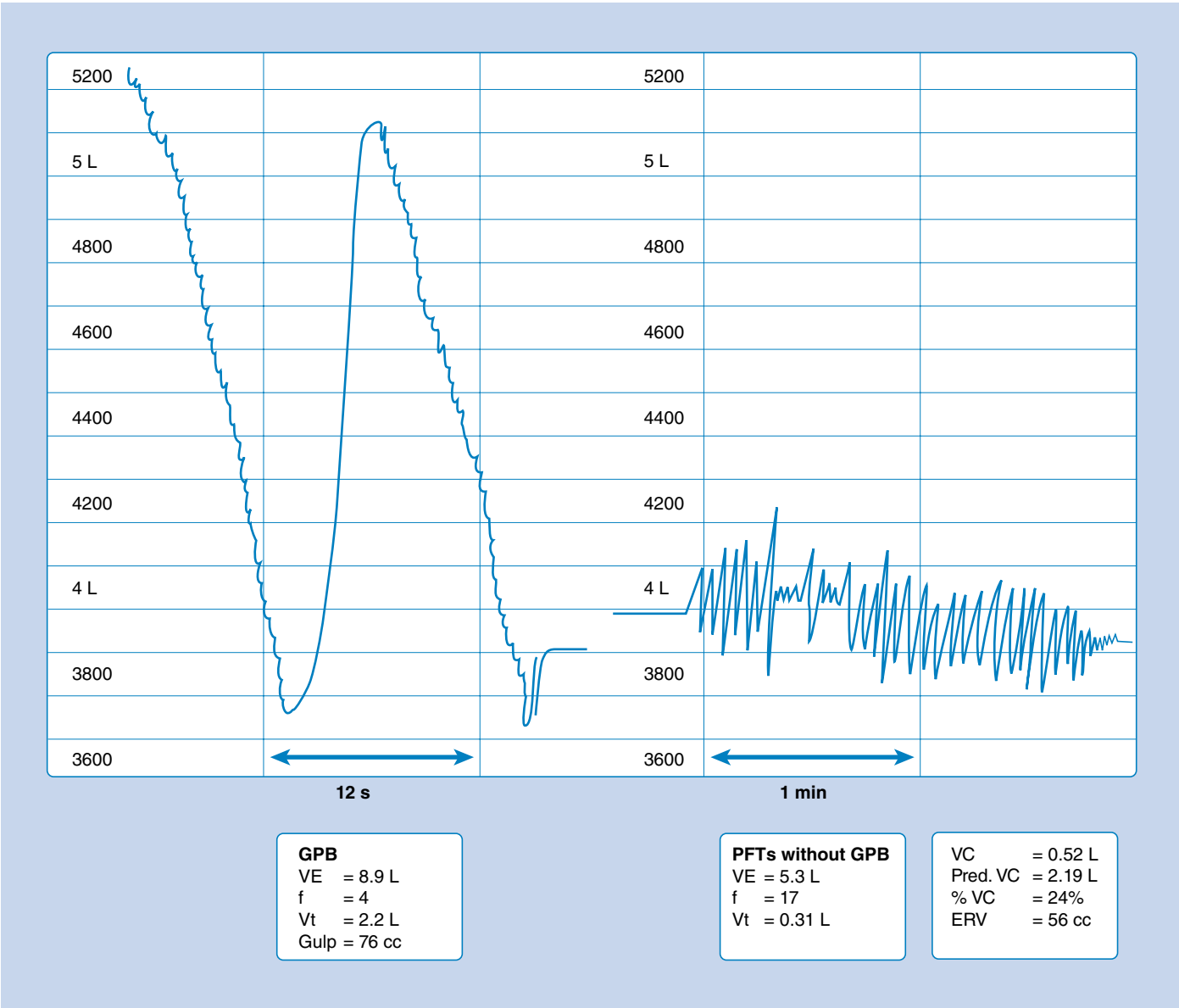
metabolic requirements of breathing and physical activity, resulting in decreased exercise tolerance. The weight relative maximal volume of oxygen consumption (VO₂ max) in obese persons can be significantly decreased compared with controls. Norman et al.¹⁸⁴ studied severely obese adolescents (BMI >40) and found that although absolute VO₂ max (mL O₂/min) was similar to controls, walk/run distance was approximately 60% of the control average. Although obese adults with COPD might have greater fatigue, functional impairment, and significantly lower 6-minute walk distance compared with normal-weight individuals with COPD, similar relative improvements in walk distance are seen after PR.²⁰⁶

Spinal Cord Injury and Pulmonary Dysfunction

In SCI, there is diminished cardiopulmonary and circulatory function as well as lower limb muscle atrophy and bone mass reduction. Patients with high cervical cord injuries, including children as young as 3 years, can learn neck breathing⁹³ or GPB as a form of voluntary respiration. This allows some time off the ventilator. In GPB (or frog breathing), air is pumped into the lungs by the patient using the tongue as a piston (Fig. 29.8). Also known as glottic press to speech pathologists, it is taught to patients who have had a laryngectomy to pump air into the esophagus for esophageal speech. GPB can be used for a number of purposes in addition to being an alternative form of breathing. GPB can improve vocal volume and flow of speech to allow the patient to call for help and to provide the deep breath needed for an assisted cough.

Future Directions and Areas of Opportunity in Pulmonary Dysfunction

Current guidelines recommend outpatient PR after hospitalization for a COPD exacerbation. However, recent data suggest very poor referral and participation rates with a dismal 1.9% of individuals hospitalized for COPD receiving PR within 6 months of their hospitalization and just 2.7% within 12 months.²²⁹ The American Association of Cardiovascular and Pulmonary Rehabilitation and the ATS are working to address this imbalance between need and utilization of PR. Major initiatives include education of individuals with pulmonary diseases as well as health care providers caring for them of the value of PR. The business model for PR does not adequately support free-standing PR centers, and thus too few PR centers exist to provide care to all who could benefit. As referral rates improve, novel methods to deliver PR are being developed utilizing telemedicine and remote patient monitoring for home- and community-based PR programs.²²⁰ Telehealth PR has been shown to be effective even for patients with severe COPD,⁴¹ and it can result in significant improvements in dyspnea, walk distance, and emotional status.²⁸² The development of remote patient monitoring and wearable technology for patients with COPD is further opening channels to provide PR to those with limited access to brick-and-mortar PR programs.⁶⁶ Insurance coverage for telehealth PR remains a challenge because it is not currently covered under Medicare or Medicaid, although some insurance companies (including Medicare and Medicaid) are reimbursing for remote patient monitoring.²⁴⁷ High rates of readmission are seen within 30 days of discharge following admission for a pulmonary exacerbation. Readmission reduction programs are addressed in an ATS Workshop



• **Fig. 29.8** Glossopharyngeal breathing (GPB) tracing and results compared with pulmonary function tests without GPB in a tetraplegic ventilator user with postpolio syndrome. *ERV*, Expiratory reserve volume; *f*, frequency; *gulp*, volume or single GPB stroke; *PFT*, pulmonary function testing; *VC*, vital capacity; *VE*, minute ventilation; *Vt*, tidal volume.

Report on current best practices and models for addressing COPD hospital readmissions,²⁰⁰ and they have been addressed in the Hospitals Readmissions Reduction Program developed by the Centers for Medicare and Medicaid Services.¹⁹⁹

Traditional PR programs and telemedicine PR programs have been shown to reduce COPD readmissions.³¹ Early referral and engagement in outpatient PR programs is aided by automatic/default orders for PR in the discharge order set when patients with pulmonary disease are discharged from acute care. Initiating PR within 90 days of hospital discharge can result in fewer rehospitalizations over the following year.²³⁵

Summary of Pulmonary Rehabilitation

PR requires a dedicated and specialist team approach to the management of disability secondary to chronic lung disease.

The rehabilitation program is ideally led by a PM&R physician with knowledge of the anatomy and pathophysiology of the cardiovascular and respiratory systems as well as exercise physiology. The PM&R evaluation and treatment plan should reflect the diverse pulmonary pathology impacting function and the comprehensive team available for management. PR reflects a continuum of care from point of entry to the health system due to pulmonary disease presentation or exacerbation, utilizing all available rehabilitation services to return patients to their highest possible level of function in their community. While, ideally, PR is delivered in a dedicated PR setting, commonly it is practiced as a component of general rehabilitation where patients have pulmonary dysfunction as a complication of their neurologic or musculoskeletal disabilities or as a medical comorbidity. General rehabilitative therapeutic approaches apply in either setting. Health policy, legislation, and regulations, including current health care delivery

systems, must consider the need for PR at all ages and for the diverse population in our society. An informed public can facilitate change in this regard. The public is more informed now because of the development of online interest groups, internet bulletin boards, and hundreds of thousands of articles pertaining to PR online.

Organ Transplantation

Introduction

The idea of organ transplantation dates back to ancient mythology and is described in Roman, Greek, Indian, Chinese, and Egyptian tales. The first written mention of an organ transplant is noted in the *Ebers Papyrus*, written circa 1550 BC, which mentions skin grafting for the treatment of burns.¹⁹⁵ Seventy years ago, the first successful organ (kidney) transplant was performed by Joseph Murray at the Peter Bent Brigham Hospital in Boston, Massachusetts. By the late 1960s, liver, heart, and pancreas transplants from deceased donors were successfully performed, and lung and intestinal organ transplant procedures began in the 1980s. The years since the first successful organ transplant have seen remarkable advances in surgical techniques, pre- and postoperative care, and development of potent immunosuppressant medications. In 1963, one of the most important landmarks in the history of organ transplantation was Tom Starzl's finding that adding prednisone to azathioprine for immunosuppression improved 1-year renal graft survival to greater than 70%.²³⁴ This finding caused a sensation and completely changed the outlook for renal transplantation. The next landmark revolutionizing transplantation was the finding in 1976 that cyclosporine substantially improved kidney transplant outcomes.³⁵ Cyclosporine soon became the standard baseline immunosuppressant. It remained this way until 1989 when it was found that tacrolimus could prevent rejection in organ allografts resistant to treatment by cyclosporine. Tacrolimus has now largely replaced cyclosporine as the baseline immunosuppressant agent.²³⁴ These discoveries have resulted in markedly reduced morbidity and mortality in posttransplantation patients. Thousands of patients in the United States and around the world have benefited immensely from these advances in medical care by not only having their lives extended but also the quality of their lives.

Due to the multiple medical and functional complexities encountered when working with an organ transplant recipient, a team approach in managing these patients is critical. The transplant rehabilitation team consists of a physiatrist, physical therapist, occupational therapist, registered nurse, social worker, case manager, clinical pharmacologist, nutritionist, psychologist, and vocational therapist. The physiatrist's role as team leader is to coordinate the interdisciplinary team. Constant communication with the primary transplant team and immediate discussion of any changes in the patient's medical status is essential for optimal patient outcomes. The physiatrist and rehabilitation team should be knowledgeable of potential short- and long-term medical complications posttransplant. The frequent and ongoing interaction with transplant recipients situates the rehabilitation team to be the early warning system for any deterioration in this patient population.

Transplant recipients require immunosuppressant medications for life—typically, a combination of a calcineurin inhibitor (tacrolimus [Prograf] or cyclosporine [Neoral]), an antimetabolite (mycophenolate mofetil [Cellcept]), and a steroid (prednisone).

The physiatrist must be aware of the potential side effects that can occur with these medications and the multiple drug interactions that can occur with this triple therapy. In addition, organ transplant recipients are considered to be at high risk for developing infections. This section will focus on the four most common transplantable organs seen in rehabilitation medicine: kidney (most commonly transplanted organ), heart, lung, and liver. For each of these transplantable organs, the pre- and postoperative rehabilitation management, postoperative medical complications, immunosuppressive and prophylactic medications utilized, and outcomes will be discussed. In addition, the key role that rehabilitation plays in restoring the organ transplant recipient's functional potential and maximizing patient quality of life will be highlighted.

Renal Transplantation

Introduction

Renal transplantation is the treatment of choice for patients with end-stage renal disease (ESRD). ESRD is total and permanent kidney failure requiring dialysis or transplantation for survival. Indications for renal transplantation and contraindications for renal transplantation are outlined in Boxes 29.4 and 29.5.

Epidemiology

The kidney is the most commonly transplanted major organ. In 2021, 25,549 kidney transplants were performed, of which 19,569 were from deceased donors and 5979 were from living donors. The number of patients with a functioning kidney transplant increased to 245,506 by the end of 2021, an increase of 39.1% over the decade. However, only approximately 30% of ESRD patients had undergone a kidney transplant, a percentage that remained virtually unchanged over the decade. Overall, White and Asian patients consistently had the highest rates of kidney transplantation. Hispanic, Black, Native American, and Native Hawaiian/Pacific Islander patients had lower rates. Kidney

• BOX 29.4 Indications for Renal Transplant

Any condition that leads to end-stage renal (kidney) disease:

- Diabetes
- Hypertension
- Prerenal (chronic or acute ischemia)
- Intrinsic renal (glomerulonephritis, focal-segmental glomerulosclerosis)
- Postrenal categories (reflux nephropathy, obstruction)

Data from Abramyan S, Hanlon M. *Kidney Transplantation*. StatPearls Publishing; 2024.

• BOX 29.5 Contraindications for Renal Transplantation

- Inability to tolerate surgery due to severe cardiac or pulmonary disease
- Active malignancy
- Active infection
- Active drug abuse
- Uncontrolled psychiatric disease

Data from Abramyan S, Hanlon M. *Kidney Transplantation*. StatPearls Publishing; 2024.

transplantation rates varied by more than fivefold across the United States, from a low of 1.2 transplants per 100 person-years in Hawaii to a high of 6.5 per 100 person-years in Utah. This may be related to the relatively high rate of kidney donation after traumatic death in Utah.²⁵⁸

Etiology of End-Stage Renal Disease

Diabetes and hypertension are the two leading causes of ESRD in the United States and worldwide. Other diseases leading to serious kidney dysfunction and for which renal transplantation may be required are grouped into prerenal (chronic or acute ischemia), intrinsic renal (glomerulonephritis, focal-segmental glomerulosclerosis), and postrenal (reflux nephropathy, obstruction) categories.⁴

Preoperative Rehabilitation Management

When prescribing a rehabilitation program for potential kidney transplant patients (preoperative or postoperative), one must consider their markedly impaired cardiorespiratory and musculoskeletal status. Physical activity, cardiorespiratory fitness, muscle mass, and bone mineral density decrease even in the early stages of chronic kidney disease (CKD) and continue to decrease with disease progression. The combined effects of uremia and physical inactivity drive the loss of muscle mass.²⁸¹ The markedly impaired exercise tolerance that these patients experience not only interferes with their ability to perform physical tasks such as climbing stairs but also interferes with their ability to perform leisure-time exercise. Their reduced physical fitness is characterized by reduced flexibility, coordination disturbances, and decreased muscular strength and endurance.⁸⁷ Decreased pretransplant physical function as measured by the Short Physical Performance Battery (which measures balance, chair stands, gait speed, and grip strength) has been shown to be associated with a longer length of hospital stay and rehospitalizations following kidney transplant.^{157,177} It has also been shown to be associated with a 2.3-fold increased risk of posttransplant mortality.¹⁷⁸

In addition, kidney transplant recipients (KTRs) have specific features that distinguish them from other SOT recipients. These need to be kept in mind when prescribing a rehab program for these patients. First, about half of all patients with ESRD worldwide are over 65 years of age. This results in a higher proportion of older KTRs compared with other SOT recipients. Next, advancements in dialysis techniques allow for a variable, sometimes very long, waiting times on dialysis. The majority of deceased-donor KTRs require a waiting time of 2 to 4 years, compared with 0 to 5 months for liver, heart, and lung transplant recipients. This chronic disease burden has a great impact on the progression of comorbid conditions and quality of life.²⁰⁸ It is felt that kidney transplant candidates are uniquely suited for rehabilitation because they experience this profound loss of functional capacity while waiting years for transplantation while on dialysis. McAdams-Demarco et al. found that 2 months of prehabilitation, consisting of weekly outpatient rehabilitation, improved physical activity by 64% and decreased posttransplant length of stay when compared with age-, sex-, and race-matched controls. In addition, they stressed the importance of optimizing metabolic and nutritional status along with promoting physical activity.¹⁷⁰ For patients with all stages of CKD, it has been shown that regular physical activity benefits cardiometabolic, neuromuscular, and cognitive function.²⁸¹

Postoperative Medical Issues

Despite the benefits of kidney transplantation for end-stage renal failure, cardiopulmonary and musculoskeletal dysfunctions are a continued concern after surgery.⁷⁵ In addition, as with any intraabdominal procedure, patients undergoing kidney transplant have impaired postoperative pulmonary function due to general anesthesia and diaphragmatic inhibition.²⁰⁴

Postoperative muscle weakness and reduced exercise tolerance are also seen after surgery.²⁶²

It is unclear whether this is a result of disease-related changes in skeletal muscle physiology or due to a reduction in physical activity posttransplantation. In addition, the patient's reduced bone mineral density continues to be a major problem after transplantation. This is a result of suboptimal kidney function and the superimposed effects of steroids on bone. A reduction in bone mineral density has been found in 60% of patients in the first 18 months after renal transplantation.⁶¹ The estimated fracture rate after transplantation is 2% per year in nondiabetic patients, 5% per year in diabetic patients, and 12% per year in pancreas-kidney recipients.

Infections are a common complication following kidney transplantation. Infectious complications are categorized as occurring in one of three time periods posttransplant: early posttransplant infections, infections during peak immunosuppression, and late-onset infections. Most infections in the first month after transplant are health care–associated infections, whereas late infections, beyond 6 to 12 months, are community-acquired infections. Early posttransplant infections occur in the first 30 days posttransplant. The majority of such infections (approximately 98%) are common postsurgical infections, including surgical site infections, pneumonias, urinary tract infections (UTIs), bacteremia, and *Clostridioides difficile* colitis. The second period of posttransplant infections occurs during peak immunosuppression (between 30 days and 6 months posttransplant) or within 3 months of treatment of rejection. They are typically opportunistic infections or pathogens that reactivate from latent infection in the recipient, such as BK polyomavirus, CMV, herpes simplex virus, varicella zoster virus, hepatitis B virus, hepatitis C virus, tuberculosis, listeria, strongyloidiasis, and Chagas disease. Late-onset infections typically present greater than 6 to 12 months posttransplant or greater than 3 months after treatment for a rejection episode. Most late-onset infections are community acquired, such as community-acquired pneumonia, respiratory viral infections, and UTIs.⁶

Postoperative Rehabilitation Management

The goal of rehabilitation following renal transplantation is to improve patients' physical and psychological health for them to achieve a level of fitness that will allow them to maintain an active lifestyle. A meta-analysis and systematic review of the effects of exercise training in kidney transplantation recipients revealed that exercise training resulted in significantly improved exercise tolerability as measured by VO₂ peak and quality of life.¹⁸⁷ In a randomized double-blind study, post-renal transplantation patients undergoing an early physiotherapy program from the first postoperative day until hospital discharge had significantly higher respiratory muscle strength at hospital discharge when compared with a control group.¹⁸⁸ A study by Zhu et al. found that early mobilization after renal transplantation had multiple benefits compared to a control group. The early mobilization group

experienced a shorter duration of indwelling drainage tube/catheter, earlier ambulation, and shorter hospital stays. The time to return of bowel sound, flatus, and defecation were all advanced in the early mobilization group. Patient satisfaction was also significantly higher (94.29% 33/35) than that in the control group (77.14%, 27/35, $P < 0.05$).²⁸³ A 12-week pilot randomized controlled trial (RCT) in KTRs concluded that both aerobic training and resistance training interventions are feasible and clinically beneficial in KTRs.¹⁰⁰ Resistance exercise training has been found to restore bone mineral density in renal transplant recipients.⁷⁸ Aerobic exercise, with or without resistance training, is effective in improving cardiorespiratory fitness in KTRs. Resistance training, with or without aerobic training, improves muscle strength, irrespective of the time after transplantation. Resistance training in stable recipients improved lower body muscular endurance assessed by the 60-second sit-to-stand test. Both aerobic and resistance training improved arterial stiffness, assessed with carotid-femoral pulse wave analysis as the primary outcome, after 3 months of training. In addition, 6 months of exercise training considerably improved autonomic regulation of the heart. It increased both vagal and sympathetic cardiac modulation while shifting the autonomic balance toward greater vagal control. The majority of RCTs indicate that exercise training improves health-related quality of life in KTRs, irrespective of the type of training or the time after transplantation.⁷³

Outcomes

The prognosis after kidney transplantation is excellent. For those transplanted in 2020, the 1-year patient survival among living donor recipients was 98.1%, and 92.6% for deceased donor recipients. The 5-year patient survival for living donor recipients was 87.2%, and 76.3% for deceased donor recipients.²⁵⁸ Despite this, kidney transplantation recipients experience a variety of challenges in adjusting to daily life activities as well as dealing with the social consequences and emotions related to their transplantation and the donor. A cross-sectional study to evaluate the different components of physical performance and quality of life in a cohort of kidney-transplanted patients revealed the presence of significant limitations in physical, social, and emotional functioning within the first 6 months after transplantation. This was felt to be due to factors related to the hospitalization and to prophylactic restrictions (use of a face mask, frequent hand washing, avoidance of contact with animals, etc.). The authors stress that a systematic assessment for every kidney transplant patient should include an evaluation of physical performance and quality of life.⁸¹

Older KTRs (ages 65 to 80 years) often feel constrained by slow recuperation and overwhelmed by unexpected comorbidities, medication-related side effects, and health decline. There is a need to prepare and support older recipients for self-management responsibilities, clarify their expectations of posttransplant risks and outcomes, and aid throughout the prolonged recovery after kidney transplantation.¹⁹⁷ It has been shown that kidney transplant patients who underwent an empowerment education program showed a decrease in uncertainty and an improvement in self-care ability and compliance.¹³⁰ In addition, patients also have to deal with the challenge of complex drug regimens. A study aimed to evaluate the behavioral and physiologic outcomes of pharmaceutical care in this population revealed that consistent posttransplant pharmaceutical care is effective to substantially improve knowledge following transplantation.²⁷⁶ Guidance was

consistently provided to patients so they could gain knowledge of their medications and side effects as well as understand the importance of rejection prevention. Interventions included face-to-face interviews, checkups for laboratory examinations, discovery of drug-related problems, and pharmaceutical consultation. The authors concluded that pharmaceutical care should be started as early as possible during the pretransplant period and continue in a long-term follow-up.

Patients with kidney failure treated with dialysis or kidney transplantation experience difficulties maintaining employment due to the condition itself and the treatment. A systematic review and meta-analysis found that patients with kidney failure had a low employment rate during dialysis and pre- and posttransplant. The employment rate for dialysis patients was 26.3%, pretransplant patients 36.9%, and posttransplant patients 38.2%. In the meta-analysis, the strongest predictor of posttransplant employment was pretransplant employment. Younger patients and those with higher educational levels were more likely to be employed posttransplant. The authors concluded that it is important to support kidney failure patients through a combination of clinical and social measures to ensure that they remain in work.¹³⁴

Future

Since the first successful kidney transplant in 1954, outcomes have steadily improved through advances in immunosuppressive medications, tissue typing, and organ preservation techniques. Kidney transplantation is associated with much better long-term outcomes than any other modality of renal replacement and should be the goal for the majority of patients with ESRD. Despite this, the growth of kidney transplantation has not kept up with the growth in the ESRD population. The organ shortage has led to patients waiting many years on dialysis with increased mortality. According to the US Renal Data System 2023 Annual Data Report, by the end of 2021 72,864 patients with ESRD were on the waitlist for a kidney transplant. One year after being on the kidney transplant waitlist, 12.3% of patients had received a living donor transplant, 15.2% had received a deceased donor transplant, 1.7% had died, and 3.6% had been removed from the waitlist. Thus 67.2% were still waitlisted and receiving dialysis 1 year after being placed on the waitlist. Five years after being waitlisted for a kidney transplant, 46.3% of patients had received a transplant, whereas 18.5% had died.²⁵⁸ The main challenge for the future will be to increase the number of kidney transplants and reduce the gap between supply and demand. Strategies include promotion of living donations, changes made to the kidney allocation system in an attempt to allocate available kidneys with increased equity and efficiency, and the use of organs from donors with hepatitis C or HIV.⁵⁸ Other challenges include increasing the focus on improving quality of life and employment opportunities for patients following kidney transplantation.

Heart Transplantation

Epidemiology

Congestive heart failure (CHF) affects approximately 6.7 million Americans over the age of 20 years, and this number is expected to increase to 8.5 million by the year 2030.³⁷ Cardiac transplantation offers the survival and quality of life benefit and remains the gold standard in treatment of select patients with end-stage heart failure where medical management fails. The biggest challenge in

the heart transplantation domain is donor heart availability.²⁶⁴ Christian Barnard at Groote Schuur Hospital in Cape Town, South Africa, performed the first human heart transplant on December 3, 1967.⁶² In the United States, Norman Shumway at Stanford Hospital in California performed the first heart transplant on January 6, 1968.²⁴ According to the United Network for Organ Sharing, as of 2021, over 31,000 heart transplants had been performed in the United States over 10 years, and 4545 were performed in 2023 alone.¹⁹⁰

Owing to advances in medical therapy for patients with CHF as well as the increased use of mechanical circulatory support implantable cardiac devices, the annual mortality rate while on the transplant waitlist has progressively declined from 15% to 20% to 7% to 10% since 2006. One-year survival for candidates on the waitlist increased from 34.1% for 1987-1990 to 67.8% for 2011-2017. The new allocation system implemented in 2018 resulted in improved waitlist survival. In addition, long-term outcomes after cardiac transplantation have improved with advances in candidate selection criteria, surgical techniques, infection prophylaxis, immunosuppressive agents, postoperative care, and rehabilitation.^{18,253} The median age of transplant recipients is 57 years, but the proportion of recipients at extremes of age has increased.

Other important changes in the demographics of heart transplant recipients over the past 2 decades include increased recipient BMI, increased proportion of females, and increasing proportion of transplant patients with diabetes, previous dialysis, hypertension, previous cardiac surgery, presence of mechanical circulatory support, and previous malignancy.¹²⁹ The median survival after heart transplantation is 12.4 years. If recipients survive past 1 year, they have a greater than 50% chance to survive at 14 years.¹⁶⁸

Adult heart transplant recipient survival is around 90% at 1 year, 85% at 3 years, and 80% at 5 years.¹⁰⁸ The 30-day operative mortality for cardiac transplantation spans from 5% to 10% because of graft failure, multisystem organ failure, and infection. Graft failure, acute rejection, and infection are the leading causes of death during the first year. Cardiac allograft vasculopathy and malignancy are the most significant causes of death after 1 year.²⁵³ The joint guidelines for cardiac transplantation candidacy from the American College of Cardiology and the American Heart Association are shown in Box 29.6. The absolute and relative contraindications are listed in Box 29.7.^{10,117,166}

Rehabilitation Before Heart Transplantation

The inciting etiology that leads to CHF includes a broad differential, including ischemic heart disease, valve dysfunction, idiopathic cardiomyopathy, or viral myocarditis. Evaluations by both a cardiologist and a physiatrist should be integrated to better understand the disease process and its probable impact on the patient's quality of life pre- and posttransplantation. The prioritization of appropriate candidates for transplantation is based on medical comorbidities as well as the expected gains in quality of life compared with the optimization of medical and surgical resources available. Geographic distance between donor and potential recipient is also a consideration. Patients awaiting transplantation may be reassessed every 3 months for quantification of circulatory impairments, exercise performance, comorbid diagnoses, and overall health. The most successful outcomes depend on the identification of candidates who have the physical capacity to maximally benefit from a new heart and the rehabilitation process.²²⁸ During the period of heart procurement, a myriad of implantable devices

• BOX 29.6 Indications for Cardiac Transplantation

- Refractory cardiogenic shock requiring intraaortic balloon pump counterpulsation or left ventricular assist device
- Cardiogenic shock requiring continuous intravenous inotropic therapy
- Peak $\dot{V}O_2$ ($\dot{V}O_{2\max}$) <10 mL/kg/min
- NYHA class III or IV despite maximized medical and resynchronization therapy
- Recurrent life-threatening left ventricular arrhythmias despite an implantable cardiac defibrillator, antiarrhythmic therapy, or catheter-based ablation
- End-stage congenital heart failure with no evidence of pulmonary hypertension
- Refractory angina without potential medical or surgical therapeutic options

NYHA, New York Heart Association.

• BOX 29.7 Absolute and Relative Contraindications to Heart Transplantation

Absolute Contraindications to Heart Transplantation

- Advanced irreversible renal failure with creatinine >2 or creatinine clearance <30 to 50 mL/min without plans for concurrent renal transplant
- Advanced irreversible liver disease
- Advanced irreversible pulmonary parenchymal disease or FEV₁ <1 L/min
- Advanced irreversible pulmonary artery hypertension (pulmonary artery systolic pressure >60 mm Hg, pulmonary vascular resistance >4 to 5 Wood units despite vasodilators) attributable to risk of acute right ventricular failure soon after transplant from insufficient accommodation of the donor heart to high pulmonary vascular resistance pressures
- History of solid organ or hematologic malignancy within the past 5 years as a result of probability of recurrence.

Relative Contraindications to Heart Transplantation Resulting From the Reversibility of the Disease Process or Lack of Direct Impact on the Transplanted Organ

- Severe peripheral vascular disease
- Severe cerebrovascular disease
- Severe osteoporosis
- Severe obesity (body mass index >35 kg/m²) or cachexia
- Acute pulmonary embolism
- Active infection (excluding left ventricular assist device-related infections)
- Advanced age (>70 years old)
- Psychological instability (e.g., posttraumatic stress disorder)
- Active or recent (within 6 months) substance abuse (alcohol, cocaine, opioids, tobacco products, etc.)
- Diabetes mellitus with end-stage organ damage
- Lack of social support or sufficient resources to permit ongoing access to immunosuppressive medication and frequent medical follow-up

can improve tolerance for activity and perhaps allow preliminary rehabilitation.²⁵³

Complications After Cardiac Transplantation

Coordination of care between the cardiac transplant surgeon, cardiologist, and physiatrist is essential for success. The cardiac rehabilitation team must be familiar with the complications of heart transplantation to address issues that might affect the recipient's functional performance. The main complication posttransplant is allograft failure from rejection. Signs of acute rejection include fulminant CHF, abrupt onset of new peripheral edema, new arrhythmias (such as premature atrial contractions),

and sudden reduction in exercise capacity. Secondary complications include problems related to immunosuppression, such as infection, neurotoxicity, renal toxicity, hypertension, and various metabolic abnormalities. Early cardiac failure still accounts for up to 20% of perioperative deaths of heart transplant patients, with the most important causes resulting from dysfunction of the donor myocardium from acute rejection, PH, and ischemic injury.²⁵³ Right-sided heart failure is a significant cause of morbidity and mortality in the setting of increased afterload resulting in ventricular failure from pulmonary vascular resistance. The heart often requires inotropic support during the period immediately after transplant. Owing to autonomic denervation, the transplant recipient may be subject to multiple types of arrhythmias, including atrial fibrillation, flutter, supraventricular tachycardia, or bradyarrhythmias.²⁵³ Cyclosporine-induced renal vasoconstriction superimposed on chronic renal hypoperfusion may result in hypertension through glomerular arteriolar vasoconstriction. Tacrolimus (Prograf, Advagraf, Protopic) is associated with a lower incidence of hypertension compared with cyclosporine, which may be used under the direction of the transplant team. Blood pressures should be closely monitored, intervening with antihypertensive agents as needed. In most circumstances, this can be achieved without interruption of the exercise therapy regimen. Moderate to severe hypertension can affect up to 90% of cardiac transplant patients. Diuretics and beta blockers should be used cautiously because of volume depletion/hypotension and blunting of the heart rate response to exercise.²⁵³

Careful assessment of neurologic and mental status as well as motor function is an essential part of the psychiatric consultation in the postoperative phase. Neurologic complications that can occur acute posttransplantation include metabolic encephalopathy, central nervous system infection, seizures, and psychosis. Strokes can occur from particulate or air embolism and cerebral hypoperfusion during the surgery. One of the leading causes of death in post-cardiac transplant patients is infection, including mediastinitis, pneumonia, UTI, or intravenous catheter-induced sepsis. *Escherichia coli*, *Pseudomonas aeruginosa*, and *Staphylococcus* have been shown to be the cause of the majority of bacterial infections in this patient population. However, CMV is the most significant cause of morbidity and mortality related to infection. CMV is associated with acute rejection, acceleration of coronary artery disease, and lymphoproliferative disease. CMV prophylaxis and active treatment include agents such as ganciclovir (Cytovene, Cymevene). *Aspergillus* is an opportunistic pathogen with the highest mortality and can cause serious pneumonia in 5% to 10% of recipients during the first 3 months posttransplant.²⁵³

The incidence of *Pneumocystis carinii* pneumonia can range from approximately 1% to 10%. *Toxoplasmosis gondii* infection is a result of reactivation of latent disease in the seropositive donor heart and can cause significant infections of the central nervous system, which require treatment with pyrimethamine (Daraprim), sulfadiazine (Lantrisol, Neotrizine), or clindamycin (Cleocin).²⁵³

Physiology of the Transplanted Heart

An understanding of the physiology of the transplanted heart is essential to the physiatrist prescribing therapy regimen. In a normal heart, chronotropic and inotropic regulation is dictated by the autonomic nervous system, with its sympathetic division responsible for increasing venous return, stroke volume, and cardiac output. Orthotopic heart transplantation involves extracting the native heart at the level of atrioventricular junction, transecting

the aorta and pulmonary artery above the semilunar valves. Immediately after the surgery, right- and left-sided filling pressures within the heart are elevated, which results in myocardial anoxia and PH that will eventually normalize over a few weeks.¹²⁷ Postoperatively, left ventricular ejection fraction and myocardial contractility are within normal limits or slightly depressed. With the loss of vagal tone, the denervated heart has a higher than normal resting heart rate, which is usually 15 to 25 beats per minute above average consistent with the intrinsic rate of the sinoatrial node of the transplanted heart. This resting tachycardia is associated with a reduced stroke volume to maintain a near-normal resting cardiac output. Vagal maneuvers such as carotid massage or Valsalva does not affect the heart rate. Owing to its denervation, a transplanted heart increases to a maximum heart rate as well as return to baseline more slowly than a normal heart by a process that is principally mediated by the rise and fall of circulating catecholamines, although a minority of patients demonstrate signs of partial cardiac reinnervation several months after surgery.²³³

Despite this, cardiac output in the transplanted heart increases with total body activity mainly by the Frank-Starling mechanism, which stimulates venous return increasing cardiac preload, resulting in greater stroke volume.¹²⁷ Thus the response of the transplanted heart to exercise is preload dependent at the onset of increased exertion, but it switches to a catecholamine-driven process through prolonged exercise that is unable to make sharp fluctuations in heart rate response. When the cardiac transplant recipient begins exercise, the Bainbridge reflex mediates a small rise in heart rate as a response to increased pressure in the right ventricular veins that can continue for up to 5 minutes, which can also contribute to a slow return to preexercise heart rate. The peak heart rate and systolic blood pressure attained during maximal exercise is lower in cardiac transplant recipients, only reaching to 80% of normal.^{45,127} Because of this chronotropic incompetence, it has historically been presumed that heart transplant recipients would not be able to fully adapt to high-intensity interval training and would be best served by enrollment in a steady-state intensity exercise regimen.¹⁸⁵ The patient must therefore be educated to begin any workout with a slow warmup period and a graded progression of activity intensity.

The transplanted heart compensates for output demand mainly by increasing stroke volume. At the onset of exercise, stroke volume in the transplanted heart can rapidly increase to 20%.¹⁴⁹

Successive increases in stroke volume and cardiac output during sustained submaximal exercise result from inotropic responses to circulating catecholamines.¹⁴⁹ Table 29.6 summarizes the effects of cardiac transplant on various cardiovascular variables. Owing to this catecholamine-induced tachycardic response to increased exertion, exercise prescriptions based on target heart rates are not suggested. More useful determinants of exercise intensity include blood pressure reserve (the difference between systolic and diastolic pressure), the Borg scale of perceived exertion (a categorical scale from 1 to 10, with 10 representing the maximal exertion), and the dyspnea index (a scale from 0 to 4, with 4 being the maximum shortness of breath that prevents counting or speaking).^{145,217} Resting hypertension is a common finding that may be associated with premonitory elevations in vascular resistance secondary to hypersensitivity to circulating catecholamines in the setting of chronic heart failure.¹²⁷ Diastolic blood pressure can decrease early in submaximal exercise because of reduced peripheral resistance. The peak systolic blood pressure is less than that of individuals without cardiac transplants, but diastolic blood pressure remains essentially

TABLE 29.6 Effect of Cardiac Transplantation on Selected Cardiovascular Variables

Condition	Heart Rate	Stroke Volume	Systolic	Diastolic	Pulmonary Arterial Pressure	O ₂ (Oxygen Consumption)	Serum Lactate
Rest	Greater than normal	Less than normal; little (Bainbridge reflex) or no positional change	Greater than normal	Greater than normal	Slightly greater than normal (although usually lower than pretransplantation)	—	Greater than normal
Submaximal exercise	Little or no immediate increase; delayed slow increase	Increase initially resulting from the Frank-Starling mechanism; late increases as a result of circulating	Greater than normal	Fall initially as a result of reduced peripheral resistance	Greater than normal (rate of change is greater than normal)	Less than normal (absolute value)	Greater than normal
Maximal exercise	Blunted peak (less than predicted for age); peak cardiac output about 25% less than normal	Peak only 40% to 50% greater than most	Peak is less than normal	About the same	—	Less than normal (absolute value); relative anaerobic threshold is slightly higher than normal	Not markedly different

unchanged. Cyclosporine can induce hypertension through its effect on renal function, endothelial vasodilation, or sympathetic activity to a greater extent in cardiac transplant patients than either liver or lung transplant patients.¹²⁷

After heart transplant, patients consume less oxygen during submaximal exercise than normal controls. Oxygen consumption at the anaerobic threshold is also considerably lower than that of age-matched normal individuals. According to Braith and Edwards, the decrement in peak oxygen consumption seen in transplant recipients is partly caused by changes in skeletal muscle. Skeletal myopathy associated with the heart failure syndrome produces atrophy, decreased mitochondrial counts, and decreased oxidative enzymes. Corticosteroids also promote muscle atrophy affecting primarily type II fibers, and cyclosporine further decreases oxidative enzymes.³⁸

Therapeutic Exercise After Cardiac Transplantation

Aerobic cardiovascular conditioning programs and endurance training improve the ability of the heart transplant recipient to accomplish sustained involvement in higher levels of ADLs.¹⁰⁷ Regular exercise training can be effective in improving exercise capacity, particularly a regimen of at least 12 weeks in duration, involving supervision, and commencing within 1 year after the transplant surgery. Enrollment into a supervised moderate-intensity exercise training program early after cardiac transplantation has shown to improve VO₂ max (by an average of 24% after 2 to 3 months), maximum power output, and body composition when compared with those recipients who did not enroll, but overall exercise capacity is still reduced compared with normal age-controlled patients.^{75,113,186}

Both cardiac hemodynamic and peripheral physiologic factors most likely contribute to this reduction in exercise capacity following heart transplantation. The cardiac factors include

chronotropic incompetence, possible impaired left ventricular function, or greater arteriovenous oxygen difference. Peripheral factors include decreased muscle mass, anabolic resistance as a result of reduced strength and oxidative capacity, or impaired vasodilatory capacity. Other issues to consider as possible etiology for decreased exercise capacity include advanced age of the patient, donor heart age, operative ischemic time, or premorbid deconditioning and comorbidities.¹⁸⁵ Only a few RCTs have investigated the effect of exercise in heart transplant recipients. Most of these studies compared some form of rehabilitation or exercise program at moderate intensity with a control group without a specific form of an exercise strategy.

Resistance training has been shown to increase skeletal muscle mass and strength and should not begin until 6 to 8 weeks after transplantation, permitting time for sternal healing and corticosteroid tapering.^{145,233,245} Studies demonstrate that heart transplant patients may be able to tolerate high-intensity interval training with benefits of overall improved exercise capacity, coronary endothelial function, and blood pressure management, but future large-scale studies are needed to confirm these effects and determine the mechanisms by which they occur. High intensity interval training exercise results in beneficial effects, such as more substantial increase in peak VO₂ and lower extremity strength.^{109,139,145,185,186,232} Early in the history of cardiac transplantation, it was considered inadvisable to start an exercise protocol immediately after the surgery. More recent research suggests, however, that early mobilization and low-level exercise should begin in the hospital after extubation barring no medical contraindications.^{128,232,233}

Inpatient Cardiac Rehabilitation

Acute inpatient rehabilitation setting is often the preferred setting for postoperative heart transplant recipients as soon as relative medical stability is achieved. Ideal candidates for acute inpatient

rehabilitation admission are those that have significant change from functional baseline, need complex medical management with close follow-up by a physician at least 3 days per week, can tolerate in-tense rehabilitation services with at least two therapy disciplines (physical, occupational, speech therapies), and have appropriate functional goals. Moreover, heart transplant recipients often require frequent bloodwork to monitor therapeutic levels of immunosuppressant medications, renal function, electrolytes, and blood counts. Other aspects of recovery, including nutrition optimization, management of symptoms of anxiety, depression, and encephalopathy, can also impact the course of rehabilitation and may require a multidisciplinary approach. The treating physiatrist should be in close communication with the heart transplantation team to coordinate medical treatment, manage any adverse effects or medical complications, and ensure follow-up care and patient/caregiver training. The goal of inpatient rehabilitation is to allow patients to safely transition to the outpatient setting. This can be accomplished by optimizing medical treatment, improving functional independence, and ensuring appropriate services and follow-up in the community.

All patients following heart transplantation should be referred to an outpatient cardiac rehabilitation program before or soon after their discharge from the inpatient setting.²¹⁷ Although almost every cardiac transplant patient faces episodes of acute cellular rejection (ACR), it is only rarely necessary to limit the exercise workout during episodes of moderate rejection. The risk of ACR is greatest within the first 3 months following transplantation. Greater than 60% of transplant recipients will have one or more episodes of rejection in the first 6 months. Risk factors for ACR include younger age at transplantation, female sex, Black race, and greater HLA mismatch.⁷²

When patients show signs of new arrhythmias, hypotension, or fever, the exercise regimen can be adjusted to allow for proper medical management. Clinical and physiologic monitoring of the patient and regular review of personal life and family goals are essential to maximize the patient's prognosis, life plans, and family functioning. Patient and family education are crucial in transplantation rehabilitation (Box 29.8).¹¹⁴

Lung Transplantation

In 2022, there were 2743 lung transplants and 45 heart/lung transplants performed in the United States. There were 3161 new adult candidates added to the lung transplant waitlist in 2022.²⁵⁹ Survival of lung transplant patients is improving but remains inferior to that of recipients of other SOTs, owing to the great number of complications in the first postoperative year. For lung transplants performed in 2021, the 1-year posttransplant mortality was 12.2%; for those performed in 2017, the 5-year posttransplant mortality was 40.4%.²⁵⁹ A significant factor in deciding which lung transplant procedure to perform is the underlying lung disease (Box 29.9). Lung conditions associated with chronic pulmonary infections (e.g., CF, bronchiectasis) must be managed with BLT to avoid spillover of infection from the native lung left in situ to the transplanted lung. While other indications, including interstitial lung disease, COPD, and pulmonary arterial hypertension, can be treated with single lung transplant, long-term survival is enhanced with BLT, and that is the procedure of choice if there is availability of lungs for transplantation.^{249,268} Older patients with secondary PH also appear to have better outcomes from bilateral transplant. Updates to the international guidelines for outlining candidacy for LTx were issued in 2021 by

• BOX 29.8 Transplantation Rehabilitation: Patient-Family Education

- Basic immune system
- Purpose of immunosuppression
- Activities related to immunosuppression
 - Avoiding crowds during high immunosuppression
 - Wearing mask in hospital and clinic
 - Care of cuts and wounds
 - Mouth care/dental visits
 - Notifying transplant team of exposure to diseases (e.g., influenza, chickenpox, measles)
- Other activities
 - Exercise (walking, bicycling, swimming if no T-tube or open wounds)
 - Restrictions on lifting and driving after major abdominal surgery (3 to 6 months)
 - Sexual activity/birth control
 - Care of T-tube (usually removed after 3 months)
- Medications—proper administration, side effects, and purpose
 - Immunosuppressants
 - Prophylactics (antivirals, antibiotics, antifungals, antacids)
 - Others (e.g., antihypertensives, multivitamins)
- Home monitoring responsibilities (how, why, and what to do if abnormal)
 - Blood pressure readings
 - Temperature
 - Stool for occult blood
 - Urine for glucose or blood
 - Daily weight
 - Quality of urine (cloudy, dark, normal)
 - Quality of stools (tan/clay color, black, maroon, normal)
- Dietary
 - Restrictions (fat, sugar, salt)
 - Balanced diet, low-calorie snacks
 - Expected increase in appetite and fat deposition attributable to steroids
- Signs and symptoms of infection and rejection—what to do and whom to call
 - Symptoms are so similar for these two that notifying the transplant team is paramount
- Medical follow-up
 - Biopsies (usually 3 and 6 months, then yearly; and with dysfunction)
 - Frequent transplant clinic visits (blood tests, radiographic tests)
 - Routine checkups with referring and transplant physicians
 - Cancer monitoring (annual Pap smear, self-breast examination, testicular examination, mammogram, stool for occult blood)
 - Routine dental visits (prophylactic antibiotics before visit)

• BOX 29.9 Common Primary Diagnoses in Lung Transplant Rehabilitation Patients

- Pulmonary vascular disease
 - Primary pulmonary hypertension
 - Eisenmenger syndrome
 - Cardiomyopathy with pulmonary hypertension
- Obstructive lung disease
 - Emphysema—idiopathic
 - Emphysema—alpha₁-antitrypsin deficiency
 - Cystic fibrosis
 - Bronchiectasis
 - Posttransplant obliterative bronchiolitis
- Rejection
 - Acute, chronic
- Side effects of immunosuppressive therapy
- Psychosocial issues

• BOX 29.10 Indications and Contraindications for Lung Transplantation

Indications for Lung Transplantation

- BODE Index^a of 7 to 10 or one the following:
 - Previous hospitalization for an acute hypercapnic exacerbation ($P_{CO_2} > 50$ mm Hg)
 - Pulmonary hypertension or cor pulmonale despite oxygen therapy
 - $FEV_1 < 20\%$ and either $DLCO < 20\%$ or homogeneous distribution of emphysema

Absolute Contraindications for Lung Transplantation

- Malignancy in the past 2 years, with the exception of cutaneous squamous and basal cell tumors
- Advanced disease deemed untreatable of another major organ system
- Coronary artery disease not amenable to percutaneous intervention or bypass grafting or associated with significant impairment of left ventricular function is an absolute contraindication to lung transplantation, but heart-lung transplantation could possibly be considered.
- Noncurable chronic extrapulmonary infection, including chronic active viral hepatitis B, hepatitis C, and human immunodeficiency virus
- Pronounced chest wall/spinal deformity
- Noncompliance or inability to fully participate in medical therapy and clinic visits
- Untreatable psychiatric or psychological condition associated with the inability to cooperate with medical therapy
- Absence of a reliable social support system
- Substance addiction (e.g., alcohol, tobacco, or narcotics) within the previous 6 months

Relative Contraindications for Lung Transplantation

- Age > 65 years
- Critical or unstable medical condition (e.g., shock, mechanical ventilation, or extracorporeal membrane oxygenation)
- Severely limited functional status with poor rehabilitation potential
- Colonization with highly resistant or highly virulent bacteria, fungi, or mycobacteria
- Body mass index > 30 kg/m²
- Severe or symptomatic osteoporosis
- Mechanical ventilation
- Medical conditions that have not resulted in end-stage organ damage should be optimized before transplantation. Patients with coronary artery disease may undergo percutaneous intervention before transplantation or coronary artery bypass graft concurrent with the procedure.

^aBODE Index consists of measures of health quality of life that includes body mass index, the degree of airflow obstruction, and dyspnea and exercise capacity.

the International Society for Heart and Lung Transplantation.¹⁴⁷ The indications and absolute and relative contraindications determined by this consensus report are summarized in Box 29.10.

Pretransplant Rehabilitation: Assessment, Education, and Conditioning

Understanding the natural history of chronic lung disease that patients often medically and functionally deteriorate before being eligible for lung transplant, and as well as the reality for many patients that they spend months or years on a lung transplant list, frequently necessitates pretransplant PR. PR professionals should be knowledgeable of the indications and basic criteria for transplant eligibility and can serve as a referral source to the regional transplant center. Most transplant programs prefer to see patients earlier in the course of their lung disease and follow patients over time to optimize the timing of the transplant. Transplant

programs recognize the value of PR in optimizing patients for success with the transplant surgery and refer for pretransplant PR. Indeed, the astute lung transplant listing committee will engage the PR team in the decision to list a patient or not. Patient factors integral to the PR team's skill set include strength, endurance, compliance, nutritional, emotional, and cognitive status. All these factors impact a patient's suitability for and success with LTx. Despite PR programs pre- and posttransplantation being considered optimal management for lung transplant patients, there remains a paucity of randomized controlled studies looking at functional outcomes. Pretransplant PR is associated with improvement in patient's medical status, strength, and function as well as improved outcomes following LTx.^{3,271} It has been noted that a pretransplant PR program will have positive effects on exercise capacity, specifically in functional domains of physical functioning, vitality, social functioning, and mental health.⁸⁵

It is important to appreciate that patients listed for lung transplant have advanced/end-stage lung disease and often significant comorbidities and do not present like standard PR patients. While the pretransplant PR program has the same basic exercise elements, including resistance strength training, postural and balance exercises, breathing retraining and chest therapies, and progressive aerobic endurance training, lung transplant patients typically have more advanced disease and disability and require more individualized attention. Patients should undergo the comprehensive pre-participation PR team evaluation as described earlier, including 6MWT and CPET if able to tolerate. Patients requiring oxygen for activity and exercise will not be eligible for metabolic testing due to the need for supplemental oxygen during ESTs. Due to weakness and fatigue, individual PR sessions may require more frequent rest breaks, and the length of the session can be extended to accommodate this. Focus on energy conservation training and pacing is essential. Nearly all patients pending lung transplant will be oxygen dependent, and attention is essential to ensure optimal oxygen delivery during the PR sessions as well as in the community. Although goal oxygen saturation is greater than 88% with activity and PR training, it is not unusual for saturations to dip into the 70% range; every effort should be made to limit such desaturations. Functional progress may be slower, and the rate of exercise progression built into the PR program should reflect this. A greater emphasis on nutrition evaluation and therapy is also required because many patients may be poorly nourished contributing to functional limitations. Anxiety, depression, and adjustment disorder are frequent in patients pending lung transplant, and there is often a greater need for psychological and supportive therapies. Insurance will cover outpatient PR for a limited period of time, and while it is not known exactly when a suitable organ will be available for transplant, coordination between the transplant team, patient, and PR team should aim to engage patients in PR up to the time of the transplant. Emphasis on home and community-based PR is essential to support the gains achieved at the PR center.

Posttransplant Rehabilitation: Medical Complications

Posttransplant PR starts immediately when the patient is deemed medically stable postop. ICU early mobilization programs for surgical populations have been shown to minimize functional loss, reduce ICU and surgical floor length of stay, and patients return to the community at a higher functional level.⁶⁴ Many patient's following lung transplant will require inpatient rehabilitation.

This is best initiated in the acute hospital setting and continued in a dedicated inpatient rehabilitation facility-level setting familiar for managing post-lung transplant patients. The most significant difference pre- to posttransplant is supplemental oxygen, which should be no longer needed postoperatively. Patient characteristics supporting the need for acute inpatient PR are similar to other inpatient rehab programs, including a multidisciplinary rehabilitation team need, medical complexities precluding discharge to home, and the ability to tolerate and make progress with at least 3 hours of PR daily. Length of stay can vary from 1 to 3 weeks on average. Ongoing engagement of the lung transplant team is essential for optimal medical management, readmission reduction, and care coordination post PR. Goals of inpatient PR include medical stability, functional safety with ADLs and ambulation, and patient and family/caregiver education. Successful transition to the community requires a dedicated transplant social worker/case manager.

Homecare services provide a bridge between inpatient and outpatient PR. The goal is to enroll patients in outpatient PR as soon as possible. Medical clearance from the lung transplant team is required. Patients on outpatient PR following lung transplant are immune suppressed, and infection prevention is paramount. Hand hygiene is essential, and respiratory protection with a mask is often required initially. As mentioned, typically oxygen therapy is not required posttransplant, and patients steadily progress with strength and endurance. Overall function can rapidly improve and, in parallel, the patient's emotional state. Goals of post-lung transplant outpatient PR include improved strength and endurance for all ADLs and community participation, improved nutrition and emotional status, education on self-care and post-transplant medical requirements, and integration of a lifelong healthy lifestyle, including ongoing exercise in the community. Return to avocational activities is frequently achieved, but although close to 80% of lung transplant patients had worked at one point prior to their transplant, only 15% were working at the time of transplant, and just 28% worked within 5 years of LTx.¹⁷⁵ A recent review of employment following heart and LTx indicates improved employment rates associated with younger age, higher education levels, and a history of pretransplant employment.¹²¹ Engagement of the vocational therapist as part of the rehabilitation team is essential to support return to vocational activities. As the PR team is engaged in the lung transplant recipient's care over the course of many months, the rehabilitation professional plays an integral part in patient monitoring for the early and late medical complications that can arise after LTx.⁹⁰ Improvements in surgical techniques, lung preservation, immunosuppression, and management of infections have significantly improved the short-term mortality. Bronchiolitis obliterans syndrome (BOS) and non-CMV infection account for the major causes of death after the first year.

Pulmonary complications after LTx have been described in detail^{133,246} and are summarized here:

Primary graft dysfunction. This is a syndrome of acute lung injury that occurs within 72 hours of a lung transplant, characterized clinically by pulmonary edema and pathologically by diffuse alveolar damage. Mild cases can resolve within 48 hours, but severe cases can persist for several weeks.

Airway-related and immediate postoperative complications. These include airway stenosis at the site of the anastomosis, which is the most common airway complication resulting from LTx and is seen in approximately 10% of cases. It develops typically around 3 months after surgery. Anastomotic dehiscence, due

to ischemia, usually occurs 2 to 4 weeks after the transplantation and is reported in 1% and 10% of cases. Patients present with increasing dyspnea, pneumothorax, pneumomediastinum, subcutaneous emphysema, and persistent air leak from the chest tubes.

Infections. Although early-onset CMV infection is now very rare (3.3% of cases) due to prophylactic use of valganciclovir, delayed-onset CMV pneumonitis is more common (13.7% to 25%) with increased risk of mortality and of BOS in those who recover. *Aspergillus fumigatus* is the most common fungal infection after lung transplant causing tracheobronchitis and invasive pulmonary aspergillosis. Bactrim prophylaxis has reduced the incidence of *Pneumocystis jirovecii* (formerly known as *P. carinii*) infection, but it must be continued lifelong in lung transplant recipients; if stopped, 90% develop infection as evidenced by fever, dyspnea, and hypoxemia.

Immunologic complications (rejection). ACR occurs when recipient lymphocytes react with donor antigens and is seen in 35% of recipients within the first year of lung transplant. It accounts for less than 4% of deaths within the first month of transplant, but it is the most significant risk factor for the development of BOS. ACR presents as dyspnea, cough, fever, and hypoxemia with a reduction in PFTs. BOS is characterized by irreversible, progressive airflow obstruction caused by the presence of obliterative bronchiolitis. It is a manifestation of CLAD, usually occurs within the first 3 months of LTx, and is responsible for the majority of deaths in the first year after transplant. It is the greatest threat to long-term graft and patient survival.

Neoplasms. Posttransplant lymphoproliferative disorder (PTLD) ranges from benign polyclonal proliferation of lymphoid tissue to non-Hodgkin lymphoma and occurs in 2% to 8% of patients receiving lung transplants. Malignancy occurs at a higher rate in the native lung of patients receiving single-lung transplants with an incidence between 7% and 9%.

Iatrogenic complications. Transbronchial biopsy, performed to evaluate rejection, causes focal areas of bleeding producing localized nodules and may persist for 1 month after biopsy. Toxicity to everolimus, an mTOR inhibitor used to prevent graft rejection, includes cytopenias, dyslipidemia, pneumonitis, and renal and liver toxicity. Recurrence of the primary disease after LTx is most commonly seen in sarcoidosis in approximately 35% of cases and is seen as early as 2 weeks and as late as 2 years after the transplantation.

Nonpulmonary complications after LTx have been described in detail^{123,160} and are summarized here:

Renal failure. Some renal dysfunction is seen in over 25% at 1 year and close to 40% at 5 years after lung transplant. Chronic renal failure (creatinine >2.5 mg/dL) is seen in nearly 7% at 1 year and 11% at 5 years after transplant and increases the risk of death by four to five times. The cause of renal failure is due to calcineurin inhibitor nephrotoxicity. Other factors that worsen renal function include hypertension, hyperlipidemia, and diabetes. Management of kidney disease includes the lowest dose of calcineurin inhibitor tolerated and aggressive management of associated comorbidities. Chronic renal failure may progress and require dialysis or renal transplantation.

Diabetes. This develops in over 24% by 1 year and over 33% by 5 years after lung transplant. Risk factors include glucocorticoid and calcineurin inhibitor use, older age of recipients, obesity with BMI above 30, and frequent acute rejection

treated with high-dose steroids. Treatment is recommended for fasting glucose levels greater than 126 mg/dL and hemoglobin A1c levels above 6.5%.

Cardiovascular disease. Immunosuppressants contribute to the development of cardiovascular risk factors, including hypertension, hyperlipidemia, diabetes, and renal disease. Hypertension is seen in over 50% of patients at 1 year and in over 85% at 5 years following lung transplant. Hyperlipidemia is seen in over 20% of patients at 1 year and over 50% at 5 years. However, CVD accounts for less than 5.3% of deaths in patients who survive longer than 1 year due to stringent pretransplant screening for CVD. Of interest, the use of statins within the first year of transplant to lower cholesterol also decreases acute rejection episodes and prevents BOS with better PFTs and improved mortality, which is likely due to the antiinflammatory properties of statins.

Osteoporosis. This is most often due to the use of corticosteroids and other immunosuppressive medications. Pretransplant risk factors include tobacco use, prior corticosteroid use, and limited mobility. Patients with CF have increased risk due to hypogonadism, malnutrition, and impaired absorption of vitamin D and calcium. Osteoporosis is seen in 32% to 54% of pre-lung transplant patients. Bone loss is accelerated in the first 3 to 6 months after lung transplant. Within the first year of lung transplant, bone mineral density drops 4% to 12%, and the fracture rate ranges from 6% to 18%. Prevention and treatment of osteoporosis include calcium and vitamin D supplementation, regular weight-bearing exercises, estrogen and HRT (recognizing an increased risk of thromboembolic disease with HRT), calcitriol, and antiresorptive therapy with bisphosphonates. The greatest effect of bisphosphonates to reduce bone loss is seen when they are started before transplantation. All pretransplant candidates should be screened for osteoporosis by dual-energy x-ray absorptiometry scan; anyone with osteopenia or osteoporosis should begin bisphosphonates and continue for 6 to 12 months after transplantation.

Avascular necrosis of the femoral head. This is seen in up to 10% of lung transplant recipients, is related to steroid use, and presents as hip and/or knee pain. Magnetic resonance imaging scans secure the diagnosis, and joint-sparing procedures such as core decompression may prevent total hip replacement.

Hematologic complications. Immunosuppressants, antivirals, and antibiotics suppress the bone marrow leading to cytopenia, the most common hematologic complication in lung transplant. Renal dysfunction also worsens anemia; erythropoietin with iron supplementation improves hemoglobin levels in anemic patients following lung transplant.

Thromboembolic complications. Deep venous thrombosis and PE are reported in 8.6% to 29% of lung transplant recipients. Risk factors include advanced age, diabetes, pneumonia, immobility, and indwelling vascular catheters. Treatment includes intravenous heparin followed by long-term warfarin anticoagulation. There is increased risk for pulmonary infarction due to altered pulmonary blood supply from the bronchial artery posttransplant. Placement of an inferior vena cava filter and lifelong warfarin therapy are often recommended.

Gastrointestinal complications. Ileus and colonic perforation are the most common gastrointestinal problems immediately posttransplant. At least 60% of patients have long-term complaints, including nausea, vomiting, gastroesophageal reflux disease (GERD), diarrhea, constipation, and abdominal pain. Nausea is normally due to medications, but gastroparesis is

common following lung transplant, seen in up to 25%. GERD is also common and can contribute to aspiration and BOS. Severe gastrointestinal complications include appendicitis, pancreatitis, cholecystitis, and diverticulitis.

Neurologic complications. These include severe headaches, seizures, strokes, and confusion in up to 26% of patients. Calcineurin inhibitor toxicity includes confusion, tremors, paresthesias, seizure, blindness, and encephalopathy.

Malignancy. There is a three- to fourfold increase in the risk of malignancy following lung transplant compared with the general population. The prevalence of malignancy in lung transplant patients is 3.7% at 1 year, 12.4% at 5 years, and 25% in 10-year survivors. Malignancy accounts for nearly 10% of deaths after the first year following transplant. PTLT is the most common cancer in the first 2 years after transplantation, and skin cancers are the most common malignancy after the second year. Recommendations are for adherence to general cancer screening guidelines and evaluation by a dermatologist annually.

PTLT. This disorder is due to abnormal lymphoid proliferation and ranges from benign polyclonal hyperplasia to malignant monoclonal lymphoma. PTLT is seen in up to 20% of post-lung transplant patients and usually involves the thorax and the transplanted lung. It most often occurs within the first year after transplant and is related to reactivation of the Epstein-Barr virus. Treatment includes lowering immunosuppression and immunotherapy.

As patients survive longer after LTx, it is expected that pulmonary and nonpulmonary complications will be seen with increasing frequency. These complications can significantly impact strength, endurance, function, and quality of life; the rehabilitation team plays a central role in early recognition and rehabilitative management.

Postoperative Exercise Considerations

As discussed previously, early mobilization in the surgical ICU improves outcomes of patients undergoing cardiothoracic surgery.⁶⁴ Progressive activity should be initiated as soon as the patient is deemed medically stable by the surgical/ICU teams progressing from ROM exercises and breathing/chest therapies to transfers, standing, and gait training. Progressive mobilization continues throughout the acute care stay with a goal of safety and minimizing requirements for assistance with ADLs and hallway ambulation. Some patients achieve independence with these activities, as well as stair climbing, and can be discharged home from the acute care hospital. For many lung transplant recipients, functional progress is slow, and inpatient rehab, preferably at an IRE, is indicated to achieve functional goals. Progression to gentle endurance training with use of a treadmill or cycle ergometer is possible. To reduce infection risk, the lung transplant recipient may be privately roomed while on the rehabilitation unit, and strict adherence to infection prevention policies is essential.

Following LTx, studies have shown substantial improvements in pulmonary function and exercise capacity, but peak exercise and quality of life related to physical functioning remain impaired. In other words, lung transplant recipients have an increase in exercise capacity below what would be expected considering the improvement in lung function, indicating that impaired skeletal muscle strength, cardiovascular deconditioning, and other peripheral factors play an important role in the limitation

of posttransplantation exercise capacity.²³ Peak exercise in the post-lung transplant period may be reduced by up to 40% to 60% of predicted values. This exercise limitation in lung transplant patients appears to be multifactorial. Evidence suggests that although the transplanted lungs are denervated, the ventilatory response to exercise seems to be unaffected. Heart rate increases during incremental exercise to approximately 60% to 70% of age-predicted maximum, which would typically meet the patient's workload demands. However, in patients with dual heart and lung transplants, resultant impairment in inotropic and chronotropic responses secondary to cardiac denervation can partially account for reduced exercise capacity. Over time, approximately 30% to 50% of lung transplant patients will develop comorbid conditions, including diabetes, hyperlipidemia, hypertension, and osteoporosis, which can be prevented or controlled with exercise.¹⁴³ A major element of reduced exercise capacity after LTx is limb muscle dysfunction, including atrophy, weakness, and alterations in muscle composition, with a reduction in type I muscle fibers likely stemming from premorbid sedentary lifestyle compounded by multiple hospitalizations. In lung transplant recipients, studies demonstrate a diminished capacity for oxidative metabolism manifested by activity-related early-onset lactic acidosis, decreased calcium uptake and release, early drop in muscle pH, and reduced peak oxygen consumption (VO₂ peak) in the peripheral muscles. This phenomenon may play a central role in limiting posttransplant peak exercise capability. Reduced oxidative capacity may also occur secondary to the effects of immunosuppressants, which can impair mitochondrial functioning.^{143,169,270}

Referral to outpatient PR is essential for all patients after LTx. Participation in supervised rehabilitation programs improves skeletal muscle dysfunction and daily physical activity, which in turn improves overall quality of life and aids in reducing the risk of comorbid conditions after transplantation. There is literature to support the implementation of structured exercise training programs to optimize functional exercise capacity, skeletal muscle strength, and lumbar bone mineral density in lung transplant patients.²⁷¹ A small RCT of 40 patients compared lung transplant recipients assigned after hospital discharge to either a structured 3-month exercise training program or a control intervention and showed that after 1 year, the group that participated in the structured exercise training program had significant improvements in 6-minute walking distance, quadriceps force, self-reported physical functioning, and 24-hour ambulatory blood pressures.¹⁴³ Further studies are needed to develop standardized guidelines for exercise prescription in the lung transplant population. Challenges in standardization of exercise training for adults following lung transplant surgery is discussed in a recent Cochrane review.¹⁰¹ Lifelong exercise and adherence to healthy lifestyles is essential for longevity, well-being, and quality of life for individuals following lung transplant. It is the role of the PR team to instill and support this goal.

Liver Transplantation

Introduction

Liver transplantation is a standard proven therapy for irreversible liver disease. It is recognized as an effective and necessary treatment of chronic and acute hepatic failure. Advances in surgical techniques, immunosuppressive drugs, and pre- and postoperative care have resulted in liver transplantation

becoming a well-established procedure that has saved innumerable lives worldwide.

Epidemiology

Liver transplant activity continues to increase in the United States. According to the Organ Procurement and Transplantation Network (OPTN)/Scientific Registry of Transplant Recipients (SRTR) 2022 Annual Data Report, in 2022, the number of liver transplants reached an all-time high of 9527, representing a 52% increase over the previous decade (2012-2022).¹⁴² Of these transplants, 8924 (93.7%) were from deceased donors and 603 (6.3%) were from living donors. Liver transplant recipients were 94.5% adult and 5.5% pediatric. The overall size of the liver transplant waitlist contracted, with more patients being removed than added, although 10,548 adult patients remained on the waitlist at the end of 2022. Simultaneous liver-kidney transplant was the most common multiorgan combination, with 800 liver-kidney transplants performed in 2022. Among adults added to the liver waitlist in 2021, 39.9% received a deceased donor liver transplant within 3 months, 45.7% within 6 months, and 54.5% within 1 year.

Etiology

Alcohol-related liver disease is currently the most common indication for liver transplantation.⁵⁶

Nonalcoholic steatohepatitis (NASH) is the second most common indication for liver transplantation.⁷⁰ NASH includes a spectrum of nonalcoholic fatty liver disease, ranging from steatosis alone to NASH with accompanying cirrhosis. These liver diseases are linked to metabolic syndrome with increased BMI and obesity. Since currently there is no effective treatment for NASH or fibrosis, liver transplantation has been rising due to NASH. Also, patients diagnosed with NASH with and without cirrhosis have an increased risk of developing hepatocellular carcinoma (HCC). Until 2015, decompensated cirrhosis due to chronic hepatitis C infection was the most common indication for liver transplantation. However, it is now the third most common indication for liver transplantation, having been replaced by alcohol-related liver disease and NASH.⁷⁰ Other less common indications for liver transplantation include the following:

- Hepatitis B infection previously resulted in increasing numbers of chronic liver disease, leading to liver transplantation. However, the advent of the hepatitis B immunoglobulins and the introduction of antivirals have resulted in decreased rates of liver transplantation due to hepatitis B.¹³²
- Autoimmune hepatitis (AIH) can lead to liver cirrhosis and failure, even with long-term corticosteroids and immunosuppression therapy. Liver transplantation is indicated in acute liver failure secondary to AIH or cases of chronic decompensated cirrhosis due to AIH.¹¹⁵
- Primary biliary cirrhosis patients with decompensated cirrhosis or severe pruritis refractory to other medical interventions require liver transplantation.¹⁴⁸
- Primary sclerosing cholangitis currently has no effective medical therapy. Liver transplantation is considered an effective treatment modality among patients who develop decompensation or those who develop perihilar cholangiocarcinoma (within certain criteria) or recurrent bouts of bacterial cholangitis.⁵⁵
- HCC is one of the leading causes of cancer-related death worldwide, being the fifth most common cancer and the

third most common cause of cancer-related mortality. The incidence of HCC has been rising in the United States over the last 20 years. Liver transplantation is an optimal treatment option because it eliminates HCC and the underlying liver disease. The Milan criteria have been adopted by many transplant societies worldwide as the criteria to determine whether patients with HCC can move forward with liver transplantation. (The criteria are one lesion ≥ 2 cm and ≤ 5 cm, or up to three lesions, each ≥ 1 cm and ≤ 3 cm.) However, many believe that the Milan criteria may be too strict in regard to its size requirements for lesions. This has led to a number of expanded criteria for liver transplantation (Box 29.11).¹⁵³ Absolute and relative contraindications to liver transplantation are outlined in Box 29.12.

• BOX 29.11 Indications for Liver Transplantation

- Acute liver failure
- Hepatitis A/B
- Intoxication (e.g., acetaminophen, death cap)
- Wilson disease
- Budd-Chiari syndrome
- Chronic liver failure: noncholestatic cirrhosis
- Hepatitis B/C
- Autoimmune hepatitis
- Alcohol-induced cirrhosis
- Chronic liver failure: cholestatic cirrhosis
- Primary biliary cirrhosis
- Primary sclerosing cholangitis
- Secondary biliary cirrhosis
- Chronic liver failure: metabolic
- Wilson disease
- Hemochromatosis
- Alpha₁-antitrypsin deficiency
- Amyloidosis
- Cystic fibrosis
- Tyrosinemia
- Chronic liver failure: vascular
- Budd-Chiari syndrome
- Other indications
- Primary oxalosis
- Glycogen storage diseases
- Hyperlipidemia
- Polycystic liver disease
- Malignant disease
- Hepatocellular carcinoma (within Milan criteria)
- Fibrolamellar carcinoma
- Hepatoblastoma
- Epithelioid hemangioendothelioma
- Cholangiocellular adenocarcinoma
- Neuroendocrine liver metastases
- Benign liver tumors
- Adenomatosis
- Liver transplantation in pediatric patients
- Biliary atresia
- Byler disease
- Alagille syndrome
- Neonatal hepatitis/neonatal viral hepatitis
- Autoimmune hepatitis
- Hepatoblastoma

From Farkas S, Hackl C, Schlitt HJ. Overview of the indications and contraindications for liver transplantation. *Cold Spring Harb Perspect Med*. 2014;4(5):a015602.

• BOX 29.12 Contraindications to Liver Transplantation

Absolute

- Severe cardiac and/or pulmonary diseases and severe pulmonary hypertension (mean pulmonary arterial pressure >45 mm Hg)
- Alcohol addiction without motivation for alcohol abstinence and untreated/ongoing substance abuse
- Hepatocellular carcinoma with extrahepatic metastases
- Current extrahepatic malignancies (eventually reevaluation after successful therapy)
- Sepsis

Relative

- Untreated alcohol abuse and other drug-related addiction
- Cholangiocellular carcinoma
- Hepatic metastatic neuroendocrine tumors, metastatic hemangioendothelioma
- Morbid obesity
- Persistent nonadherence

From Graziadei I, Zoller H, Fickert P, et al. Indications for liver transplantation in adults: recommendations of the Austrian Society for Gastroenterology and Hepatology (ÖGGH) in cooperation with the Austrian Society for Transplantation, Transfusion and Genetics (ATX). *Wien Klin Wochenschr*. 2016;128 (19-20):679-690.

Patient Selection

The liver allocation system implemented by the OPTN in February 2002 is based primarily on the severity of liver disease as assessed by end-stage liver disease (MELD) and pediatric end-stage liver disease (PELD) survival models for all patients with chronic liver disease.¹³⁸ The MELD score, which is based on biochemical variables, has been shown in retrospective and prospective studies to be highly predictive of 3-month mortality in patients with chronic liver disease. The MELD score is limited to 40 points and is calculated on the basis of the following variables: serum creatinine, serum bilirubin, and international normalized ratio. The MELD score ranges from 6 (less ill) to 40 (gravely ill) and is used for transplant candidates aged 12 years and older. The PELD score is used for transplant candidates younger than 12 years of age.²⁹

Rehabilitation Management

Patient classification

In assessing the patient who is to undergo a liver transplant, it is important to recognize that patients may present dramatically differently in the extent of their disability depending on whether they have acute fulminant liver failure or chronic liver failure. Those with acute fulminant liver failure (e.g., secondary to acetaminophen overdose) often are otherwise healthy patients and present with limited functional deficits. In contrast, those with chronic liver failure (e.g., secondary to chronic hepatitis) have lived with their disease for many years and often present with severe deconditioning, sarcopenia, and fatigue. Muscle strength in these patients has been found to be 30% of age-predicted levels.²⁹ Sarcopenia and physical deconditioning are the end results of impaired dietary intake, chronic inflammation, altered macronutrient and micronutrient metabolism, and low physical activity. They are all associated with increased mortality in end-stage liver disease.⁷⁷ Patients with chronic liver disease have also been found to have a VO_2 max that is 40% less than predicted for sedentary healthy individuals of the same age and sex.³⁰ Patients with

cirrhosis develop a global physical dysfunction associated with a multiplicity of factors, including muscle wasting, undernutrition and malnutrition, and functional impairment defined as frailty. Studies in this vulnerable population have demonstrated that the presence of frailty is independently predictive of adverse outcomes both pre- and posttransplantation. It is felt that its incorporation into clinical practice could result in improved clinical decision making, particularly regarding the identification of candidates for physical and nutritional interventions.²⁰⁹

Preoperative rehabilitation management

Pre- and postoperative management of the liver transplant recipient is a team effort that requires close collaboration between the surgeon, physiatrist, hepatologist, nephrologist, and other members of the rehabilitation team. It is critical to institute rehabilitation therapy as soon as possible to maintain relatively healthy levels of physical functioning and to improve survival before and after transplantation. A program of graded isometric exercises aimed at restoring muscle mass and strength, as well as ROM exercises and gentle stretching to prevent contractures, should be used preoperatively. The introduction of a respiratory physiotherapeutic program for liver transplantation candidates has also been found to significantly improve respiratory parameters preoperatively. The program consists of diaphragmatic breathing exercises, diaphragmatic isometric exercises, and upper limb and abdominal strengthening exercises.¹⁵² Preoperative clinical trials addressing the usefulness of exercise in patients with cirrhosis have shown that it improves the metabolic syndrome, sarcopenia, cardiopulmonary endurance, health-related quality of life, and the hepatic venous pressure gradient.⁷⁷ A systematic review found that prehabilitation in patients actively listed for orthotopic liver transplant (OLT) improves aerobic and functional capacity. Significant improvements were observed in VO_2 peak, 6-minute walking distance, hand grip strength, liver frailty index, and quality of life. No serious adverse events were reported, and adherence to the training programs was high. The authors concluded that prehabilitation is safe and feasible in the OLT candidate and that all patients awaiting OLT, especially the most physically frail, should be enrolled in predefined prehabilitation programs.¹¹⁸ A retrospective study evaluating the efficacy of a structured in- and outpatient exercise training program for cirrhotic patients who were referred for liver transplant evaluation compared groups who underwent the exercise program pretransplant (258 patients) vs. those who did not (200 patients). The results revealed a trend toward lower 90-day readmission rate in the exercise training group (17.9% vs. 20%) and shorter length of stay (14 vs. 17 days).⁹

Postoperative medical complications

In managing the individual who has undergone liver transplant, one must be aware of the potential postoperative complications. The most common complications in the liver transplant recipient include¹³¹ acute graft rejection, vascular thrombosis, biliary leak or stricture, infection, and adverse effects of immunosuppressant drugs. Most life-threatening complications associated with liver transplantation occur within the perioperative period. These include primary graft malfunction, acute rejection episodes, severe infections, and technical complications such as hepatic artery thrombosis and biliary leaks. Acute hepatic allograft rejection typically occurs within the first 7 to 14 days, and the risk is highest (40%) during the first 3 to 6 months after transplantation.¹⁵⁶ Rejection is most commonly manifested by fever, malaise, right upper quadrant pain or tenderness, graft enlargement, and

diminished graft function. A rise in bilirubin and transaminase levels is observed. With early detection, most acute rejection episodes can be treated successfully by augmentation of existing immunosuppressive medications or high doses of steroids. Post-transplant infections (bacterial and fungal infections) are more common in the first month due to severe immunosuppression. After 1 year posttransplant, nonhepatic-related causes of death include malignancy (22%), CVD (11%), infection (9%), and renal failure (6%), whereas liver allograft failure accounts for less than one-third of deaths.² Post-liver transplant patients have an increased risk of acquiring COVID-19; however, they do not have a heightened risk of developing severe COVID-19 infection nor do they have an increased mortality.¹⁴¹

Postoperative rehabilitation management

Rehabilitation therapies should be instituted immediately after the transplant patient has been stabilized. Therapy is aimed at restoring muscle mass and strength and reversing the musculo-skeletal changes resulting from inactivity and bedrest. Therapy begins with graded isometric and ROM exercises. Aerobic conditioning exercises are especially important because a lower mean VO_2 max is a significant predictor of poorer short-term survival. The introduction of an intensive early rehabilitation program for liver transplant recipients has been found to be well tolerated and feasible in the ICU. In a prospective randomized study evaluating liver transplant recipients over a period of 1 year, patients in the usual treatment group receiving one physiotherapy session a day were compared with those in an early and intensive rehabilitation group who received two sessions a day. Patients in the early intensive rehabilitation group sat on the edge of their beds sooner (2.6 vs. 9.7 days; $P = 0.048$), and their intestinal transit resumed earlier than patients in the usual treatment group.¹⁶² In another study to evaluate the impact of early mobilization, it was found that following pediatric liver transplantation, the early mobilization patient group regained the ability to walk without a rolling walker sooner and had a shorter length of hospital stay.²⁵⁵

A physical exercise program for liver transplant patients has also been shown to lead to an improvement in functional capacity. In a study by Garcia et al., patients in the exercise group showed a 19.4% increase in the distance walked as well as an increase in their resting energy expenditure compared with a control group. The authors concluded that the exercise program promoted significant improvements in functional capacity and felt these findings had positive implications for the control of metabolic diseases, which are common in patients after liver transplantation.⁸⁹ Combining supervised resistance training with aerobic exercises has also been found to be beneficial. In a study by Moya-Nájera et al. 6 months after liver transplantation, patients were randomized into two groups: intervention group (IG) and control group (CG). The IG completed a 6-month program of moderate-to high-intensity exercises, and the CG followed usual care recommendations. The IG, compared with the CG, had significant improvements in aerobic capacity, hip extension, elbow flexion, overall maximal strength, physical functioning, and vitality health-related quality of life.¹⁷⁴

Fatigue

Fatigue and reduced physical activity are major problems following liver transplantation.²⁶⁰ Fatigue is present in a considerable portion of recipients after liver transplantation, and almost half of them have severe fatigue.¹¹¹ A 12-week rehabilitation program, consisting of supervised physical exercise training and counseling

on physical activity, has been shown to be well tolerated and promising in reducing fatigue and improving physical fitness. Furthermore, after the program, aerobic capacity and knee flexion strength were higher, and body fat was lower than at baseline.²⁶¹

Nutrition

The importance of good nutrition combined with a rehabilitation program is critical to the successful outcome for a liver transplant recipient. Protein-energy malnourishment is commonly encountered in patients with end-stage liver disease who undergo liver transplantation. Malnutrition may further increase morbidity, mortality, and costs in the posttransplantation setting. Severely malnourished patients have a higher incidence of sepsis and death due to sepsis.²⁷⁷ It is essential to provide sufficient nutritional support during all phases of liver transplantation. Oral nutrition is preferred, but enteral nutrition may be required to provide the needed energy intake. Perioperative nutritional interventions in liver transplant recipients include synbiotics, micronutrients, branched-chain amino acid supplementation, immune-nutrition formulas, fluid and electrolyte balance, and the offering of nocturnal meals.¹⁰³

Outcomes

According to the OPTN/SRTR 2022 Annual Data Report, patient survival after deceased donor liver transplant continues to improve, with 5.1% mortality observed at 6 months, 6.8% at 1 year, 12.7% at 3 years, 19.8% at 5 years, and 35.7% at 10 years. Five-year patient survival rates after living donor liver transplant exceeded those of deceased donor liver transplant.¹⁴² A meta-analysis to evaluate alcohol relapse rate and its predictors after liver transplantation found posttransplant alcohol relapse occurred in about one-fifth of patients who underwent alcohol-related liver transplantation. Psychiatric comorbidities, pretransplant abstinence of less than 6 months, unmarried status, and smoking were associated with alcohol relapse after liver transplantation. Psychiatric comorbidities represented the strongest predictor of alcohol relapse. The authors concluded that close monitoring of psychiatric comorbidities and pretransplant alcohol abstinence for at least 6 months may decrease alcohol relapse after liver transplantation.⁵⁷

In light of the improved survival rates, there has been an increased emphasis on returning to normal activity in the workplace and on restoring social quality of life following liver transplantation. The assessment of quality of life represents an ancillary tool to evaluate the efficacy of SOT in addition to graft and patient survival rates and complications. A systematic review of publications between 2001 and 2016 revealed that employment rates ranged from 26% to 80% pretransplant and 18% to 44% posttransplant. Improved survival following liver transplantation was not reflected in a return to employment, and retirement was common. The authors felt that interventions to minimize physical deconditioning and postoperative depression could improve the rate of return to employment.²⁶⁵ Kang et al. found that up to 50% of liver recipients remain unemployed mostly due to fatigue and muscle weakness. Therefore they stress the importance of rehabilitation therapy aimed at reducing fatigue and increasing muscle mass after liver transplantation.¹²⁵ Emerging evidence suggests that continued patient information, structured psychosocial support systems, and caregiver involvement may further help improve long-term health-related quality of life.²

Future

Due to the enormous progress made in technical skills as well as improvements in immunosuppressive medications and the

management of postoperative complications, patients are living longer following liver transplantation. This prolonged longevity has brought about new concerns, such as the long-term effects of immunosuppression, as they relate to effects on the cardiovascular system, infections, and propensity for malignancy. In addition, despite improved longevity, improvements in quality of life and return to employment have not kept pace. Moreover, the demand for transplants continues to exceed the available supply. The consequence of this disparity in organ need vs. availability is that many patients die on the waitlist every year. In the United States, one of four patients who are listed for liver transplant either die on the waitlist (12%) or become too sick to undergo the transplant (13%).²⁴⁸ Expanding living donor transplantation remains a goal in countries where deceased donor liver transplant predominates, such as North America and Europe. However, transplant from living donors is more complex and requires careful dissection. A living donor graft is partial, unlike the whole graft provided from a deceased donor.²⁴¹ Another means of increasing the available deceased donors is by reducing organ discard rates. Advances such as machine perfusion systems, which preserve the graft by improving its oxygenation, offer great promise. Additionally, using organs from viremic donors is now possible with the use of highly effective antiviral therapies in the recipient. Finally, xenotransplantation (transplanting livers from one species to another) has progressed recently, but many barriers remain.²⁴⁸

Rehabilitation Management of Diabetes Mellitus

Epidemiology

Globally, diabetes caused 1.5 million deaths in 2012 and an additional 2.2 million deaths from related conditions, such as cardiovascular and kidney diseases. Of these deaths, 43% occur before the age of 70 years. It is the eighth leading cause of death among both sexes worldwide.²⁷⁵

According to the National Diabetes Statistics Report, an estimated 38.4 million people had diabetes as of 2021. This comprises 11.6% of the US population. The prevalence of diabetes in adults over 65 years of age is 29.2%. For US adults age 18 years or older, for both sexes, diagnosis of diabetes was more prevalent in the American Indian and Alaska Native groups (13.6%), closely followed by non-Hispanic Blacks (12.1%), Hispanics (11.7%), non-Hispanic Asians (9.1%), and non-Hispanic Whites (6.9%). Factors associated with increased prevalence of diabetes included level of education, family income, and geographic proximity to metropolitan areas. The incidence of diabetes in adults age 18 years or older was similar in 2021 (5.8 per 1000 adults) to 2000 (6.2 per 1000 adults). However, for children and adolescents, the incidence of both type 1 and especially type 2 diabetes increased significantly in 2002 to 2018. The main risk factors for diabetes-related complications in adults include smoking, overweight and obesity, physical inactivity, increased hemoglobin A1c, hypertension, and hyperlipidemia.

Comorbidities and Complications

In 2020, there were 16.8 million emergency department visits for adults with diabetes, and these included 267,000 for hyperglycemic crisis, 240,000 for diabetic ketoacidosis (DKA), and 202,000 for hypoglycemia.

In 2020, there were 7.86 million hospital discharges with diabetes as one of the listed diagnoses in adults, with 1.68 million for major CVDs, including ischemic heart disease and stroke, and 160,000 for lower extremity amputation. The data for 2017-2020 revealed CKD in close to 40% of adults with diagnosed diabetes. Additionally, diabetes is the leading cause of vision disability with 10.1% reported severe vision difficulty or blindness in individuals with diabetes in 2021. Diabetes was the eighth leading cause of death in the United States in 2021. The total estimated costs of diabetes in the United States in 2022 was \$413 billion. These costs include both direct medical costs and indirect costs related to loss of productivity, unemployment, and premature mortality. Given the prevalence of diabetes mellitus in the United States and globally, coupled with the significant impairments with which the condition is associated, it is not surprising that there is an increasing demand for the physiatrist to become well versed in the diagnosis, management, and rehabilitation of patients with diabetes mellitus.

Pathophysiology and Diagnosis

Diabetes can be grouped into four categories:

- Type 1 diabetes is caused by destruction of beta cells by an autoimmune process. This usually leads to complete deficiency of insulin.
- Type 2 diabetes is characterized by a dysfunction of insulin secretion by beta cells in the setting of insulin resistance and metabolic syndrome
- Specific types of diabetes are due to other causes:
 - Monogenic diabetes syndromes. Examples of these include neonatal diabetes and maturity-onset diabetes of the young
 - Diseases of exocrine pancreas, such as CF and pancreatitis
 - Drug or chemical-induced diabetes, such as diabetes associated with glucocorticoid use, treatment of HIV/AIDS, or organ transplantation
- Gestational diabetes mellitus

At the time of the diagnosis, it is not always easy to discern type 1 and type 2 diabetes based on age alone because both can occur in adults and children. Presence and extent of beta-cell deficiency can be inferred by screening for autoantibodies to insulin, glutamic acid decarboxylase, islet antigen 2, and zinc transporter 8. The presence of these markers can help determine the type of diabetes, assess the risk for DKA and other complications, and assist in treatment selection.

Prediabetes and Diabetes Type 2

For the general population, screening for prediabetes and diabetes type 2 should begin at 35 years of age. However, those with a BMI of 25 kg/m² or more or 23 kg/m² or more in individuals of Asian descent and with one or more risk factors should be tested for diabetes regardless of their age. Risk factors include having a first-degree relative with diabetes, high-risk race/ethnicity (e.g., Black, Latino, Native American, Asian, Pacific Islander), history of CVD, hypertension, those with low high-density lipoprotein (<35 mg/dL) and high triglyceride (>250 mg/dL), polycystic ovary syndrome, sedentary lifestyle, gestational diabetes, or HIV. Prediabetes is a risk factor for diabetes. It is characterized by impaired fasting glucose (100 to 125 mg/dL) and/or impaired glucose tolerance with a 2-hour glucose level from 140 to 199 mg/dL, and hemoglobin A1c of 5.7% to 6.4%. Those individuals with prediabetes should be counseled on risks of developing diabetes

and risk of CVD and be provided education on strategies to lower these risks.⁸⁰ The American Diabetes Association (ADA) guidelines for diagnosis of diabetes mellitus include:

- Hemoglobin A1c level $\geq 6.5\%$
- Fasting plasma glucose (FPG) ≥ 126 mg/dL
- Oral glucose tolerance test blood glucose at 2 hours ≥ 200 mg/dL
- Random plasma glucose test blood glucose ≥ 200 mg/dL
- Classic symptoms of hyperglycemia (polyuria, polydipsia, unexpected weight loss) or hyperglycemic crisis with a random plasma glucose of ≥ 200 mg/dL (≥ 11.1 mmol/L)
- Prediabetes: hemoglobin A1c of 5.7% to 6.4% or FPG of 100 to 125 mg/dL or 2-hour plasma glucose of 140 to 199 mg/dL following an oral glucose tolerance test¹⁵

Prevention Guidelines

Lifestyle Modifications

Current recommendations for the prevention of type 2 diabetes include lifestyle modifications, which include maintenance of healthy weight, proper and healthy nutrition intake, and participation in aerobic and resistance exercise programs.⁸ Some suggest there are additional benefits to a more comprehensive lifestyle modification program that incorporates behavioral components, such as tobacco cessation, cooking lessons, use of food diaries, and attendance at annual diabetic education classes.²¹⁸ Such programs should include an interdisciplinary team with physicians, case managers, nurses, exercise advisors and trainers, behavioral therapists and counselors, psychologists, dietitians, and physiotherapists.²¹⁸ Participation in such a comprehensive program has been shown to effectively reduce body weight and BMI in high-risk individuals and prevent type 2 diabetes, although advantages are less clear for people already diagnosed with the condition.²¹⁸ In fact, participation in combined programs of dietary modification, aerobic exercise, and resistance training by at-risk individuals has been shown to reduce the incidence of type 2 diabetes by up to 58%.⁸

Treatment Guidelines

Exercise and Diabetes Mellitus

Given current recommendations for lifestyle modifications as the first line of diabetes prevention and treatment, the physiatrist is in a prime position to take leadership of management of patients with diabetes mellitus. Rehabilitation physicians are uniquely trained in musculoskeletal anatomy and function, exercise physiology, and the use of physical activity as a treatment for disease. Further, the physiatrist is versed in the team approach to patient care, working closely with physiotherapists, exercise trainers, case managers, and nurses, as well as the individualization of patient care, taking into consideration the patient's unique medical needs, impairments, goals, and social background.

Exercise is a major part of lifestyle changes for prevention and treatment of diabetes mellitus. The ADA recommends both aerobic and resistance training for adults with diabetes. This should include at least 150 minutes of moderate to vigorous aerobic exercise and two to three sessions of resistance training per week. The benefits of regular exercise include diabetes risk factor modifications, such as weight reduction, blood pressure reduction, improved insulin sensitivity and blood glucose levels, and improved lipoprotein numbers. Following exercise recommendations has been shown to decrease cardiovascular mortality and all-cause mortality.²⁷⁹

Unfortunately, physical inactivity is prevalent in general in the United States. According to the CDC National Diabetes Statistics Report website, less than one-fourth of individuals with diabetes are exercising at or above the recommended 150 minutes per week.⁵²

Assessment of Physical Activity

The American Heart Association recommends that physical activity (bodily movement produced by skeletal muscles that result in energy expenditure) and other cardiovascular risk factors be regularly assessed by clinicians. It is important to consider an individual's overall physical activity profile, including all structured (e.g., exercise, planned activities) and incidental (e.g., ADLs, vocational, transport, unplanned activities) physical activity. Further, the dimensions (mode or type, frequency, duration, intensity) and domains (occupational, domestic, transportation, leisure) should be documented with each assessment. Both subjective and objective methods of evaluation exist for clinical use. Physical activity questionnaires (PAQs) such as the Global PAQ, Short Recall PAQ, and quantitative history PAQs may be completed via personal or telephone interviews or self-reported. Additionally, physical activity diaries or logs such as the Bouchard Physical Activity Record are frequently used. Objective methods of quantifying physical activity, such as indirect calorimetry, the doubly labeled water method, direct observation, heart rate monitoring, or use of motion sensors (accelerometers and pedometers), may also be used by the clinician depending on availability.²³⁹

Physiologic Effects of Exercise in Type 2 Diabetes Mellitus

Individuals with diabetes mellitus typically present with limitations in physical fitness and activity, which contributes to the high cardiovascular mortality seen in this population. Type 2 diabetes is associated with impaired glucose and fat metabolism as a result of impaired insulin sensitivity and lipid oxidation, as well as impaired myocardial and skeletal muscle perfusion likely resulting from vascular endothelial dysfunction.²¹⁰ Risk for type 2 diabetes increases with physical inactivity and occurs more frequently in individuals with obesity, hypertension, and dyslipidemia. The objectives of physical activity in the treatment of type 2 diabetes include improvement of diabetes-associated abnormalities in glucose, lipid, and blood pressure control and facilitation of weight loss and maintenance.^{60,124} During exercise, glucose uptake into actively contracting skeletal muscles is increased, which is balanced by hepatic gluconeogenesis and release of fatty acids. Additionally, muscle contraction appears to circumvent diabetes-associated insulin resistance by inducing GLUT-4 translocation via a separate mechanism, which facilitates uptake of circulating glucose into the muscle. Insulin sensitivity following exercise can last up to 72 hours. Regular activity also facilitates beta-cell function, insulin sensitivity, vascular function, and even gut microbiota.^{60,124,161,238} The degree of glycemic reduction is related to the duration and intensity of the exercise, preexercise glucose levels, and the individual's state of physical training.⁶⁰

Over the long term, both aerobic and resistance training result in increased muscle lipid storage, oxidation, and use. Type 2 diabetes mellitus has been associated with a decrease in lipid oxidation and preferential use of carbohydrate for energy. Chronic benefits of exercise also include reduction of total and low-density lipoprotein cholesterol, elevation of high-density lipoprotein levels, reduction in systolic blood pressure, and reduction of depressive symptoms in both sexes of all age groups. Further,

physical activity has been shown to improve vascular endothelial function, which may reduce incidence of complicating atherosclerotic cardiovascular or peripheral artery disease, prevent the onset of diabetic peripheral neuropathy, and improve autonomic function in patients with diabetic autonomic neuropathy. Overall, physical activity and physical fitness are associated with lower all-cause and cardiovascular mortality.⁶⁰ The exact mechanisms by which physical activity improves cardiovascular function in patients with type 2 diabetes remains unclear, but recent evidence shows that exercise is associated with reductions in inflammatory markers C-reactive protein (CRP) and interleukin-6 (IL-6) levels in individuals with type 2 diabetes mellitus. CRP is a strong independent predictor of CVD and the outcome of acute coronary syndrome. CRP has long been identified to be generated in the liver in response to elevation in inflammatory cytokines, such as IL-6, but recent studies have shown production of CRP in both adipose tissue and atherosclerotic plaque.¹⁰⁶

Preexercise Assessment

Low- to moderate-intensity exercise is likely safe and does not require special clearance. For those with a history of CVD or other diabetes-related complications, medical clearance is needed. Stress testing can be considered by the clinician to assess cardiovascular risk and overall level of fitness. Those with CVD may benefit from a more supervised training program, such as formal cardiac rehabilitation. Several potential complications should be kept in mind before prescribing an exercise program and guide patient education accordingly. Electrocardiogram stress testing may be indicated for patients over 40 years of age with or without CVD risk factors other than diabetes mellitus; for patients over 30 years of age and with type 1 or 2 diabetes mellitus greater than 10 years' duration, hypertension, cigarette smoking, dyslipidemia, proliferative and pre-proliferative retinopathy, and nephropathy; and for patients with diagnosed or suspected coronary artery disease, cerebrovascular disease, peripheral artery disease, autonomic neuropathy, or advanced nephropathy with renal failure.^{59,60}

Those with hypoglycemia and hyperglycemia risk may require medication dose adjustments, frequent blood glucose checks, counseling on carbohydrate intake prior to exercise, and hydration. Due to the risk of DKA in type 1 diabetes, testing for ketones is necessary with levels of blood glucose above 250 mg/dL. Exercise should be delayed if moderate to large amounts of ketones are present.

Types of Exercises for Patients with Type 2 Diabetes Mellitus

The beneficial effects of physical exercise in the management of type 2 diabetes mellitus may result from aerobic or resistance activities. ADA and ACSM recommendations are outlined in Table 29.7.⁶⁰ Aerobic exercise improves insulin sensitivity in adults with type 2 diabetes, which may be related to improved mitochondrial function. This can occur with or without weight loss.

Resistance exercise training leads to 10% to 15% improvement in strength, bone density, insulin sensitivity, blood pressure, and lipids. As it turns out, a combination of aerobic and resistance training appears to be even more beneficial than each modality alone. According to the results of a meta-analysis comparing aerobic training, resistance training, and combined training across 14 trials, which included 915 participants, combined training leads to a larger reduction in hemoglobin A1c, glucose, and lipids.^{124,219}

TABLE 29.7 Exercise Prescription Guidelines for Individuals With Type 2 Diabetes Mellitus

	Frequency	Intensity	Duration	Mode	Progression
Aerobic	At least 3 days/week with no more than 2 consecutive days per session	At least moderate intensity, that is, 40–60% of $\text{VO}_{2\text{ max}}$ (e.g., brisk walking)	At least 150 min/week; 30 min/day for 5 days/week of moderate-intensity activity <i>or</i> 20 min/day for 3 days/week of vigorous-intensity activity	Any form of aerobic exercise that uses large muscle groups and results in sustained heart rate elevation	Gradual to reduce injury risk and increase compliance
Resistance	At least twice weekly on nonconsecutive days; ideally three times per week, along with aerobic activities	Moderate (50% of 1-repetition maximum) or vigorous (75–80% of 1-repetition maximum)	5–10 exercises involving major muscle groups (upper body, lower body, core), 10–15 repetitions per set, minimum of 1 set but as many as 4 sets	Resistance machines and free weights yield equivalent gains in strength and muscle mass	Slowly to prevent injury and increase compliance

From Colberg SR, Sigal RJ, Fernhall B, et al. Exercise and type 2 diabetes: the American College of Sports Medicine and the American Diabetes Association: joint position statement executive summary. *Diabetes Care*. 2010;33(12):2692-2696.

TABLE 29.8 Diabetic Complications Requiring Exercise Modification

Complication	Exercise Alteration
Hyperglycemia (blood glucose >300 mg/dL without ketosis)	Patient may participate but ensure clinical stability and maintenance of hydration
Hypoglycemia	Glucose monitoring before and after physical activity (PA); patients taking insulin or insulin secretagogues are advised to ingest carbohydrates before any PA if blood glucose is <100 mg/dL before PA; insulin users should consume up to 15 g of carbohydrates before exercise
Vascular disease (cardiovascular disease and peripheral artery disease)	Patients with angina and type 2 diabetes mellitus should start PA in a supervised cardiac rehabilitation setting; patients with peripheral artery disease and intermittent claudication may participate in low to moderate walking, arm crank, and cycling
Peripheral neuropathy	Patients may participate in moderate weight-bearing exercise because moderate walking does not increase risk of diabetic foot ulceration; patients with foot injury or diabetic foot ulcer should be restricted to nonweight-bearing activities; closely examine feet daily
Autonomic neuropathy	Patients may proceed only with physician approval, possibly following stress testing given elevated risk for silent myocardial infarction
Retinopathy	Avoid high-intensity aerobic or resistance PA and head-down activities because these may elevate intraocular pressures; avoid jumping and jarring activities that may increase risk of hemorrhage
Nephropathy	Patients may proceed following physician evaluation and possibly stress testing to detect associated coronary artery disease; start at low intensity and volume because aerobic and muscle capacity are likely to be reduced

From Colberg SR, Sigal RJ, Fernhall B, et al. Exercise and type 2 diabetes: the American College of Sports Medicine and the American Diabetes Association: joint position statement executive summary. *Diabetes Care*. 2010;33(12):2692-2696.

Flexibility exercise helps with mobilization of joints that can otherwise be limited by mechanisms of glycation, which result from aging and chronic hyperglycemia. While flexibility training does not impact glucose control, it can significantly improve participation and performance in exercise programs.

Balance exercises can be incorporated into other training modalities and can lower fall risk especially in those with peripheral neuropathy.¹²⁴

Although the guidelines provide a solid foundation upon which to build an exercise prescription for patients with type 2 diabetes, it is important to individualize exercise programs to enhance benefits, promote compliance, and reduce risk of injury. Exercise prescription should be personalized based on an

individual's habits, preferences, motivation, and tolerance, rather than a general prescription with duration, intensity, and frequency.

A clinician developing an exercise regimen should not only consider which dose (type, intensity, duration, volume) provides the greatest health benefits but also review the risk profile relative to that dose. In other words, the clinician must balance the benefits with the risks.²² Additionally, it is imperative to assess for the presence of any diabetes-associated complications (Table 29.8) that may limit the patient's ability to participate in a desired exercise program. Of particular concern is the potential for delayed hypoglycemia (nocturnal hypoglycemia), a possibly fatal complication of the exercising individual with diabetes mellitus.

Vigorous exercise severely depletes skeletal muscle glycogen stores, which the liver and muscles attempt to replenish postexercise by extracting circulating blood glucose, resulting in profound hypoglycemia. This condition may occur 6 to 12 hours after exercise, but may develop as late as 28 hours following physical activity. Signs and symptoms may include seizures, cardiac arrhythmias, altered mental status, altered consciousness, and death. The condition is especially threatening because it commonly develops nocturnally, during the sleeping hours.¹⁰⁴ Those with hypoglycemia and hyperglycemia risk may require medication dose adjustments, frequent blood glucose checks, counseling on carbohydrate intake prior to exercise, and hydration. Due to the risk of DKA in type 1 diabetes, testing for ketones is necessary with levels of blood glucose above 250 mg/dL. Exercise should be delayed if moderate to large amounts of ketones are present.¹²⁴

Oral Pharmacologic Treatment

The American College of Physicians (ACP) recommends the addition of an oral pharmacologic agent in the treatment of patients with type 2 diabetes when lifestyle modifications, including diet, exercise, and weight loss, have failed to adequately improve hyperglycemia. The ACP recommends initiating therapy with metformin for most patients.²⁰⁵ Metformin, however, is contraindicated in patients with CHF, renal insufficiency, COPD, age older than 80 years, and patients receiving renal transplantation given the possible risk of renal insufficiency.⁹⁵ The ACP updated the clinical guidelines to include newer pharmacologic agents to treat type 2 diabetes. The ACP recommended adding a sodium-glucose cotransporter-2 (SGLT-2) inhibitor (such as empagliflozin) or a glucagon-like peptide-1 (GLP-1) agonist (such as semaglutide) to metformin and lifestyle modifications in adults with type 2 diabetes whose glycemic control is still inadequate. The SGLT-2 inhibitor reduces the risk for all-cause mortality, major adverse cardiovascular events, progression of CKD, and hospitalization due to CHF. The GLP-1 agonist reduces the risk for all-cause mortality, major adverse cardiovascular events, and stroke. Additionally, the ACP recommends against adding a dipeptidyl peptidase-4 (DPP-4) inhibitor (such as sitagliptin) to metformin and lifestyle modification because DPP-4 inhibitor agents offer no benefit in reduction of all-cause mortality, major adverse cardiovascular events, stroke, CHF hospitalization, or CKD progression when compared to the SGLT-2 inhibitor or GLP-1 agonist. Metformin and lifestyle modifications are the first line of treatment of type 2 diabetes in most patients. When another agent is needed, clinicians need to consider benefits, adverse effects, comorbidities, and associated medication costs. For those patients with type 2 diabetes who are also affected by CHF or CKD, SGLT-2 inhibitors may be favored. GLP-1 agonists may be more appropriate for those with increased stroke risk and when weight reduction is necessary. When considering hemoglobin A1c levels, the goal level is between 7% and 8%. When adequate glycemic control is achieved with metformin and either SGLT-2 inhibitor or GLP-1 agonist, the other agents, such as sulfonylureas and long-acting insulins, should be decreased or discontinued when possible to reduce the risk of hypoglycemia.²⁰³

Diabetes Mellitus Treatment With Insulin

The ADA recommends initiation of insulin and/or other therapies for treatment of persistent hyperglycemia at or above 180 mg/dL, confirmed on two occasions for all hospitalized, noncritically ill patients. The goal of glucose control is 140 to 180 mg/dL.

For those hospitalized, hyperglycemia is defined as blood glucose levels above 140 mg/dL. Hemoglobin A1c levels must be checked for all individuals who do not have an A1c level from the previous 3 months, and a value at or above 6.5 suggests that the patient had diabetes prior to the hospitalization.

Blood glucose monitoring should be performed with a point-of-care (POC) blood glucose meter before meals during the hospital stay. For those on intravenous insulin therapy, more frequent monitoring is required, ranging from every 30 minutes to every 2 hours. Patients with diabetes with or without automated insulin delivery and those using a personal continuous glucose monitor should continue to use these during their hospitalization. However, POC blood glucose measurements are still required for insulin dosing and hypoglycemia evaluation.

Insulin therapy, including basal, prandial, and correction insulin formulations, is the preferred method of glucose management in most hospitalized patients with adequate nutritional intake. Using only a correctional insulin scale without basal insulin is discouraged in the hospital.¹⁴

Three types of insulin include basal, nutritional, and correctional insulin. Basal insulins include formulations such as glargine (once daily) and neutral protamine Hagedorn (at bedtime or morning and bedtime). Basal insulin is required even when patients are not eating to account for gluconeogenesis, especially in those with type 1 diabetes to prevent DKA. Nutritional insulin covers carbohydrate intake from food and other sources. Rapid-acting insulin, such as aspart and lispro, or short-acting insulin (regular) can be used for this purpose. Correctional insulin is needed to regulate high blood glucose to target range and utilizes rapid-acting or short-acting insulins.²¹⁵ Dosing insulin is based on weight and glucose levels. Generally, half of the patient's total daily dose should be given as basal or long-acting insulin, and the rest as premeal nutritional insulin. Although adequate glucose control is important to reduce and prevent complications and decrease overall hospital stay for patients with diabetes, current guidelines favor a more moderate glycemic target than in the past to avoid hypoglycemic episodes.¹³⁷

Management of Diabetes in the Inpatient Setting

For many physicians, management of patients with type 2 diabetes mellitus is likely to start in the inpatient setting, following a diabetes-associated complication, such as stroke, coronary syndrome, lower limb amputation, or a prolonged hospitalization complicated by the comorbidity of diabetes.²⁶⁷

For hospitalized individuals, dysglycemia is associated with adverse outcomes and increased morbidity and mortality. Diabetes and hyperglycemia management provided by a highly trained diabetes specialty team may reduce the hospital length of stay, prevent readmissions, and improve clinical outcomes. Diabetes education and a discharge plan that provides access to medication, supplies, and follow-up can also improve clinical outcomes. For the hospitalized adults, insulin is the preferred treatment option for hyperglycemia. Insulin and possibly other agents should be started for persistent hyperglycemia at or above 180 mg/dL. The recommended blood glucose goal is 140 to 180 mg/dL for most critically ill patients. Tighter control may be acceptable for those individuals without significant hypoglycemia. In noncritical patients, the goal of blood glucose is 100 to 180 mg/dL.

Glucose testing in the hospital should be done with POC blood glucose monitoring before meals. For those who are not eating, glucose monitoring should be done every 4 to 6 hours.

When intravenous insulin is administered, more frequent glucose testing is necessary. When patients use a continuous glucose monitor at home, they can continue using it during the hospital stay if clinically appropriate. However, POC glucose measurements still need to be performed to assess for hypoglycemia and make decisions regarding insulin dosing.

If oral medications are held in the hospital and the plan is to restart them as outpatient, the clinician should plan to resume these 1 to 2 days prior to discharge.¹⁴ Patient education on diabetes self-management can begin during the hospitalization. When an individual with diabetes is unable to complete this type of training due to cognitive or physical impairments, a caregiver can be identified and participate in education and training prior to discharge from the hospital. It is recommended to review and address several areas of knowledge regarding diabetes management prior to discharge¹⁴:

- A follow-up with appropriate clinician to provide diabetes care
- Appropriate durable medical equipment, medications, and supplies
- Knowledge of diagnosis, glucose monitoring, and goals; how to address emergencies
- Recognition and treatment of hyperglycemia and hypoglycemia
- Nutrition
- Diabetes medication management, including insulin and other agents
- Proper disposal of insulin pens, syringes, needles, lancets

Treatment of Diabetic Peripheral Neuropathy

Diabetic peripheral neuropathy is a diffuse, symmetric, sensorimotor length-dependent peripheral nerve injury that afflicts up to 40% of individuals with diabetes mellitus.⁶⁰ The condition results in limb pain, impaired sleep and overall quality of life, and may predispose to foot injuries, diabetic foot ulcers, and lower limb amputations. Management options include gabapentinoids (such as gabapentin and pregabalin), serotonin-norepinephrine reuptake inhibitors (SNRIs) (such as duloxetine), tricyclic antidepressants (TCAs) (such as amitriptyline), sodium channel blockers (such as carbamazepine, valproic acid, lamotrigine), and SNRI/opioid agents (such as tramadol). However, it is important to note that complete resolution of symptoms is not always achieved. About 30% of pain reduction is considered to be successful in clinical trials; therefore appropriate patient counseling and management of expectations are necessary. Treating other comorbid conditions, such as mood and sleep disorders, that can often be seen in individuals with diabetes can help achieve improvement in the overall quality of life.

Side effect profile and possible adverse reactions need to be considered when selecting from the abovementioned pharmacologic agent groups. For instance, gabapentinoids can cause peripheral edema. TCAs can have anticholinergic side effects, so patients can be at risk for urinary retention, constipation, or orthostatic hypotension. SNRIs can cause nausea, fatigue, and dizziness. Valproic acid can cause teratogenic effects, hepatotoxicity, and bone marrow suppression. Renal dose adjustment may be necessary for many of these medications. Due to similar efficacy of these agents, medication choice to treat painful diabetic peripheral neuropathy should be guided by potential side effects, comorbid conditions, cost, and patient preferences. Due to lack

of long-term efficacy and potential for severe adverse outcomes, including dependency and overdose, the use of opioids for treatment of painful diabetic neuropathy is not recommended. Patients may prefer topical, nontraditional, and nonpharmacologic treatments, which can further guide selection of treatment agents. Topical agents that may help with pain relief include capsaicin and glyceryl trinitrate spray. Ginkgo biloba is an example of a nontraditional agent. Cognitive behavioral therapy, exercise, tai chi, and mindfulness are examples of nonpharmacologic interventions.²⁰¹

Treatment of Diabetic Gastroparesis

The physiatrist should ensure that individuals with diabetes mellitus who are participating in rehabilitation or exercise programs receive adequate nutrition to meet metabolic demands of the prescribed physical activity. One potential complication may prove to be a serious threat to the nutritional status of the patient with diabetes mellitus. Diabetic gastroparesis affects 5.2% of patients with type 1 and 1% with type 2 and is characterized by delayed gastric emptying leading to nausea, vomiting, early satiety, and early postprandial fullness. Patients with diabetic gastroparesis using insulin or insulin secretagogues who are unable to consume adequate carbohydrates are therefore placed at risk for profound hypoglycemia. Thus it is important to be vigilant about the diagnosis and management of this uncommon yet potentially deadly complication. Any patient with suspected gastroparesis should immediately undergo gastric emptying scintigraphy, which remains the standard of diagnosis. Medications that further impair gastric emptying, such as opiates and anticholinergics, should be discontinued. Should gastroparesis persist, pharmacologic agents may be used. Metoclopramide, a D2 dopamine receptor antagonist, remains the only FDA-approved medication for treatment of gastroparesis. Domperidone (Motilium), also a D2 dopamine antagonist, has shown equal efficacy to metoclopramide and may be considered for treatment. Erythromycin, a motilin-stimulating antibiotic, has been proven effective in the management of diabetic gastroparesis when administered intravenously. Symptomatic management of nausea and vomiting with phenothiazines, promethazine, or ondansetron may provide additional relief. At present, intrapyloric botulinum toxin injection has shown no symptomatic improvement and is therefore not recommended. Gastric electrical stimulation, by contrast, is associated with symptomatic benefits. For patients failing conservative treatment, surgical management such as venting gastrostomy, gastrojejunostomy, pyloroplasty, and gastrectomy may be considered.⁴⁴

Summary

With an aging population, there is an increasing proportion of individuals living with chronic medical conditions, including pulmonary pathology, organ failure requiring transplantation, and diabetes mellitus with multiple diabetic complications. Given the complexity of these conditions, management cannot be compartmentalized and handled by only one medical specialist. Additionally, chronic medical diseases often directly or indirectly lead to impairments in patient function; therefore it is imperative for the physiatrist to become knowledgeable and involved in the management of the patient with chronic medical conditions.

Key References

3. Abidi Y, Kovats Z, Bohacs A, et al. Lung transplant rehabilitation—a review. *Life (Basel)*. 2023;13(2):506.
9. Al-Judaibi B, Alqalami I, Sey M, et al. Exercise training for liver transplant candidates. *Transplant Proc*. 2019;51(10):3330-3337.
12. American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription*. Lippincott Williams & Wilkins; 2000.
13. American Thoracic Society, American College of Chest Physicians. ATS/ACCP statement on cardiopulmonary exercise testing. *Am J Respir Crit Care Med*. 2003;167(2):211-277.
14. American Diabetes Association Professional Practice Committee. Diabetes care in the hospital: standards of care in diabetes-2024. *Diabetes Care*. 2024;47(1):S295-S306.
15. American Diabetes Association. *Blood Glucose and A1: Understanding Diabetes Diagnosis*. Accessed June 30, 2024. <https://diabetes.org/about-diabetes/diagnosis>.
19. Bach JR, Smith WH, Michaels J, et al. Airway secretion clearance by mechanical exsufflation for post-poliomyelitis ventilator-assisted individuals. *Arch Phys Med Rehabil*. 1993;74(2):170-177.
20. Bach JR. Noninvasive respiratory management of patients with neuromuscular disease. *Ann Rehabil Med*. 2017;41(4):519-538.
25. Beling J. Lung volume reduction surgery and pulmonary rehabilitation improve exercise capacity and reduce dyspnea during functional activities in people with emphysema. *Cardiopulm Phys Ther J*. 2009;20(2):5-12.
31. Bhatt SP, Patel SB, Anderson EM, et al. Video telehealth pulmonary rehabilitation intervention in chronic obstructive pulmonary disease reduces 30-day readmissions. *Am J Respir Crit Care Med*. 2019;200(4):511-513.
48. Celli BR, Cote CG, Marin JM, et al. The body-mass index, airflow obstruction, dyspnea, and exercise capacity index in chronic obstructive pulmonary disease. *N Engl J Med*. 2004;350(10):1005-1012.
50. Celli BR. Pulmonary rehabilitation in COPD. In: Stoller JK, ed. *Pulmonary Rehabilitation*. Wolters Kluwer; 2024.
51. Centers for Disease Control and Prevention. *Data and Statistics—Chronic obstructive Pulmonary Disease (COPD)*. CDC; 2014.
52. Centers for Disease Control and Prevention. *National Diabetes Statistics Report*. CDC; 2024.
64. Corcoran JR, Herbsman JM, Bushnik T, et al. Early rehabilitation in the medical and surgical intensive care units for patients with and without mechanical ventilation: an interprofessional performance improvement project. *PM&R*. 2017;9(2):113-119.
66. Coutu FA, Iorio OC, Ross BA. Remote patient monitoring strategies and wearable technology in chronic obstructive pulmonary disease. *Front Med (Lausanne)*. 2023;10:1236598.
73. De Smet S, Van Craenenbroeck AH. Exercise training in patients after kidney transplantation. *Clin Kidney J*. 2021;14(2):Sii15-Sii24.
75. Didsbury M, McGee RG, Tong A, et al. Exercise training in solid organ transplant recipients: a systematic review and meta-analysis. *Transplantation*. 2013;95(5):679-687.
77. Duarte-Rojo A, Ruiz-Margáin A, Montaña-Loza AJ, et al. Exercise and physical activity for patients with end-stage liver disease: improving functional status and sarcopenia while on the transplant waiting list. *Liver Transpl*. 2018;24(1):122-139.
80. ElSayed NA, Aleppo G, Aroda VR, et al. Classification and diagnosis of diabetes: standards of care in diabetes-2023. *Diabetes Care*. 2023;46(1):S19-S40.
87. Fuhrmann I, Krause R. Principles of exercising in patients with chronic kidney disease, on dialysis and for kidney transplant recipients. *Clin Nephrol*. 2004;61:S14-S25.
89. Garcia AM, Veneroso CE, Soares DD, et al. Effect of a physical exercise program on the functional capacity of liver transplant patients. *Transplant Proc*. 2014;46(6):1807-1808.
90. Garrido G, Dhillon GS. Medical course and complications after lung transplantation. In: Sher Y, Maldonado JR, eds. *Psychosocial Care of End-Stage Organ Disease and Transplant Patients*. Springer; 2018:279-288.
92. Gerow M, Bruner PJ. *Exercise-Induced Asthma*. StatPearls Publishing; 2024.
94. Global Initiative for Chronic Obstructive Lung Disease. *Pocket Guide to COPD Diagnosis, Management and Prevention—A Guide for Health Care Professionals*. GOLD; 2018.
98. Gozzi-Silva SC, Teixeira FME, Duarte AJS, et al. Immunomodulatory role of nutrients: how can pulmonary dysfunctions improve? *Front Nutr*. 2021;8:674258.
101. Gutierrez-Arias R, Martinez-Zapata MJ, Gaete-Mahn MC, et al. Exercise training for adult lung transplant recipients. *Cochrane Database Syst Rev*. 2021;7(7):CD012307.
103. Hammad A, Kaido T, Aliyev V, et al. Nutritional therapy in liver transplantation. *Nutrients*. 2017;9(10):1126.
105. Hartman JE, Klooster K, Koster TD, et al. Long-term follow-up after bronchoscopic lung volume reduction valve treatment for emphysema. *ERJ Open Res*. 2022;8(4):00235-2022.
108. Heidenreich P, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145(18):e895-e1032.
109. Hermann TS, Dall CH, Christensen SB, et al. Effect of high intensity exercise on peak oxygen uptake and endothelial function in long-term heart transplant recipients. *Am J Transplant*. 2011;11(3):536-541.
112. Horwood CR, Mansour D, Abdel-Rasoul M, et al. Long-term results after lung volume reduction surgery: a single institution's experience. *Ann Thorac Surg*. 2019;107(4):1068-1073.
118. Jetten WD, Hogenbirk RNM, Van Meeteren NLU, et al. Physical effects, safety and feasibility of prehabilitation in patients awaiting orthotopic liver transplantation, a systematic review. *Transpl Int*. 2022;35(10):10330.
123. Kanade R, Kler A, Banga A. Non-pulmonary complications after lung transplantation: part II. *Indian J Thorac Cardiovasc Surg*. 2022;38(2):S290-S299.
124. Kanaly J, Colberg S, Corcoran M, et al. Exercise/physical activity in individuals with type 2 diabetes: a consensus statement from the American College of Sports Medicine. *Med Sci Sports Exerc*. 2022;54(2):353-368.
126. Kanne JP, Little BP, Schulte JJ, et al. Long-term lung abnormalities associated with COVID-19 pneumonia. *Radiology*. 2023;306(2):e221806.
129. Khush KK, Hsich E, Potena L, et al. The International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: thirty-eighth adult heart transplantation report—2021; focus on recipient characteristics. *J Heart Lung Transplant*. 2021;40(10):1035-1049.
133. Kim HH, Jo KW, Shim TS, et al. Incidence, risk factors, and clinical characteristics of airway complications after lung transplantation. *Sci Rep*. 2023;13:667.
134. Kirkeskov L, Carlsen RK, Lund T, et al. Employment of patients with kidney failure treated with dialysis or kidney transplantation—a systematic review and meta-analysis. *BMC Nephrol*. 2021;22(1):348.
139. Kourek C, Karatzanos E, Nanas S, et al. Exercise training in heart transplantation. *World J Transplant*. 2021;11(11):466-479.
141. Kulkarni AV, Tevethia HV, Premkumar M, et al. Impact of COVID-19 on liver transplant recipients—a systematic review and meta-analysis. *EclinMed*. 2021;38:101025.
145. Lavie CJ, Arena R, Earnest CP. High-intensity interval training in patients with cardiovascular diseases and heart transplantation. *J Heart Lung Transplant*. 2013;32(11):1056-1058.
147. Leard LE, Holm AM, Valapour M, et al. Consensus document for the selection of lung transplant candidates: an update from the International Society for Heart and Lung Transplantation. *J Heart Lung Transplant*. 2021;40(11):1349-1379.
155. Long term domiciliary oxygen therapy in chronic hypoxic cor pulmonale complicating chronic bronchitis and emphysema. Report of the Medical Research Council Working Party. *Lancet*. 1981;1(8222):681-686.

157. Lorenz EC, Cheville AI, Amer H, et al. Relationship between pre-transplant physical function and outcomes after kidney transplant. *Clin Transplant*. 2017;31(5):318.
160. Lyu DM, Zamora MR. Medical complications of lung transplantation. *Proc Am Thorac Soc*. 2009;6:101-107.
162. Maffei P, Wiramus S, Bensoussan L, et al. Intensive early rehabilitation in the intensive care unit for liver transplant recipients: a randomized controlled trial. *Arch Phys Med Rehabil*. 2017;98(8):1518-1525.
167. Mannino DM, Roberts MH, Mapel DW, et al. National and local direct medical cost burden of COPD in the United States from 2016 to 2019 and projections through 2029. *Chest J*. 2024;165(5):1093-1106.
170. McAdams-DeMarco MA, Ying H, Rasmussen SVP, et al. Prehabilitation prior to kidney transplantation: results from a pilot study. *Clin Transplant*. 2018;33(1):2-19.
177. Nastasi AJ, Bryant TS, Le JT, et al. Pre-kidney transplant lower extremity impairment and transplant length of stay: a time-to-discharge analysis of a prospective cohort study. *BMC Geriatr*. 2018;18(1):246.
183. Nocturnal Oxygen Therapy Trial Group. Continuous or nocturnal oxygen therapy in hypoxemic chronic obstructive lung disease: a clinical trial. *Ann Intern Med*. 1980;93:391-398.
185. Nytrøen K, Gullestad L. Exercise after heart transplantation: an overview. *World J Transplant*. 2013;3(4):78-90.
187. Oguchi H, Tsujita M, Yazawa M, et al. The efficacy of exercise training in kidney transplant recipients: a meta-analysis and systematic review. *Clin Exp Nephrol*. 2019;23(2):275-284.
192. Panagioti M, Scott C, Blakemore A, et al. Overview of the prevalence, impact, and management of depression and anxiety in chronic obstructive pulmonary disease. *Int J Chron Obstruct Pulmon Dis*. 2014;9:1289-1306.
200. Press VG, Au DH, Bourbeau J, et al. Reducing chronic obstructive pulmonary disease hospital readmissions. An official American Thoracic Society workshop report. *Ann Am Thorac Soc*. 2019;16(2):161-170.
201. Price R, Smith D, Franklin G, et al. Oral and topical treatment of painful diabetic polyneuropathy: practice guideline update summary: report of the AAN Guideline Subcommittee. *Neurology*. 2022;98(1):31-43.
205. Qaseem A, Barry MJ, Humphrey LL, et al. Oral pharmacologic treatment of type 2 diabetes mellitus: a clinical practice guideline update from the American College of Physicians. *Ann Intern Med*. 2017;166(4):279-290.
206. Ramachandran K, McCusker C, Connors M, et al. The influence of obesity on pulmonary rehabilitation outcomes in patients with COPD. *Chron Respir Dis*. 2008;5(4):205-209.
211. Ries AL, Bauldoff GS, Carlin BW, et al. Pulmonary rehabilitation: joint ACCP/AACVPR evidence-based clinical practice guidelines. *Chest*. 2007;131(5):S4-S42.
213. Rochester CL, Alison JA, Carlin B, et al. Pulmonary rehabilitation for adults with chronic respiratory disease: an official American Thoracic Society clinical practice guideline. *Am J Respir Crit Care Med*. 2023;208(4):e7-e26.
217. Rydberg L, Barker K, Lanphere J, et al. Heart transplantation and the role of inpatient rehabilitation: a narrative review. *PM&R*. 2023;15(10):1351-1360.
220. Selzler AM, Wald J, Sedeno M, et al. Telehealth pulmonary rehabilitation: a review of the literature and an example of a nationwide initiative to improve the accessibility of pulmonary rehabilitation. *Chron Respir Dis*. 2018;15(1):41-47.
222. Shah PL, Herth FJ, van Geffen WH, et al. Lung volume reduction for emphysema. *Lancet Respir Med*. 2017;5:147-156.
224. Shaheen SO, Jameson KA, Syddall HE, et al. The relationship of dietary patterns with adult lung function and COPD. *Eur Respir J*. 2010;36(2):277-284.
227. Singh SJ, ZuWallack RL, Garvey C, et al. Spruit on behalf of the American Thoracic Society/European Respiratory Society task force on pulmonary rehabilitation. Learn from the past and create the future: the 2013 ATS/ERS statement on pulmonary rehabilitation. *Eur Respir J*. 2013;42:1169-1174.
229. Spitzer KA, Stefan MS, Priya A, et al. Participation in pulmonary rehabilitation after hospitalization for chronic obstructive pulmonary disease among Medicare beneficiaries. *Ann Am Thorac Soc*. 2019;16(1):99-106.
230. Spruit MA, Singh SJ, Garvey C, et al. An official American Thoracic Society/European Respiratory Society statement: key concepts and advances in pulmonary rehabilitation. *Am J Respir Crit Care Med*. 2013;188(8):e13-e64.
233. Squires RW. Exercise therapy for cardiac transplant recipients. *Prog Cardiovasc Dis*. 2011;53(6):429-436.
246. Tejwani V, Panchabhai TS, Kotloff RM, et al. Complications of lung transplantation: a roentgenographic perspective. *Chest*. 2016;149(6):1535-1545.
256. The US Burden of Disease Collaborators. The state of US health, 1990-2016: burden of diseases, injuries, and risk factors among US states. *JAMA*. 2018;319(14):1444-1472.
259. Valapour M, Lehr CJ, Schladt DP, et al. *OPTN/SRTR 2022 Annual Data Report: Lung*. US Department of Health and Human Services; 2022.
260. van den Berg-Emons R, van Ginneken B, Wijffels M, et al. Fatigue is a major problem after liver transplantation. *Liver Transpl*. 2006;12(6):928-933.
261. van den Berg-Emons RJG, Berbke TJ, van Ginneken CFJ, et al. Fatigue after liver transplantation: effects of a rehabilitation program including exercise training and physical activity counseling. *Phys Ther*. 2014;94(6):857-865.
264. Vucicevic D, Honoris L, Raia F, et al. Current indications for transplantation: stratification of severe heart failure and shared decision-making. *Ann Cardiothorac Surg*. 2018;7(1):56-66.
271. Wickerson L, Rozenberg D, Janaudis-Ferreira T, et al. Physical rehabilitation for lung transplant candidates and recipients: an evidence-informed clinical approach. *World J Transplant*. 2016;6(3):517-531.
273. Wirth KJ, Scheibenbogen C. Dyspnea in post-COVID syndrome following mild acute COVID-19 infections: potential causes and consequences for a therapeutic approach. *Medicina (Kaunas)*. 2022;58(3):419.
275. World Health Organization. *Global Report on Diabetes*. WHO; 2016.
280. Zammit C, Liddicoat H, Moonsie I, et al. Obesity and respiratory diseases. *Int J Gen Med*. 2010;3:335-343.
283. Zhu Q, Yang J, Zhang Y, et al. Early mobilization intervention for patient rehabilitation after renal transplantation. *Am J Transl Res*. 2021;13(6):7300-7305.

Visit Elsevier eBooks+ at [eBooks.Elsevier.com](https://ebooks.elsevier.com) for a complete list of references.

References

- National Heart, Lung and Blood Institute. Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group. NHLB; 2020.
- Åberg F. Quality of life after liver transplantation. *Best Pract Res Clin Gastroenterol.* 2020;46-47:101684.
- Abidi Y, Kovacs Z, Bohacs A, et al. Lung transplant rehabilitation—a review. *Life (Basel).* 2023;13(2):506.
- Abramyan S, Hanlon M. *Kidney Transplantation.* StatPearls Publishing; 2024.
- Adjaye-Gbewonyo D, Vahratian A, Perrine CG, et al. Long COVID in Adults: United States, 2022. NCHS Data Brief No. 480. US Department of Health and Human Services, Centers for Disease Control and Prevention, National Center for Health Statistics; 2023.
- Agrawal A, Ison M, Danziger-Isakov L. Long-term infectious complications of kidney transplantation. *CJASN.* 2022;17(2):286-295.
- Agrawal S, Gupta N, Gonuguntla HK, et al. Evidence-based review of bronchoscopic lung volume reduction. *Adv Respir Med.* 2021;89(1):43-48.
- Aguiar EJ, Morgan PJ, Collins CE, et al. Efficacy of interventions that include diet, aerobic and resistance training components for type 2 diabetes prevention: a systematic review with meta-analysis. *Int J Behav Nutr Phys Act.* 2014;11(1):2.
- Al-Judaibi B, Alqalami I, Sey M, et al. Exercise training for liver transplant candidates. *Transplant Proc.* 2019;51(10):3330-3337.
- Alraies MC, Eckman P. Adult heart transplant: indications and outcomes. *J Thorac Dis.* 2014;6(8):1120-1128.
- Ambrosino N, Strambi S. New strategies to improve exercise tolerance in chronic obstructive pulmonary disease. *Eur Respir J.* 2004;24(2):313-322.
- American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription.* Lippincott Williams & Wilkins; 2000.
- American Thoracic Society, American College of Chest Physicians. ATS/ACCP statement on cardiopulmonary exercise testing. *Am J Respir Crit Care Med.* 2003;167(2):211-277.
- American Diabetes Association Professional Practice Committee. Diabetes care in the hospital: standards of care in diabetes-2024. *Diabetes Care.* 2024;47(1):S295-S306.
- American Diabetes Association. Blood Glucose and A1: Understanding Diabetes Diagnosis. Accessed June 30, 2024. <https://diabetes.org/about-diabetes/diagnosis>.
- Ashour AM. Use of vaping as a smoking cessation aid: a review of clinical trials. *J Multidiscip Healthc.* 2023;16:2137-2144.
- American Thoracic Society. ATS statement: guidelines for the six-minute walk test. *Am J Respir Crit Care Med.* 2002;166:111-117.
- Awad MA, Shah A, Griffith BP. Current status and outcomes in heart transplantation: a narrative review. *Rev Cardiovasc Med.* 2022;23(1):11.
- Bach JR, Smith WH, Michaels J, et al. Airway secretion clearance by mechanical exsufflation for post-poliomyelitis ventilator-assisted individuals. *Arch Phys Med Rehabil.* 1993;74(2):170-177.
- Bach JR. Noninvasive respiratory management of patients with neuromuscular disease. *Ann Rehabil Med.* 2017;41(4):519-538.
- Bach JR. *Management of Patients with Neuromuscular Disease.* 1st ed. Hanley & Belfus; 2003.
- Balducci S, Sacchetti M, Haxhi J, et al. Physical exercise as therapy for type 2 diabetes mellitus. *Diabetes Metab Res Rev.* 2014;30(1):S13-S23.
- Bartels MN, Armstrong HF, Gerardo RE, et al. Evaluation of pulmonary function and exercise performance by cardiopulmonary exercise testing before and after lung transplantation. *Chest.* 2011;140(6):1604-1611.
- Baumgartner WA, Reitz BA, Gott VL, et al. Norman E. Shumway, MD, PhD: visionary, innovator, humorist. *J Thorac Cardiovasc Surg.* 2009;137(2):269-277.
- Beling J. Lung volume reduction surgery and pulmonary rehabilitation improve exercise capacity and reduce dyspnea during functional activities in people with emphysema. *Cardiopulm Phys Ther J.* 2009;20(2):5-12.
- Bělohlávek J, Dytrych V, Linhart A. Pulmonary embolism, part I: epidemiology, risk factors and risk stratification, pathophysiology, clinical presentation, diagnosis and nonthrombotic pulmonary embolism. *Exp Clin Cardiol.* 2013;18(2):129-138.
- Benten D, Stauffer K, Sterneck M. Orthotopic liver transplantation and what to do during follow-up: recommendations for the practitioner. *Gastroenterol Hepatol.* 2009;6:23-36.
- Berthon BS, Wood LG. Nutrition and respiratory health—feature review. *Nutrients.* 2015;7(3):1618-1643.
- Beyer N, Aadahl M, Strange B, et al. Exercise capacity of patients after liver transplantation. *Med Sci Sports Exerc.* 1995;6:S84.
- Beyer N, Aadahl M, Strange B, et al. Improved physical performance after orthotopic liver transplantation. *Liver Transpl Surg.* 1999;5:301-309.
- Bhatt SP, Patel SB, Anderson EM, et al. Video telehealth pulmonary rehabilitation intervention in chronic obstructive pulmonary disease reduces 30-day readmissions. *Am J Respir Crit Care Med.* 2019;200(4):511-513.
- Blau H, Mussaffi-Georgy H, Fink G, et al. Effects of an intensive 4-week summer camp on cystic fibrosis: pulmonary function, exercise tolerance, and nutrition. *Chest.* 2002;121(4):1117-1122.
- Boersma P, Black LI, Ward BW. Prevalence of multiple chronic conditions among US adults, 2018. *Prev Chronic Dis.* 2020;17:200130.
- Bolte RG. Management of pediatric asthma. *Clin Ped Emerg Med.* 2004;5:256-269.
- Borel JF, Feurer C, Gubler HU, et al. Biological effects of cyclosporin A: a new antilymphocytic agent. *Agents Actions.* 1976;6:468-475.
- Bos S, Vos R, Van Raemdonck DE, et al. Survival in adult lung transplantation: where are we in 2020? *Curr Opin Organ Transplant.* 2020;25(3):268-273.
- Bozkurt B, Ahmad T, Alexander KM, et al. Heart failure epidemiology and outcomes statistics: a report of the Heart Failure Society of America. *J Card Fail.* 2023;29(10):1412-1451.
- Braith RW, Edwards DG. Exercise following heart transplantation. *Sports Med.* 2000;30(3):171-192.
- Broekhuizen R, Creutzberg EC, Weling-Scheepers CA, et al. Optimizing oral nutritional drink supplementation in patients with chronic obstructive pulmonary disease. *Br J Nutr.* 2005;93(6):965-971.
- Brug J, Schols A, Mesters I. Dietary change, nutrition education and chronic obstructive pulmonary disease. *Patient Educ Couns.* 2004;52(3):249-257.
- Bryant MS, Bandi VD, Nguyen CK, et al. Telehealth pulmonary rehabilitation for patients with severe chronic obstructive pulmonary disease. *Fed Pract.* 2019;36(9):430-435.
- Cagle Jr SD, Landrum LS, Kennedy AM. Chronic obstructive pulmonary disease: diagnosis and management. *Am Fam Physician.* 2023;107(6):604-612.
- California Pulmonary Rehabilitation Collaborative Group. Effects of pulmonary rehabilitation on dyspnea, quality of life, and health-care costs in California. *J Cardiopulm Rehabil.* 2004;24(1):52-62.
- Camilleri M, Parkman HP, Shafi MA, et al. American College of Gastroenterology. Clinical guideline: management of gastroparesis. *Am J Gastroenterol.* 2013;108(1):18-37, quiz 38.
- Carter R, Al-Rawas OA, Stevenson A, et al. Exercise responses following heart transplantation: 5 year follow-up. *Scott Med J.* 2006;51(3):6-14.
- Casaburi R, Patessio A, Ioli F, et al. Reductions in exercise lactic acidosis and ventilation as a result of exercise training in patients with obstructive lung disease. *Am Rev Respir Dis.* 1991;143(1):9-18.
- Casaburi R. Principles of exercise training. *Chest.* 1992;101(5):S263-S267.
- Celli BR, Cote CG, Marin JM, et al. The body-mass index, airflow obstruction, dyspnea, and exercise capacity index in chronic obstructive pulmonary disease. *N Engl J Med.* 2004;350(10):1005-1012.
- Celli BR, Thomas NE, Anderson JA, et al. Effect of pharmacotherapy on rate of decline of lung function in chronic obstructive

- pulmonary disease: results from the TORCH study. *Am J Respir Crit Care Med*. 2008;178(4):332-338.
50. Celli BR. Pulmonary rehabilitation in COPD. In: Stoller JK, ed. *Pulmonary Rehabilitation*. Wolters Kluwer; 2024.
 51. Centers for Disease Control and Prevention. *Data and Statistics—Chronic obstructive Pulmonary Disease (COPD)*. CDC; 2014.
 52. Centers for Disease Control and Prevention. *National Diabetes Statistics Report*. CDC; 2024.
 53. [Deleted and reviw]
 54. Chambers DC, Cherikh WS, Harhay MO, et al. The International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: thirty-sixth adult lung and heart-lung transplantation report-2019; focus theme: donor and recipient size match. *J Heart Lung Transplant*. 2019;38(10):1042-1055.
 55. Chapman R, Fevery J, Kalloo A, et al. American Association for the Study of Liver Diseases. Diagnosis and management of primary sclerosing cholangitis. *Hepatology*. 2010;51(2):660-678.
 56. Cholakkeril G, Ahmed A. Alcoholic liver disease replaces hepatitis C virus infection as the leading indication for liver transplantation in the United States. *Clin Gastroenterol Hepatol*. 2018;16(8):1356-1351.
 57. Chuncharunee L, Yamashiki N, Thakkinsian A, et al. Alcohol relapse and its predictors after liver transplantation for alcoholic liver disease: a systematic review and meta-analysis. *BMC Gastroenterol*. 2019;19(1):150.
 58. Clayton L, Rizzolo D, Nair V. Kidney transplant wait list: review and current trends. *J Am Acad Physician Assist*. 2018;31(10):1-5.
 59. Colberg SR, Sigal RJ, Yardley JE, et al. Physical activity/exercise and diabetes: a position statement of the American Diabetes Association. *Diabetes Care*. 2016;39(11):2065-2079.
 60. Colberg SR, Sigal RJ, Fernhall B, et al. Exercise and type 2 diabetes: the American College of Sports Medicine and the American Diabetes Association: joint position statement. *Diabetes Care*. 2010;33(12):e147-e167.
 61. Colm CM, Pascual M. Update in renal transplantation. *Arch Intern Med*. 2004;164:1373-1388.
 62. Cooper DKC. Christiaan Barnard—the surgeon who dared: the story of the first human-to-human heart transplant. *Glob Cardiol Sci Pract*. 2018;2018(2):11.
 63. Cooper JD, Patterson GA, Sundaresan RS, et al. Results of 150 consecutive bilateral lung volume reduction procedures in patients with severe emphysema. *J Thorac Cardiovasc Surg*. 1996;112(5):1319-1329, discussion 1329-1330.
 64. Corcoran JR, Herbsman JM, Bushnik T, et al. Early rehabilitation in the medical and surgical intensive care units for patients with and without mechanical ventilation: an interprofessional performance improvement project. *PM&R*. 2017;9(2):113-119.
 65. Cornelius ME, Loretan CG, Wang TW, et al. Tobacco product use among adults—United States, 2020. *MMWR*. 2022;71:397-405.
 66. Coutu FA, Iorio OC, Ross BA. Remote patient monitoring strategies and wearable technology in chronic obstructive pulmonary disease. *Front Med (Lausanne)*. 2023;10:1236598.
 67. Creutzberg EC, Wouters EF, Mostert R, et al. Efficacy of nutritional supplementation therapy in depleted patients with chronic obstructive pulmonary disease. *Nutrition*. 2003;19(2):120-127.
 68. Cromwell EA, Ostrenga JS, Todd JV, et al. Cystic fibrosis prevalence in the United States and participation in the Cystic Fibrosis Foundation patient registry in 2020. *J Cyst Fibros*. 2023;22(3):436-442.
 69. Curtis JR. The long-term outcomes of mechanical ventilation: what are they and how should they be used? *Respir Care*. 2002;47(4):496-505, discussion 505-507.
 70. Dababneh Y, Mousa OY. *Liver Transplantation*. StatPearls Publishing; 2024.
 71. Daines L, Zheng B, Pfeffer P, et al. A clinical review of long-COVID with a focus on the respiratory system. *Curr Opin Pulm Med*. 2022;28(3):174-179.
 72. Davis MK, Hunt SA. State of the art: cardiac transplantation. *Trends Cardiovasc Med*. 2014;24(8):341-349.
 73. De Smet S, Van Craenenbroeck AH. Exercise training in patients after kidney transplantation. *Clin Kidney J*. 2021;14(2):Sii15-Sii24.
 74. Dekhuijzen PN, Beek MM, Folgering HT, et al. Psychological changes during pulmonary rehabilitation and target-flow inspiratory muscle training in COPD patients with a ventilatory limitation during exercise. *Int J Rehabil Res*. 1990;13(2):109-117.
 75. Didsbury M, McGee RG, Tong A, et al. Exercise training in solid organ transplant recipients: a systematic review and meta-analysis. *Transplantation*. 2013;95(5):679-687.
 76. Doucet M, Russell AP, Léger B, et al. Muscle atrophy and hypertrophy signaling in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 2007;176(3):261-269.
 77. Duarte-Rojo A, Ruiz-Margáin A, Montañón-Loza AJ, et al. Exercise and physical activity for patients with end-stage liver disease: improving functional status and sarcopenia while on the transplant waiting list. *Liver Transpl*. 2018;24(1):122-139.
 78. Eatemadololama A, Karimi MT, Rahnama N, et al. Resistance exercise training restores bone mineral density in renal transplant recipients. *Clin Cases Miner Bone Metab*. 2017;14(2):157-160.
 79. Elborn JS. Cystic fibrosis. *Lancet*. 2016;388(10059):2519-2531.
 80. ElSayed NA, Aleppo G, Aroda VR, et al. Classification and diagnosis of diabetes: standards of care in diabetes-2023. *Diabetes Care*. 2023;46(1):S19-S40.
 81. Esposito P, Furini F, Rampino T, et al. Assessment of physical performance and quality of life in kidney-transplanted patients: a cross-sectional study. *Clin Kidney J*. 2017;10(1):124-130.
 82. Fahy BF. Pulmonary rehabilitation for chronic obstructive pulmonary disease: a scientific and political agenda. *Respir Care*. 2004;49(1):28-36, discussion 36-38.
 83. Farkas S, Hackl C, Schlitt HJ. Overview of the indications and contraindications for liver transplantation. *Cold Spring Harb Perspect Med*. 2014;4(5):a015602.
 84. Fishman A, Martinez F, Naunheim K, et al. A randomized trial comparing lung-volume-reduction surgery with medical therapy for severe emphysema. *N Engl J Med*. 2003;348(21):2059-2073.
 85. Florian J, Rubin A, Mattiello R, et al. Impact of pulmonary rehabilitation on quality of life and functional capacity in patients on waiting lists for lung transplantation. *J Bras Pneumol*. 2013;39(3):349-356.
 86. Fryar CD, Carroll MD, Afful J. *Prevalence of Overweight, Obesity, and Severe Obesity Among Adults Aged 20 and Over: United States, 1960–1962 Through 2017–2018*. NCHS Health E-Stats, Centers for Disease Control and Prevention; 2020.
 87. Fuhrmann I, Krause R. Principles of exercising in patients with chronic kidney disease, on dialysis and for kidney transplant recipients. *Clin Nephrol*. 2004;61:S14-S25.
 88. Furlanetto KC, Pitta F. Oxygen therapy devices and portable ventilators for improved physical activity in daily life in patients with chronic respiratory disease. *Expert Rev Med Devices*. 2017;14(2):103-115.
 89. Garcia AM, Veneroso CE, Soares DD, et al. Effect of a physical exercise program on the functional capacity of liver transplant patients. *Transplant Proc*. 2014;46(6):1807-1808.
 90. Garrido G, Dhillon GS. Medical course and complications after lung transplantation. In: Sher Y, Maldonado JR, eds. *Psychosocial Care of End-Stage Organ Disease and Transplant Patients*. Springer; 2018:279-288.
 91. Garuti G, Cilione C, Dell Orso D, et al. Impact of comprehensive pulmonary rehabilitation on anxiety and depression in hospitalized COPD patients. *Monaldi Arch Chest Dis*. 2003;59(1):56-61.
 92. Gerow M, Bruner PJ. *Exercise-Induced Asthma*. StatPearls Publishing; 2024.
 93. Gilgoff IS, Barras DM, Jones MS, et al. Neck breathing: a form of voluntary respiration for the spine-injured ventilator-dependent quadriplegic child. *Pediatrics*. 1988;82(5):741-745.

94. Global Initiative for Chronic Obstructive Lung Disease. *Pocket Guide to COPD Diagnosis, Management and Prevention—A Guide for Health Care Professionals*. GOLD; 2018.
95. Golden SH, Hill-Briggs F, Williams K, et al. Management of diabetes during acute stroke and inpatient stroke rehabilitation. *Arch Phys Med Rehabil*. 2005;86(12):2377-2384.
96. Gonzalez C, Ferris G, Diaz J, et al. Kyphoscoliotic ventilatory insufficiency: effects of long-term intermittent positive-pressure ventilation. *Chest*. 2003;124(3):857-862.
97. Goodwin WE, Martin EC. Transplantation of the kidney. *Urol Surv*. 1963;13:229-248.
98. Gozzi-Silva SC, Teixeira FME, Duarte AJS, et al. Immunomodulatory role of nutrients: how can pulmonary dysfunctions improve? *Front Nutr*. 2021;8:674258.
99. Graziadei I, Zoller H, Fickert P, et al. Indications for liver transplantation in adults: recommendations of the Austrian Society for Gastroenterology and Hepatology (ÖGGH) in cooperation with the Austrian Society for Transplantation, Transfusion and Genetics (ATX). *Wien Klin Wochenschr*. 2016;128(19-20):679-690.
100. Greenwood SA, Koufaki P, Mercer TH, et al. Aerobic or resistance training and pulse wave velocity in kidney transplant recipients: a 12-week pilot randomized controlled trial (the Exercise in Renal Transplant [ExeRT] Trial). *Am J Kidney Dis*. 2015;66:689-698.
101. Gutierrez-Arias R, Martinez-Zapata MJ, Gaete-Mahn MC, et al. Exercise training for adult lung transplant recipients. *Cochrane Database Syst Rev*. 2021;7(7):CD012307.
102. Haas F. *Pulmonary Therapy and Rehabilitation: Principles and Practice*. 2nd ed. Williams & Wilkins; 1991.
103. Hammad A, Kaido T, Aliyev V, et al. Nutritional therapy in liver transplantation. *Nutrients*. 2017;9(10):1126.
104. Harris GD, White RD. Diabetes in the competitive athlete. *Curr Sports Med Rep*. 2012;11(6):309-315.
105. Hartman JE, Klooster K, Koster TD, et al. Long-term follow-up after bronchoscopic lung volume reduction valve treatment for emphysema. *ERJ Open Res*. 2022;8(4):00235-2022.
106. Hayashino Y, Jackson JL, Hirata T, et al. Effects of exercise on C-reactive protein, inflammatory cytokine and adipokine in patients with type 2 diabetes: a meta-analysis of randomized controlled trials. *Metab Clin Exp*. 2014;63(3):431-440.
107. Haykowsky M, Taylor D, Kim D, et al. Exercise training improves aerobic capacity and skeletal muscle function in heart transplant recipients. *Am J Transplant*. 2009;9(4):734-739.
108. Heidenreich P, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145(18):e895-e1032.
109. Hermann TS, Dall CH, Christensen SB, et al. Effect of high intensity exercise on peak oxygen uptake and endothelial function in long-term heart transplant recipients. *Am J Transplant*. 2011;11(3):536-541.
110. Hillerdal G, Löfdahl C-G, Ström K, et al. Comparison of lung volume reduction surgery and physical training on health status and physiologic outcomes: a randomized controlled clinical trial. *Chest*. 2005;128(5):3489-3499.
111. Hong K, Kim H, Lee JM, et al. Fatigue and related factors after liver transplantation. *Korean J Hepatobiliary Pancreat Surg*. 2015;19(4):149-153.
112. Horwood CR, Mansour D, Abdel-Rasoul M, et al. Long-term results after lung volume reduction surgery: a single institution's experience. *Ann Thorac Surg*. 2019;107(4):1068-1073.
113. Hsieh P-L, Wu Y-T, Chao W-J. Effects of exercise training in heart transplant recipients: a meta-analysis. *Cardiology*. 2011;120(1):27-35.
114. Hummel M, Michauk I, Hetzer R, et al. Quality of life after heart and heart-lung transplantation. *Transplant Proc*. 2001;33(7/8):3546-3548.
115. Ilyas JA, O'Mahony CA, Vierling JM. Liver transplantation in autoimmune liver diseases. *Best Pract Res Clin Gastroenterol*. 2011;25(6):765-782.
116. Jaeger RJ, Turba RM, Yarkony GM, et al. Cough in spinal cord injured patients: comparison of three methods to produce cough. *Arch Phys Med Rehabil*. 1993;74(12):1358-1361.
117. Jessup M, Abraham WT, Casey DE, et al. 2009 focused update: ACCF/AHA guidelines for the diagnosis and management of heart failure in adults: a report of the American College of Cardiology Foundation/American Heart Association Task Force on practice guidelines: developed in collaboration with the International Society for Heart and Lung Transplantation. *Circulation*. 2009;119:1977-2016.
118. Jetten WD, Hogenbirk RNM, Van Meeteren NLU, et al. Physical effects, safety and feasibility of prehabilitation in patients awaiting orthotopic liver transplantation, a systematic review. *Transpl Int*. 2022;8(35):10330.
119. Ji Z, Jareño-Esteban JJ, de Miguel-Díez J. Role of vaccines in COPD patients. *Open Respir Arch*. 2022;4(3):100191.
120. Jia S, Taylor-Cousar JL. Cystic fibrosis modulator therapies. *Ann Rev Med*. 2023;74(1):413-426.
121. Jobst S, Schaefer J, Kleiser C, et al. A systematized review of professional employment following thoracic transplantation. *Prog Transplant*. 2022;32(1):55-66.
122. Jubran A, Grant BJB, Duffner LA, et al. Long-term outcome after prolonged mechanical ventilation. A long-term acute-care hospital study. *Am J Respir Crit Care Med*. 2019;199(12):1508-1516.
123. Kanade R, Kler A, Banga A. Non-pulmonary complications after lung transplantation: part II. *Indian J Thorac Cardiovasc Surg*. 2022;38(2):S290-S299.
124. Kanaly J, Colberg S, Corcoran M, et al. Exercise/physical activity in individuals with type 2 diabetes: a consensus statement from the American College of Sports Medicine. *Med Sci Sports Exerc*. 2022;54(2):353-368.
125. Kang SH, Choi R, Han HS, et al. Fatigue and weakness hinder patient social reintegration after liver transplantation. *Clin Mol Hepatol*. 2018;24:402-408.
126. Kanne JP, Little BP, Schulte JJ, et al. Long-term lung abnormalities associated with COVID-19 pneumonia. *Radiology*. 2023;306(2):e221806.
127. Kavanagh T. Exercise rehabilitation in cardiac transplantation patients: a comprehensive review. *Eura Medicophys*. 2005;41(1):67-74.
128. Kawauchi TS, Almeida PO, Lucy KR, et al. Randomized and comparative study between two intra-hospital exercise programs for heart transplant patients. *Rev Bras Cir Cardiovasc*. 2013;28(3):338-346.
129. Khush KK, Hsieh E, Potena L, et al. The International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: thirty-eighth adult heart transplantation report—2021; focus on recipient characteristics. *J Heart Lung Transplant*. 2021;40(10):1035-1049.
130. Kim SH, You HS. The effects of an empowerment education program for kidney transplantation patients. *J Korean Acad Nurs*. 2017;47(4):445-455.
131. Kim SI, Kim YJ, Choi JY, et al. Strategies to reduce infectious complication using epidemiologic data analysis in liver transplant recipients. *Transplant Proc*. 2013;45(8):3061-3064.
132. Kim WR, Terrault NA, Pedersen RA, et al. Trends in waiting list registration for liver transplantation for viral hepatitis in the United States. *Gastroenterology*. 2009;137(5):1680-1686.
133. Kim HH, Jo KW, Shim TS, et al. Incidence, risk factors, and clinical characteristics of airway complications after lung transplantation. *Sci Rep*. 2023;13:667.
134. Kirkeskov L, Carlsen RK, Lund T, et al. Employment of patients with kidney failure treated with dialysis or kidney transplantation—a systematic review and meta-analysis. *BMC Nephrol*. 2021;22(1):348.
135. Klijn PH, Oudshoorn A, van der Ent CK, et al. Effects of anaerobic training in children with cystic fibrosis: a randomized controlled study. *Chest*. 2004;125(4):1299-1305.

136. Klooster K, Slebos DJ. Endobronchial valves for the treatment of advanced emphysema. *Chest*. 2021;159(5):1833-1842.
137. Kodner C, Anderson L, Pohlgeers K. Glucose management in hospitalized patients. *Am Fam Physician*. 2017;96(10):648-654.
138. Koffron A, Stein JA. Liver transplantation: indications, pre-transplant evaluation, surgery and post-transplant complications. *Med Clin North Am*. 2008;92:861-888.
139. Kourek C, Karatzanos E, Nanas S, et al. Exercise training in heart transplantation. *World J Transplant*. 2021;11(11):466-479.
140. Krieger LM, Krieger AJ. The intercostal to phrenic nerve transfer: an effective means of reanimating the diaphragm in patients with high cervical spine injury. *Plast Reconstr Surg*. 2000;105(4):1255-1261.
141. Kulkarni AV, Tevethia HV, Premkumar M, et al. Impact of COVID-19 on liver transplant recipients—a systematic review and meta-analysis. *EClinMed*. 2021;38:101025.
142. Kwong AJ, Kim WR, Lake JR, et al. OPTN/SRTR 2022 annual data report: liver. *Am J Transplant*. 2024;24(2/1):S176-S265.
143. Langer D, Burtin C, Schepers L, et al. Exercise training after lung transplantation improves participation in daily activity: a randomized controlled trial. *Am J Transplant*. 2012;12(6):1584-1592.
144. Lasfargues JE, Custis D, Morrone F, et al. A model for estimating spinal cord injury prevalence in the United States. *Paraplegia*. 1995;33(2):62-68.
145. Lavie CJ, Arena R, Earnest CP. High-intensity interval training in patients with cardiovascular diseases and heart transplantation. *J Heart Lung Transplant*. 2013;32(11):1056-1058.
146. Le Pimpec-Barthes F, Legras A, Arame A, et al. Diaphragm pacing: the state of the art. *J Thorac Dis*. 2016;8(4):S376-S386.
147. Leard LE, Holm AM, Valapour M, et al. Consensus document for the selection of lung transplant candidates: an update from the International Society for Heart and Lung Transplantation. *J Heart Lung Transplant*. 2021;40(11):1349-1379.
148. Lee J, Belanger A, Doucette JT, et al. Transplantation trends in primary biliary cirrhosis. *Clin Gastroenterol Hepatol*. 2007;5(11):1313-1315.
149. Leenen FH, Davies RA, Fournay A. Role of cardiac beta2-receptors in cardiac responses to exercise in cardiac transplant patients. *Circulation*. 1995;91(3):685-690.
150. Leung MKW, Bai D, Yip BHK, et al. Combined nicotine patch with gum versus nicotine patch alone in smoking cessation in Hong Kong primary care clinics: a randomised controlled trial. *BMC Public Health*. 2019;19:1302.
151. Levine SM. Transplant/Immunology Network of the American College of Chest Physicians. A survey of clinical practice of lung transplantation in North America. *Chest*. 2004;125(4):1224-1238.
152. Limongi V, dos Santos DC, da Silva AM, et al. Effects of a respiratory physiotherapeutic program in liver transplantation candidates. *Transplant Proc*. 2014;46(6):1775-1777.
153. Lingiah VA, Niazi M, Olivo R, et al. Liver transplantation beyond Milan criteria. *J Clin Transl Hepatol*. 2020;8(1):69-75.
154. Liu Y, Carlson SA, Watson KB, et al. Trends in the prevalence of chronic obstructive pulmonary disease among adults aged ≥ 18 years—United States, 2011–2021. *MMWR*. 2023;72:1250-1256.
155. Long term domiciliary oxygen therapy in chronic hypoxic cor pulmonale complicating chronic bronchitis and emphysema. Report of the Medical Research Council Working Party. *Lancet*. 1981;1(8222):681-686.
156. Lopez PM, Martin P. Update on liver transplantation: indications, organ allocation and long-term care. *Mt Sinai J Med*. 2006;73:1056-1066.
157. Lorenz EC, Cheville AI, Amer H, et al. Relationship between pre-transplant physical function and outcomes after kidney transplant. *Clin Transplant*. 2017;31(5):318.
158. Lorenzi CM, Cilione C, Rizzardi R, et al. Occupational therapy and pulmonary rehabilitation of disabled COPD patients. *Respiration*. 2004;71(3):246-251.
159. Lozano R, Naghavi M, Foreman K, et al. Global and regional mortality from 235 causes of death for 20 age groups in 1990 and 2010: a systematic analysis for the Global Burden of Disease Study 2010. *Lancet*. 2012;380(9859):2095-2128.
160. Lyu DM, Zamora MR. Medical complications of lung transplantation. *Proc Am Thorac Soc*. 2009;6:101-107.
161. MacLeod SF, Terada T, Chahal BS, et al. Exercise lowers postprandial glucose but not fasting glucose in type 2 diabetes: a meta-analysis of studies using continuous glucose monitoring. *Diabetes Metab Res Rev*. 2013;29(8):593-603.
162. Maffei P, Wiramus S, Bensoussan L, et al. Intensive early rehabilitation in the intensive care unit for liver transplant recipients: a randomized controlled trial. *Arch Phys Med Rehabil*. 2017;98(8):1518-1525.
163. Mah JK, Korngut L, Dykeman J, et al. A systematic review and meta-analysis on the epidemiology of Duchenne and Becker muscular dystrophy. *Neuromuscul Disord*. 2014;24(6):482-491.
164. Mahlmeister MJ, Fink JB, Hoffman GL, et al. Positive-expiratory-pressure mask therapy: theoretical and practical considerations and a review of the literature. *Respir Care*. 1991;36(11):1218-1230.
165. Man SF, McAlister FA, Anthonisen NR, et al. Contemporary management of chronic obstructive pulmonary disease: clinical applications. *JAMA*. 2003;290(17):2313-2316.
166. Mancini D, Lietz K. Selection of cardiac transplantation candidates in 2010. *Circulation*. 2010;122(2):173-183.
167. Mannino DM, Roberts MH, Mapel DW, et al. National and local direct medical cost burden of COPD in the United States from 2016 to 2019 and projections through 2029. *Chest J*. 2024;165(5):1093-1106.
168. Martin AK, Ripoll JG, Wilkey BJ, et al. Analysis of outcomes in heart transplantation. *J Cardiothorac Vasc Anesth*. 2020;34(2):551-561.
169. Mathur S, Reid WD, Levy RD. Exercise limitation in recipients of lung transplants. *Phys Ther*. 2004;84(12):1178-1187.
170. McAdams-DeMarco MA, Ying H, Rasmussen SVP, et al. Prehabilitation prior to kidney transplantation: results from a pilot study. *Clin Transplant*. 2018;33(1):2-19.
171. McCool FD, Rosen MJ. Nonpharmacologic airway clearance therapies: ACCP evidence-based clinical practice guidelines. *Chest*. 2006;129(1):S250-S259.
172. Morris NR, Walsh J, Adams L, et al. Exercise training in COPD: what is it about intensity? *Respirology*. 2016;21(7):1185-1192.
173. Mota S, Güell R, Barreiro E, et al. Clinical outcomes of expiratory muscle training in severe COPD patients. *Respir Med*. 2007;101(3):516-524.
174. Moya-Nájera D, Moya-Herraiz Á, Compte-Torrero L, et al. Combined resistance and endurance training at a moderate-to-high intensity improves physical condition and quality of life in liver transplant patients. *Liver Transpl*. 2017;23(10):1273-1281.
175. Munsif M, Levin K, Snell G, et al. Return to work in lung transplant recipients: an Australian perspective—predictors, health status and outcomes. *J Heart Lung Transplant*. 2019;38(4):S328-S329.
176. Murugan AT, Sharma G. Obesity and respiratory diseases. *Chron Respir Dis*. 2008;5(4):233-242.
177. Nastasi AJ, Bryant TS, Le JT, et al. Pre-kidney transplant lower extremity impairment and transplant length of stay: a time-to-discharge analysis of a prospective cohort study. *BMC Geriatr*. 2018;18(1):246.
178. Nastasi AJ, McAdams-DeMarco MA, Schrack J, et al. Pre-kidney transplant lower extremity impairment and post kidney transplant mortality. *Am J Transplant*. 2018;18(1):189-196.
179. National Heart, Lung, and Blood Institute, National Asthma Education and Prevention Program. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma. NIH Publication No. 07-4051. NHLBI; 1997/2002/2007.
180. Naunheim KS, Wood DE, Mohsenifar Z, et al. Long-term follow-up of patients receiving lung-volume-reduction surgery versus medical therapy for severe emphysema by the National Emphysema Treatment Trial Research Group. *Ann Thorac Surg*. 2006;82(2):431-443.

181. Naylor JM, McLean A, Chow C-M, et al. A modified postural drainage position produces less cardiovascular stress than a head-down position in patients with severe heart disease: a quasi-experimental study. *Aust J Physiother.* 2006;52(3):201-209.
182. Nichol KL, Baken L, Wuorenma J, et al. The health and economic benefits associated with pneumococcal vaccination of elderly persons with chronic lung disease. *Arch Intern Med.* 1999;159(20):2437-2442.
183. Nocturnal Oxygen Therapy Trial Group. Continuous or nocturnal oxygen therapy in hypoxemic chronic obstructive lung disease: a clinical trial. *Ann Intern Med.* 1980;93:391-398.
184. Norman AC, Drinkard B, McDuffie JR, et al. Influence of excess adiposity on exercise fitness and performance in overweight children and adolescents. *Pediatrics.* 2005;115(6):e690-e696.
185. Nytrøen K, Gullestad L. Exercise after heart transplantation: an overview. *World J Transplant.* 2013;3(4):78-90.
186. Nytrøen K, Rustad LA, Aukrust P, et al. High-intensity interval training improves peak oxygen uptake and muscular exercise capacity in heart transplant recipients. *Am J Transplant.* 2012;12(11):3134-3142.
187. Oguchi H, Tsujita M, Yazawa M, et al. The efficacy of exercise training in kidney transplant recipients: a meta-analysis and systematic review. *Clin Exp Nephrol.* 2019;23(2):275-284.
188. Onofre T, Junior JFF, Amorim CF, et al. Impact of an early physiotherapy program after kidney transplant during hospital stay: a randomized controlled trial. *J Bras Nefrol.* 2017;39(4):424-432.
189. Oo T, Watt JW, Soni BM, et al. Delayed diaphragm recovery in 12 patients after high cervical spinal cord injury. A retrospective review of the diaphragm status of 107 patients ventilated after acute spinal cord injury. *Spinal Cord.* 1999;37(2):117-122.
190. OPTN Organizational Metrics Dashboard. UNOS: United Network for Organ Sharing. Accessed July 28, 2024. <https://insights.unos.org/OPTN-metrics/>.
191. Örem C. Epidemiology of pulmonary hypertension in the elderly. *J Geriatr Cardiol.* 2017;14(1):11-16.
192. Panagioti M, Scott C, Blakemore A, et al. Overview of the prevalence, impact, and management of depression and anxiety in chronic obstructive pulmonary disease. *Int J Chron Obstruct Pulmon Dis.* 2014;9:1289-1306.
193. Pastva A, Estell K, Schoeb TR, et al. Aerobic exercise attenuates airway inflammatory responses in a mouse model of atopic asthma. *J Immunol.* 2004;172(7):4520-4526.
194. Patessio A, Carone M, Ioli F, et al. Ventilatory and metabolic changes as a result of exercise training in COPD patients. *Chest.* 1992;101(5):S274-S278.
195. Paul C. Skin grafting in burns. *Wounds.* 2008;20(7):199-202.
196. Pauwels RA, Buist AS, Calverley PM, et al. Global strategy for the diagnosis, management, and prevention of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2012;163(5):1256-1276.
197. Pinter J, Hanson CS, Chapman JR, et al. Perspectives of older kidney transplant recipients on kidney transplantation. *Clin J Am Soc Nephrol.* 2017;12(3):443-453.
198. Planas M, Alvarez J, García-Peris PA, et al. Nutritional support and quality of life in stable chronic obstructive pulmonary disease (COPD) patients. *Clin Nutr.* 2005;24(3):433-441.
199. Press VG, Myers LC, Feemster LC. Preventing COPD readmissions under the hospital readmissions reduction program: how far have we come? *Chest.* 2021;159(3):996-1006.
200. Press VG, Au DH, Bourbeau J, et al. Reducing chronic obstructive pulmonary disease hospital readmissions. An official American Thoracic Society workshop report. *Ann Am Thorac Soc.* 2019;16(2):161-170.
201. Price R, Smith D, Franklin G, et al. Oral and topical treatment of painful diabetic polyneuropathy: practice guideline update summary: report of the AAN Guideline Subcommittee. *Neurology.* 2022;98(1):31-43.
202. Pumar MI, Gray CR, Walsh JR, et al. Anxiety and depression—important psychological comorbidities of COPD. *J Thorac Dis.* 2014;6(11):1615-1631.
203. Qaseem A, Obley AJ, Shamliyan T, et al. Newer pharmacologic treatments in adults with type 2 diabetes: a clinical guideline from the American College of Physicians. *Ann Intern Med.* 2024;177(5):658-666.
204. Qaseem A, Snow V, Fitterman N, et al. Clinical Efficacy Assessment Subcommittee of the American College of Physicians. Risk assessment for and strategies to reduce perioperative pulmonary complications for patients undergoing noncardiothoracic surgery: a guideline from the American College of Physicians. *Ann Intern Med.* 2006;144:575-580.
205. Qaseem A, Barry MJ, Humphrey LL, et al. Oral pharmacologic treatment of type 2 diabetes mellitus: a clinical practice guideline update from the American College of Physicians. *Ann Intern Med.* 2017;166(4):279-290.
206. Ramachandran K, McCusker C, Connors M, et al. The influence of obesity on pulmonary rehabilitation outcomes in patients with COPD. *Chron Respir Dis.* 2008;5(4):205-209.
207. Ramaswamy A, Puchalski J. Bronchoscopic lung volume reduction: recent updates. *J Thorac Dis.* 2018;10(4):2519-2527.
208. Reese PP, Bloom RD, Shults J, et al. Functional status and survival after kidney transplantation. *Transplantation.* 2014;97:189-195.
209. Renau LP, Lopez JH, Iranzo MAC, et al. Physical frailty in liver transplantation. *Rev Esp Enferm Dig.* 2021;113(7):533-540.
210. Reusch JE, Bridenstine M, Regensteiner JG. Type 2 diabetes mellitus and exercise impairment. *Rev Endocr Metab Disord.* 2013;14(1):77-86.
211. Ries AL, Bauldoff GS, Carlin BW, et al. Pulmonary rehabilitation: joint ACCP/AACVPR evidence-based clinical practice guidelines. *Chest.* 2007;131(5):S4-S42.
212. Ries AL, Make BJ, Lee SM, et al. The effects of pulmonary rehabilitation in the National Emphysema Treatment Trial. *Chest.* 2005;128(6):3799-3809.
213. Rochester CL, Alison JA, Carlin B, et al. Pulmonary rehabilitation for adults with chronic respiratory disease: an official American Thoracic Society clinical practice guideline. *Am J Respir Crit Care Med.* 2023;208(4):e7-e26.
214. Rubin BK, van der Schans CP, Kishioka C, et al. Mucus and mucociliary therapy in chronic bronchitis. *Clin Pulm Med.* 1998;5(1):1-14.
215. Rushakoff RJ. Inpatient diabetes management. In: Feingold KR, Anawalt B, Blackman MR, et al., eds. *Endotext*. MDText.com, Inc.; 2000.
216. Rutten EP, Franssen FM, Engelen MP, et al. Greater whole-body myofibrillar protein breakdown in cachectic patients with chronic obstructive pulmonary disease. *Am J Clin Nutr.* 2006;83(4):829-834.
217. Rydberg L, Barker K, Lanphere J, et al. Heart transplantation and the role of inpatient rehabilitation: a narrative review. *PM&R.* 2023;15(10):1351-1360.
218. Schellenberg ES, Dryden DM, Vandermeer B, et al. Lifestyle interventions for patients with and at risk for type 2 diabetes: a systematic review and meta-analysis. *Ann Intern Med.* 2013;159(8):543-551.
219. Schwingshackl L, Missbach B, Dias S, et al. Impact of different training modalities on glycaemic control and blood lipids in patients with type 2 diabetes: a systematic review and network meta-analysis. *Diabetologia.* 2014;57(9):1789-1797.
220. Selzler AM, Wald J, Seden M, et al. Telehealth pulmonary rehabilitation: a review of the literature and an example of a nationwide initiative to improve the accessibility of pulmonary rehabilitation. *Chron Respir Dis.* 2018;15(1):41-47.
221. Shah CH, Reed RM, Wastila L, et al. Direct medical costs of COPD in the USA: an analysis of the Medical Expenditure Panel Survey 2017-2018. *Appl Health Econ Health Policy.* 2023;21(6):915-924.

222. Shah PL, Herth FJ, van Geffen WH, et al. Lung volume reduction for emphysema. *Lancet Respir Med*. 2017;5:147-156.
223. Shahar E, Whitney CW, Redline S, et al. Sleep-disordered breathing and cardiovascular disease: cross-sectional results of the Sleep Heart Health Study. *Am J Respir Crit Care Med*. 2001;163(1):19-25.
224. Shaheen SO, Jameson KA, Syddall HE, et al. The relationship of dietary patterns with adult lung function and COPD. *Eur Respir J*. 2010;36(2):277-284.
225. Shaul DB, Danielson PD, McComb JG, et al. Thoracoscopic placement of phrenic nerve electrodes for diaphragmatic pacing in children. *J Pediatr Surg*. 2002;37(7):974-978, discussion 974-978.
226. Silverman EK, Sandhaus RA. Clinical practice: alpha1-antitrypsin deficiency. *N Engl J Med*. 2009;360(26):2749-2757.
227. Singh SJ, ZuWallack RL, Garvey C, et al. Spruit on behalf of the American Thoracic Society/European Respiratory Society task force on pulmonary rehabilitation. Learn from the past and create the future: the 2013 ATS/ERS statement on pulmonary rehabilitation. *Eur Resp J*. 2013;42:1169-1174.
228. Solomon NA, McGiven JR, Alison PM, et al. Changing donor and recipient demographics in a heart transplantation program: influence on early outcome. *Ann Thorac Surg*. 2004;77(6):2096-2102.
229. Spitzer KA, Stefan MS, Priya A, et al. Participation in pulmonary rehabilitation after hospitalization for chronic obstructive pulmonary disease among Medicare beneficiaries. *Ann Am Thorac Soc*. 2019;16(1):99-106.
230. Spruit MA, Singh SJ, Garvey C, et al. An official American Thoracic Society/European Respiratory Society statement: key concepts and advances in pulmonary rehabilitation. *Am J Respir Crit Care Med*. 2013;188(8):e13-e64.
231. Spruit MA, Troosters T, Trappenburg JC, et al. Exercise training during rehabilitation of patients with COPD: a current perspective. *Patient Educ Couns*. 2004;52(3):243-248.
232. Squires RW, Bonikowske AR. Cardiac rehabilitation for heart transplant patients: considerations for exercise training. *Prog Cardiovasc Dis*. 2022;70:40-48.
233. Squires RW. Exercise therapy for cardiac transplant recipients. *Prog Cardiovasc Dis*. 2011;53(6):429-436.
234. Starzl TE, Todo S, Fung J, et al. FK 506 for human liver, kidney and pancreas transplantation. *Lancet*. 1989;2:1000-1004.
235. Stefan MS, Pekow PS, Priya A, et al. Association between initiation of pulmonary rehabilitation and rehospitalizations in patients hospitalized with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 2021;204(9):1015-1023.
236. Stempel DA. The pharmacologic management of childhood asthma. *Pediatr Clin N Am*. 2003;50:609-629.
237. Stewart I, Jacob J, George PM, et al. Residual lung abnormalities after COVID-19 hospitalization: interim analysis of the UKILD post-COVID-19 study. *Am J Respir Crit Care Med*. 2023;207(6):693-703.
238. Strasser B, Pesta D. Resistance training for diabetes prevention and therapy: experimental findings and molecular mechanisms. *Biomed Res Int*. 2013;805217.
239. Strath SJ, Kaminsky LA, Ainsworth BE, et al. Guide to the assessment of physical activity: clinical and research applications: a scientific statement from the American Heart Association. *Circulation*. 2013;128(20):2259-2279.
240. Substance Abuse and Mental Health Services Administration. 2015 National Survey on Drug Use and Health: Detailed Tables. 2016. Accessed September 8, 2016.
241. Sugawara Y, Makuuchi M. Liver transplantation for hepatitis B-related cirrhosis: recent advances. *J Hepatobiliary Pancreat Surg*. 2006;13(5):378-381.
242. Syamlal G, Doney B, Mazurek JM. Chronic obstructive pulmonary disease prevalence among adults who have never smoked, by industry and occupation—United States, 2013–2017. *MMWR*. 2019;68:303-307.
243. Takaoka ST, Weinacker AB. The value of preoperative pulmonary rehabilitation. *Thorac Surg Clin*. 2005;15(2):203-211.
244. Tang NL, Chung ML, Elia M, et al. Total daily energy expenditure in wasted chronic obstructive pulmonary disease patients. *Eur J Clin Nutr*. 2002;56(4):282-287.
245. Tegtbur U, Busse MW, Jung K, et al. Time course of physical reconditioning during exercise rehabilitation late after heart transplantation. *J Heart Lung Transplant*. 2005;24(3):270-274.
246. Tejwani V, Panchabhai TS, Kotloff RM, et al. Complications of lung transplantation: a roentgenographic perspective. *Chest*. 2016;149(6):1535-1545.
247. Telehealth.HHS.gov. Telehealth and Remote Patient Monitoring. Accessed October 21, 2024. <https://telehealth.hhs.gov/providers/preparing-patients-for-telehealth/telehealth-and-remote-patient-monitoring>.
248. Terrault N, Francoz C, Berenguer M. Liver transplantation 2023: status report, current and future challenges. *Clin Gastroenterol Hepatol*. 2023;21(8):2150-2166.
249. Thabut G, Christie JD, Ravaud P, et al. Survival after bilateral versus single lung transplantation for patients with chronic obstructive pulmonary disease: a retrospective analysis of registry data. *Lancet*. 2008;371(9614):744-751.
250. Thabut G, Mal H. Outcomes after lung transplantation. *J Thorac Dis*. 2017;9(8):2684-2691.
251. Thomson CC, Clark S, Camargo CA, et al. Body mass index and asthma severity among adults presenting to the emergency department. *Chest*. 2003;124(3):795-802.
252. Toder DS. Respiratory problems in the adolescent with developmental delay. *Adolesc Med*. 2000;11(3):617-631.
253. Toyoda Y, Guy TS, Kashem A. Present status and future perspectives of heart transplantation. *Circ J*. 2013;77(5):1097-1110.
254. Trulock EP, Christie JD, Edwards LB, et al. Registry of the International Society for Heart and Lung Transplantation: twenty-fourth official adult lung and heart-lung transplantation report—2007. *J Heart Lung Transplant*. 2007;26(8):782-795.
255. Tsuboi N, Hiratsuka M, Kaneko S, et al. Benefits of early mobilization after pediatric liver transplantation. *Pediatr Crit Care Med*. 2019;20(2):e91-e97.
256. The US Burden of Disease Collaborators. The state of US health, 1990-2016: burden of diseases, injuries, and risk factors among US states. *JAMA*. 2018;319(14):1444-1472.
257. US Census Bureau. American Community Survey. US Census Bureau; 2022.
258. US Renal Data System. 2023 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2023.
259. Valapour M, Lehr CJ, Schladt DP, et al. OPTN/SRTR 2022 Annual Data Report: Lung. US Department of Health and Human Services; 2022.
260. van den Berg-Emons R, van Ginneken B, Wijffels M, et al. Fatigue is a major problem after liver transplantation. *Liver Transpl*. 2006;12(6):928-933.
261. van den Berg-Emons RJG, Berbke TJ, van Ginneken CFJ, et al. Fatigue after liver transplantation: effects of a rehabilitation program including exercise training and physical activity counseling. *Phys Ther*. 2014;94(6):857-865.
262. van den Ham EC, Kooman JP, Schols AM, et al. Similarities in skeletal muscle strength and exercise capacity between renal transplant and hemodialysis patients. *Am J Transplant*. 2005;5:1957-1965.
263. Varraso R. Nutrition and asthma. *Curr Allergy Asthma Rep*. 2012;12(3):201-210.
264. Vucicevic D, Honoris L, Raia F, et al. Current indications for transplantation: stratification of severe heart failure and shared decision-making. *Ann Cardiothorac Surg*. 2018;7(1):56-66.
265. Wacławski ER, Noone P. Systematic review: impact of liver transplantation on employment. *Occup Med*. 2018;68(2):88-95.

266. Wadell K, Sundelin G, Henriksson-Larsén K, et al. High intensity physical group training in water—an effective training modality for patients with COPD. *Respir Med.* 2004;98(5):428-438.
267. Weeks DL, Daratha KB, Towle LA. Diabetes prevalence and influence on resource use in Washington state inpatient rehabilitation facilities, 2001 to 2007. *Arch Phys Med Rehabil.* 2009;90(11):1937-1943.
268. Weiss ES, Allen JG, Merlo CA, et al. Survival after single versus bilateral lung transplantation for high-risk patients with pulmonary fibrosis. *Ann Thorac Surg.* 2009;88(5):1616-1625, discussion 1625-1626.
269. Whitson BA, Hayes Jr D. Indications and outcomes in adult lung transplantation. *J Thorac Dis.* 2014;6(8):1018-1023.
270. Wickerson L, Mathur S, Brooks D. Exercise training after lung transplantation: a systematic review. *J Heart Lung Transplant.* 2010;29(5):497-503.
271. Wickerson L, Rozenberg D, Janaudis-Ferreira T, et al. Physical rehabilitation for lung transplant candidates and recipients: an evidence-informed clinical approach. *World J Transplant.* 2016;6(3):517-531.
272. Wijkstra PJ, Avendaño MA, Goldstein RS. Inpatient chronic assisted ventilatory care: a 15-year experience. *Chest.* 2003;124(3):850-856.
273. Wirth KJ, Scheibenbogen C. Dyspnea in post-COVID syndrome following mild acute COVID-19 infections: potential causes and consequences for a therapeutic approach. *Medicina (Kaunas).* 2022;58(3):419.
274. Wood LG, Garg ML, Smart JM, et al. Manipulating antioxidant intake in asthma: a randomized controlled trial. *Am J Clin Nutr.* 2012;96(3):534-543.
275. World Health Organization. *Global Report on Diabetes.* WHO; 2016.
276. Xu XF, Feng T, Tian YF, et al. Pharmaceutical care in kidney transplant recipients: behavioral and physiologic outcomes at 12 months. *Transplant Proc.* 2018;50(8):2451-2456.
277. Yadav SK, Choudhary NS, Saraf N, et al. Nutritional status using subjective global assessment independently predicts outcome of patients waiting for living donor liver transplant. *Indian J Gastroenterol.* 2017;36(4):275-281.
278. Yeung JC, Keshavjee S. Overview of clinical lung transplantation. *Cold Spring Harb Perspect Med.* 2014;4(1):a015628.
279. Zahalka SJ, Abushamat LA, Scalzo RL, et al. The role of exercise in diabetes. In: Feingold KR, Anawalt B, Blackman MR, et al., eds. *Endotext.* MDText.com, Inc.; 2000.
280. Zammit C, Liddicoat H, Moonsie I, et al. Obesity and respiratory diseases. *Int J Gen Med.* 2010;3:335-343.
281. Zelle DM, Klaasen G, van Adrichem E, et al. Physical inactivity: a risk factor and target for intervention in renal care. *Nat Rev Nephrol.* 2017;13:152-166.
282. Zhang L, Maitinuer A, Lian Z, et al. Home based pulmonary tele-rehabilitation under telemedicine system for COPD: a cohort study. *BMC Pulm Med.* 2022;22(1):284.
283. Zhu Q, Yang J, Zhang Y, et al. Early mobilization intervention for patient rehabilitation after renal transplantation. *Am J Transl Res.* 2021;13(6):7300-7305.
284. Zu Wallack RL, Patel K, Reardon JZ, et al. Predictors of improvement in the 12-minute walking distance following a six-week outpatient pulmonary rehabilitation program. *Chest.* 1991;99(4):805-808.